

Industrial Commission of North Dakota

Oil and Gas Division

Spill / Incident Report

Date/Time Reported : Jan 11 2015 / 16:26

State Agency person :

Responsible Party : OASIS PETROLEUM NORTH AMERICA LLC

Well Operator : OASIS PETROLEUM NORTH AMERICA LLC

Date/Time of Incident : 1/6/2015 12:00:00 AM

NDIC File Number : 23230

Facility Number :

Well or Facility Name : ASH FEDERAL 5300 11-18T

Type of Incident : Valve/Piping Connection Leak

Field Name : BAKER

County : MCKENZIE

Section : 18

Township : 153

Range : 100

Quarter-Quarter :

Quarter :

Distance to nearest residence : 3 Mile

Distance to nearest water well : 3 Mile

Release Oil : 1 Barrels

Release Brine :

Release Other :

Recovered Oil : 1 Barrels

Recovered Brine :

Recovered Other :

Has/Will the incident be reported to the NRC? : No

Was release contained : Yes - On Constructed Well Site

Description of other released substance :

Immediate risk evaluation : None just a little surface contamination, cleaned up with a skid steer
Followup Report Requested Y/N : N



SUNDRY NOTICE AND REPORTS ON WELLS - FORM
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
23230

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed February 8, 2013
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Ash Federal 5300 11-18T					
Footages 800 F N L 350 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective February 8, 2013 the well referenced above is on pump.

Tubing: 2-7/8" L-80 tubing @ 10127

Pump: 2-1/2" x 2.0" x 24' insert pump @ 10196

Company Oasis Petroleum North America LLC		Telephone Number 281 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Heather McCowan</i>		Printed Name Heather McCowan	
Title Regulatory Assistant		Date April 4, 2013	
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date <i>April 10, 2013</i>	
By <i>[Signature]</i>	
Title PETROLEUM ENGINEER	

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
23230

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Ash Federal 5300 11-18T		Spacing Unit Description T153N R100W Section 17 & 18	
Operator Oasis Petroleum North America LLC		Telephone Number (281) 404-9491	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		Pool Bakken	
State TX		Zip Code 77002	
Permit Type ☐ Wildcat ☒ Development ☐ Extension			

LOCATION OF WELL

At Surface 800 F N L	350 F WL	Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W	County McKenzie
Spud Date August 22, 2012	Date TD Reached September 15, 2012	Drilling Contractor and Rig Number Nabors B22		KB Elevation (Ft) 2078	Graded Elevation (Ft) 2053	
Type of Electric and Other Logs Run (See Instructions) N/A						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	Type	String Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8		2200	13 1/2	36				0
Vertical Hole	Intermediate	7		11101	8 3/4	29 & 32				2150
Lateral1	Liner	4 1/2	10197	20447	6	11.6				

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD Ft)		Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
			Top	Bottom						
Lateral1	20482 14938	Perforations	11101	20482 14938	10335		10/26/2012			
sidetrack1	20482		13332	20482						

PRODUCTION

Current Producing Oper Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11, '101'-20,482'						Name of Zone (If Different from Pool Name) Three Forks				
Date Well Completed (SEE INSTRUCTIONS) November 7, 2012			Producing Method Flowing		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In) Producing		
Date of Test 11/20/2012	Hours Tested 24	Choke Size 68 /64	Production for Test		Oil (Bbls) 2734	Gas (MCF) 2940	Water (Bbls) 3564	Oil Gravity-API (Corr.) 42.6 °	Disposition of Gas Flared	
Flowing Tubing Pressure (PSI)			Flowing Casing Pressure (PSI) 594		Calculated 24-Hour Rate	Oil (Bbls) 2734	Gas (MCF) 2940	Water (Bbls) 3564	Gas-Oil Ratio 1075	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Drill Stem Test								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
	cmt							
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 10/26/2012	Stimulated Formation Three Forks	Top (Ft) 11101	Bottom (Ft) 20482	Stimulation Stages 36	Volume 92216	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 3700348	Maximum Treatment Pressure (PSI) 9214		Maximum Treatment Rate (BBLS/Min) 42.2	
Details 100 mesh-144,777 40/70 sand- 1,426,577 20/40 ceramic-2,128,994						

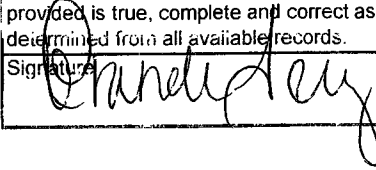
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.			Email Address bterry@oasispetroleum.com	Date 01/16/2013
Signature 	Printed Name Brandi Terry	Title Regulatory Specialist		



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No.
23230
NDIC CTB No.
1 23230

220275

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number ASH FEDERAL 5300 11-18T	Qtr-Qtr LOT1	Section 18	Township 153N	Range 100W	County #N/A McKenzie
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Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
---	------------------------------------	----------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9573	% Purchased 100%	Date Effective November 7, 2012
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Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Field Address	City	State	Zip Code
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Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective November 7, 2012
----------------------------------	------------------------------------	----------------------	------------------------------------

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
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The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Blackstone Crude Oil LLC	25%	November 7, 2012
Other Transporters Transporting From This Lease	% Transported	Date Effective
Comments		

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date November 9, 2012
Signature <i>Annette Terrell</i>	Printed Name Annette Terrell	Title Marketing Assistant

Above Signature Witnessed By: Signature <i>Dina Barron</i>		Printed Name Dina Barron	Title Mktg. Contracts Administrator
--	--	-----------------------------	--

Industrial Commission of North Dakota
Oil and Gas Division
Verbal Approval To Purchase and Transport Oil

Well or Facility No
23230

Tight Hole **Yes**

OPERATOR

Operator OASIS PETROLEUM NORTH AMERICA LL	Representative Terry Nelson	Rep Phone (406) 489-3036
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WELL INFORMATION

Well Name ASH FEDERAL 5300 11-18T	Inspector Richard Dunn
Well Location QQ Sec Twp Rng LOT1 18 153 N 100 W	County MCKENZIE
Footages 800 Feet From the N Line	Field BAKER
350 Feet From the W Line	Pool BAKKEN
Date of First Production Through Permanent Wellhead	This Is Not The First Sales

PURCHASER / TRANSPORTER

Purchaser BANNER PIPELINE COMPANY LLC	Transporter BANNER TRANSPORTATION CO., LLC
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TANK BATTERY

Single Well Tank Battery Number :

SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD	DATE
15000	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	

DETAILS

Must E-Mail or Call Inspector at 701-770-3554/rsdunn@nd.gov on first date of sales and report amount sold, date sold, and first date of production through the permanent wellhead. Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date	11/6/2012
Date Approved	11/6/2012
Approved By	Richard Dunn



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No. **23230**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 10, 2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Waiver from tubing/packer requirement

Well Name and Number Ash Federal 5300 11-18T					
Footages 800 F N L 350 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a waiver from the tubing/pkr requirement included in NDAC 43-02-03-21: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.
The following assurances apply:

1. The well is equipped with new 29# & 32# casing at surface with an API burst rating of 11,220 psi
2. The frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure.
3. Damage to the casing during the frac would be detected immediately by monitoring equipment.
4. The casing is exposed to significantly lower rates and pressures during flow back than during the frac job.
5. The frac fluid and formation fluids have very low corrosion and erosion rates.
6. Production equipment will be installed as soon as possible after the well ceases flowing.
7. A 300# gauge will be installed on the surface casing during the flowback period.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 		Printed Name Brandi Terry	
Title Regulatory Specialist		Date October 10, 2012	
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date October 15, 2012	
By 	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

23230

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
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☒ Notice of Intent	Approximate Start Date October 10, 2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Ash Federal 5300 11-18T					
Footages 800 F N L 350 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective immediately, we request CONFIDENTIAL STATUS for the above referenced well.

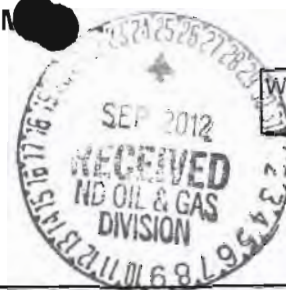
Ends 4-11-2013

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Brandi Terry</i>	Printed Name Brandi Terry		
Title Regulatory Specialist	Date October 10, 2012		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 10-11-2012	
By <i>David Talar</i>	
Title Engineering Technician	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
23230

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date September 27, 2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	
Approximate Start Date	

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☒ Reclamation
☐ Other	<u>Reserve pit reclamation</u>

Well Name and Number Ash Federal 5300 11-18T					
Footages 800 F N L 350 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Excel Industries, Inc			
Address P.O. Box 159	City Miles City	State MT	Zip Code 59301

DETAILS OF WORK

Oasis Petroleum North America LLC plans to reclaim the reserve pit for the above referenced wells as follows:
The NDIC field inspector, Rick Dunn and the surface owners, were notified on 09-21-2012.
Surface owners: Stewart and Alicia Kline, 13794 46th Street NW, Williston, ND 58801
There is no fluid in the pit to dispose of. Spread cuttings out in pit and backfill with clay, fold edges of liner over pit to completely cover pit. Slope and contour wellsite to ensure proper drainage.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Brandi Terry</i>	Printed Name Brandi Terry		
Title Regulatory Specialist	Date September 24, 2012		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 9-26-12	
By <i>[Signature]</i>	
Title <i>[Signature]</i>	



RYAN DIRECTIONAL SERVICES, INC.
A NABORS COMPANY

19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Monday, October 01, 2012

State of North Dakota

Subject: **Surveys**

Re: **Oasis**
Ash Federal 5300 11-18T
McKenzie County, ND

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc. Other information required by your office is as follows:

<i>Surveyor Name</i>	<i>Surveyor Title</i>	<i>Borehole Number</i>	<i>Start Depth</i>	<i>End Depth</i>	<i>Start Date</i>	<i>End Date</i>	<i>Type of Survey</i>	<i>TD Straight Line Projection</i>
Mccammond, Mike	MWD Operator	O.H.	2183'	14837'	08/25/12	09/07/12	MWD	14938'
Mccammond, Mike	MWD Operator	ST 1	13238'	20420'	09/08/12	09/18/12	MWD	20482'

A certified plat on which the bottom hole location is oriented both to the surface location and to the lease lines (or unit lines in case of pooling) is attached to the survey report. If any other information is required please contact the undersigned at the letterhead address or phone number.

Douglas Hudson
Well Planner



19510 Oil Center Blvd
Houston, TX 77073
Bus 281.443.1414
Fax 281.443.1676

Monday, October 01, 2012

State of North Dakota

Subject: **Survey Certification Letter**

Re: **Oasis**
Ash Federal 5300 11-18T
McKenzie County, ND

I, Mike Mccammond, certify that; I am employed by Ryan Directional Services, Inc.; that I did on the conduct or supervise the taking of the following MWD surveys:

on the day(s) of 8/25/2012 thru 9/7/2012 from a depth of 2183' MD to a depth of 14837' MD and Straight line projection to TD 14938' MD;

on the day(s) of 9/8/2012 thru 9/18/2012 from a depth of 13238' MD to a depth of 20420' MD and Straight line projection to TD 20482' MD;

that the data is true, correct, complete, and within the limitations of the tool as set forth by Ryan Directional Services, Inc.; that I am authorized and qualified to make this report; that this survey was conducted at the request of Oasis for the Ash Federal 5300 11-18T; in McKenzie County, ND.

Mike Mccammond

Mike Mccammond

MWD Operator

Ryan Directional Services, Inc.

Report #: **1**

Date: _____



RYAN DIRECTIONAL SERVICES

A NABORS COMPANY

SURVEY REPORT

Ryan Job # **5451**Kit # **9**

Customer: **Oasis Petroleum**
 Well Name: **Ash Federal 5300 11-18T**
 Block or Section: **Section 18 T-153N R-100W**
 Rig #: **Nabors B-22**
 Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Mike McCammond**
 Directional Drillers: **D Bohn/M Bader**
 Survey Corrected To: **True North**
 Vertical Section Direction: **90.29**
 Survey Correction: **8.47**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	2183	0.00	0.00		2183.00	0.00	0.00	0.00	0.00
1	2261	1.20	173.10	89.00	2260.99	0.10	-0.81	0.10	1.54
2	2354	0.90	183.40	93.00	2353.98	0.18	-2.51	0.17	0.38
3	2448	0.70	203.60	96.00	2447.97	-0.08	-3.77	-0.10	0.36
4	2541	0.40	249.50	100.00	2540.97	-0.61	-4.40	-0.63	0.55
5	2634	0.80	271.30	102.00	2633.96	-1.56	-4.50	-1.59	0.49
6	2727	1.00	274.10	105.00	2726.95	-3.02	-4.43	-3.05	0.22
7	2821	1.10	288.60	107.00	2820.93	-4.70	-4.08	-4.72	0.30
8	2914	1.20	294.00	109.00	2913.91	-6.44	-3.40	-6.45	0.16
9	3007	1.50	299.60	109.00	3006.89	-8.39	-2.41	-8.40	0.35
10	3101	1.40	305.10	111.00	3100.86	-10.41	-1.14	-10.41	0.18
11	3194	1.30	305.50	113.00	3193.83	-12.20	0.13	-12.20	0.11
12	3287	1.20	304.30	114.00	3286.81	-13.87	1.29	-13.86	0.11
13	3381	1.20	313.70	118.00	3380.79	-15.40	2.52	-15.39	0.21
14	3474	1.20	308.00	120.00	3473.77	-16.88	3.80	-16.86	0.13
15	3568	1.10	310.30	123.00	3567.75	-18.35	4.99	-18.32	0.12
16	3661	1.00	303.00	125.00	3660.74	-19.71	6.01	-19.68	0.18
17	3754	1.00	310.00	125.00	3753.72	-21.02	6.97	-20.99	0.13
18	3848	1.00	311.40	125.00	3847.71	-22.27	8.04	-22.23	0.03
19	3941	0.80	310.70	127.00	3940.70	-23.38	9.00	-23.33	0.22
20	4034	0.80	314.00	131.00	4033.69	-24.34	9.87	-24.29	0.05
21	4127	0.80	316.20	132.00	4126.68	-25.26	10.79	-25.21	0.03
22	4221	0.80	320.90	134.00	4220.67	-26.13	11.78	-26.07	0.07
23	4314	0.50	322.80	136.00	4313.66	-26.79	12.60	-26.73	0.32
24	4407	0.40	321.20	136.00	4406.66	-27.24	13.18	-27.18	0.11
25	4500	0.40	324.30	138.00	4499.66	-27.64	13.70	-27.57	0.02
26	4594	0.50	331.60	138.00	4593.65	-28.03	14.32	-27.96	0.12
27	4687	0.80	321.50	140.00	4686.65	-28.63	15.19	-28.56	0.34
28	4780	0.80	340.50	138.00	4779.64	-29.26	16.31	-29.18	0.28
29	4873	0.60	1.50	138.00	4872.63	-29.47	17.41	-29.38	0.35
30	4966	0.60	12.50	140.00	4965.63	-29.35	18.37	-29.26	0.12
31	5059	0.60	38.30	141.00	5058.62	-28.95	19.23	-28.85	0.29
32	5153	0.40	59.70	141.00	5152.62	-28.37	19.78	-28.27	0.29
33	5246	0.30	183.10	143.00	5245.62	-28.10	19.70	-28.00	0.66
34	5339	0.70	141.20	145.00	5338.61	-27.75	19.01	-27.66	0.56
35	5432	0.20	152.80	147.00	5431.61	-27.32	18.43	-27.23	0.54
36	5526	0.10	56.20	147.00	5525.61	-27.18	18.33	-27.08	0.25
37	5619	0.30	339.60	149.00	5618.61	-27.19	18.60	-27.10	0.32
38	5712	0.30	264.60	149.00	5711.61	-27.52	18.80	-27.43	0.39
39	5806	0.60	296.80	149.00	5805.61	-28.21	19.00	-28.11	0.41
40	5899	0.50	310.60	150.00	5898.60	-28.95	19.49	-28.85	0.18
41	5992	0.50	341.80	150.00	5991.60	-29.39	20.14	-29.29	0.29
42	6086	0.80	344.20	152.00	6085.59	-29.70	21.16	-29.60	0.32
43	6179	1.20	1.20	154.00	6178.58	-29.87	22.76	-29.75	0.53
44	6273	1.60	3.90	156.00	6272.55	-29.77	25.05	-29.64	0.43
45	6366	1.50	10.20	158.00	6365.52	-29.48	27.54	-29.34	0.21
46	6458	1.90	20.20	159.00	6457.48	-28.75	30.16	-28.60	0.54
47	6552	2.20	21.60	161.00	6551.42	-27.57	33.30	-27.40	0.32
48	6645	2.70	22.70	163.00	6644.33	-26.08	36.98	-25.89	0.54
49	6738	1.30	359.40	165.00	6737.27	-25.26	40.06	-25.06	1.71
50	6832	1.20	345.30	168.00	6831.25	-25.53	42.07	-25.32	0.34
51	6925	1.60	358.30	170.00	6924.22	-25.83	44.31	-25.61	0.55
52	7018	1.50	4.30	172.00	7017.19	-25.79	46.82	-25.55	0.20
53	7111	1.50	1.50	174.00	7110.16	-25.68	49.26	-25.43	0.08
54	7205	1.40	7.40	176.00	7204.13	-25.51	51.62	-25.25	0.19
55	7298	1.50	7.70	177.00	7297.10	-25.21	53.96	-24.94	0.11
56	7391	1.60	21.90	176.00	7390.06	-24.58	56.37	-24.29	0.43
57	7484	1.60	32.50	177.00	7483.03	-23.41	58.67	-23.11	0.32
58	7578	1.60	32.10	177.00	7576.99	-22.02	60.89	-21.71	0.01
59	7671	1.50	39.50	179.00	7669.96	-20.56	62.93	-20.25	0.24
60	7764	1.70	38.80	179.00	7762.92	-18.94	64.94	-18.61	0.22
61	7857	1.00	37.00	179.00	7855.89	-17.59	66.66	-17.25	0.75
62	7951	0.80	48.50	179.00	7949.88	-16.61	67.75	-16.27	0.29
63	8044	0.80	58.20	179.00	8042.87	-15.58	68.53	-15.23	0.15
64	8137	0.80	53.90	179.00	8135.86	-14.50	69.25	-14.15	0.06
65	8231	0.60	46.40	181.00	8229.86	-13.62	69.98	-13.27	0.23

Report #: **1**
Date: _____

 RYAN DIRECTIONAL SERVICES
A NABORS COMPANY
SURVEY REPORT

Ryan Job # **5451**
Kit # **9**

Customer: **Oasis Petroleum**
Well Name: **Ash Federal 5300 11-18T**
Block or Section: **Section 18 T-153N R-100W**
Rig #: **Nabors B-22**
Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Mike McCammond**
Directional Drillers: **D Bohn/M Bader**
Survey Corrected To: **True North**
Vertical Section Direction: **90.29**
Survey Correction: **8.47**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
66	8324	0.50	50.80	185.00	8322.85	-12.96	70.57	-12.60	0.12
67	8417	0.50	46.40	183.00	8415.85	-12.35	71.10	-11.99	0.04
68	8511	0.60	51.60	181.00	8509.85	-11.67	71.69	-11.31	0.12
69	8604	0.80	68.00	183.00	8602.84	-10.69	72.24	-10.33	0.30
70	8697	0.90	59.40	185.00	8695.83	-9.46	72.85	-9.10	0.17
71	8791	1.00	69.20	186.00	8789.82	-8.07	73.52	-7.69	0.20
72	8884	0.80	66.70	188.00	8882.80	-6.71	74.07	-6.34	0.22
73	8977	0.80	70.50	190.00	8975.79	-5.51	74.54	-5.13	0.06
74	9071	0.90	79.80	190.00	9069.78	-4.16	74.89	-3.78	0.18
75	9164	0.80	79.40	192.00	9162.77	-2.81	75.14	-2.43	0.11
76	9257	0.80	75.00	194.00	9255.77	-1.54	75.43	-1.16	0.07
77	9351	0.80	79.80	195.00	9349.76	-0.27	75.71	0.12	0.07
78	9444	0.70	92.20	197.00	9442.75	0.94	75.80	1.32	0.20
79	9538	0.70	79.20	199.00	9536.74	2.08	75.89	2.46	0.17
80	9631	0.60	92.70	194.00	9629.74	3.12	75.97	3.51	0.20
81	9724	0.70	83.50	201.00	9722.73	4.17	76.01	4.56	0.16
82	9818	0.70	78.00	203.00	9816.72	5.30	76.20	5.69	0.07
83	9911	0.50	78.00	203.00	9909.72	6.25	76.40	6.64	0.22
84	10004	0.50	50.20	203.00	10002.71	6.96	76.75	7.35	0.26
85	10098	0.60	45.70	203.00	10096.71	7.63	77.35	8.02	0.12
86	10191	0.70	50.70	206.00	10189.70	8.41	78.05	8.81	0.12
87	10283	0.70	30.90	203.00	10281.70	9.13	78.89	9.53	0.26
88	10315	0.80	42.50	186.00	10313.69	9.38	79.22	9.78	0.57
89	10346	3.20	87.70	186.00	10344.67	10.39	79.42	10.79	8.70
90	10377	8.10	88.40	188.00	10375.51	13.44	79.51	13.84	15.81
91	10408	13.90	85.90	188.00	10405.93	19.34	79.84	19.74	18.77
92	10440	17.90	85.90	186.00	10436.70	28.08	80.47	28.49	12.50
93	10471	18.50	85.20	186.00	10466.15	37.73	81.22	38.14	2.06
94	10502	21.50	82.70	190.00	10495.28	48.26	82.35	48.68	10.06
95	10533	25.40	85.50	190.00	10523.71	60.52	83.60	60.94	13.08
96	10564	28.50	86.20	190.00	10551.34	74.53	84.61	74.96	10.05
97	10595	33.90	87.50	192.00	10577.85	90.55	85.48	90.98	17.55
98	10626	38.10	87.60	190.00	10602.92	108.75	86.25	109.19	13.55
99	10657	40.10	87.50	190.00	10626.98	128.27	87.09	128.72	6.45
100	10688	45.50	88.40	192.00	10649.72	149.31	87.84	149.76	17.53
101	10720	50.00	89.60	190.00	10671.23	172.99	88.24	173.44	14.33
102	10751	51.20	89.60	190.00	10690.90	196.94	88.41	197.39	3.87
103	10782	56.70	88.90	190.00	10709.14	221.99	88.74	222.44	17.84
104	10813	61.10	90.40	190.00	10725.15	248.52	88.89	248.98	14.79
105	10844	64.40	91.30	190.00	10739.34	276.08	88.48	276.53	10.95
106	10875	67.80	90.60	190.00	10751.90	304.41	88.01	304.86	11.16
107	10906	71.40	89.30	190.00	10762.70	333.46	88.04	333.91	12.26
108	10937	74.70	88.50	190.00	10771.74	363.10	88.61	363.56	10.93
109	10969	77.90	88.10	192.00	10779.32	394.17	89.54	394.63	10.07
110	11000	80.20	88.40	192.00	10785.21	424.58	90.47	425.05	7.48
111	11031	82.60	88.40	192.00	10789.84	455.22	91.32	455.68	7.74
112	11048	84.80	88.10	192.00	10791.71	472.10	91.84	472.57	13.06
113	11116	91.70	87.60	208.00	10793.78	539.97	94.39	540.45	10.17
114	11190	92.10	88.60	204.00	10791.33	613.87	96.84	614.37	1.45
115	11284	90.40	88.80	204.00	10789.28	707.81	98.97	708.32	1.82
116	11378	88.20	90.00	206.00	10790.43	801.78	99.96	802.30	2.67
117	11470	87.80	89.20	206.00	10793.64	893.72	100.60	894.24	0.97
118	11562	89.10	88.80	206.00	10796.13	985.66	102.20	986.19	1.48
119	11656	91.90	90.90	206.00	10795.31	1079.64	102.45	1080.17	3.72
120	11749	91.10	92.80	208.00	10792.87	1172.57	99.45	1173.09	2.22
121	11842	89.80	94.30	210.00	10792.14	1265.41	93.69	1265.90	2.13
122	11937	88.70	94.40	210.00	10793.38	1360.16	86.49	1360.62	1.16
123	12031	88.00	94.00	212.00	10796.09	1453.90	79.60	1454.33	0.86
124	12124	88.00	94.20	213.00	10799.34	1546.64	72.96	1547.03	0.21
125	12218	88.00	93.60	213.00	10802.62	1640.40	66.57	1640.76	0.64
126	12310	87.60	93.30	212.00	10806.15	1732.19	61.04	1732.52	0.54
127	12403	89.70	91.80	217.00	10808.34	1825.08	56.90	1825.40	2.77
128	12495	89.50	91.60	215.00	10808.98	1917.05	54.17	1917.35	0.31
129	12585	89.60	90.60	217.00	10809.69	2007.04	52.44	2007.33	1.12
130	12676	90.00	89.90	217.00	10810.01	2098.04	52.05	2098.33	0.89

Report #: **1**
 Date: _____



Ryan Job # **5451**
 Kit # **9**

Customer: **Oasis Petroleum**
 Well Name: **Ash Federal 5300 11-18T**
 Block or Section: **Section 18 T-153N R-100W**
 Rig #: **Nabors B-22**
 Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Mike McCammond**
 Directional Drillers: **D Bohn/M Bader**
 Survey Corrected To: **True North**
 Vertical Section Direction: **90.29**
 Survey Correction: **8.47**
 Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
131	12708	90.80	89.70	219.00	10809.78	2130.04	52.16	2130.33	2.58
132	12769	90.00	89.40	219.00	10809.36	2191.03	52.64	2191.32	1.40
133	12863	90.20	89.70	221.00	10809.19	2285.02	53.38	2285.32	0.38
134	12956	90.20	89.90	222.00	10808.87	2378.02	53.70	2378.32	0.22
135	13050	90.60	89.20	224.00	10808.21	2472.01	54.44	2472.31	0.86
136	13144	89.90	90.10	222.00	10807.80	2566.00	55.01	2566.31	1.21
137	13238	88.90	90.40	224.00	10808.79	2659.99	54.60	2660.30	1.11
138	13331	89.70	90.90	224.00	10809.92	2752.98	53.55	2753.29	1.01
139	13425	92.10	91.40	226.00	10808.45	2846.95	51.66	2847.25	2.61
140	13519	91.60	93.20	226.00	10805.41	2940.84	47.89	2941.12	1.99
141	13618	89.30	94.20	228.00	10804.63	3039.66	41.50	3039.90	2.53
142	13711	89.10	94.30	228.00	10805.93	3132.42	34.61	3132.64	0.24
143	13742	89.20	94.90	230.00	10806.39	3163.33	32.13	3163.54	1.96
144	13805	90.70	95.10	230.00	10806.45	3226.12	26.64	3226.30	2.40
145	13899	89.20	94.90	228.00	10806.53	3319.80	18.44	3319.93	1.61
146	13992	88.00	94.90	230.00	10808.80	3412.47	10.50	3412.57	1.29
147	14086	88.00	94.30	230.00	10812.08	3506.15	2.97	3506.21	0.64
148	14180	90.10	93.20	230.00	10813.64	3599.95	-3.18	3599.98	2.52
149	14274	90.00	92.30	231.00	10813.56	3693.87	-7.69	3693.87	0.96
150	14368	90.50	92.40	233.00	10813.15	3787.80	-11.54	3787.79	0.54
151	14462	89.20	92.00	235.00	10813.39	3881.75	-15.15	3881.72	1.45
152	14555	89.50	92.80	235.00	10814.45	3974.68	-19.05	3974.63	0.92
153	14649	87.60	91.70	235.00	10816.83	4068.59	-22.73	4068.53	2.34
154	14743	90.80	91.70	235.00	10818.14	4162.54	-25.52	4162.46	3.40
155	14837	94.00	91.30	237.00	10814.20	4256.42	-27.98	4256.34	3.43
Projection	14938	94.00	91.30		10807.16	4357.16	-30.27	4357.06	0.00

Report #: **2**
Date: **8-Sep-12**

 RYAN DIRECTIONAL SERVICES
A NABORS COMPANY
SURVEY REPORT

Ryan Job # **5451**
Kit # **9**

Customer: **Oasis Petroleum**
Well Name: **Ash Federal 5300 11-18T**
Block or Section: **Section 18 T-153N R-100W**
Rig #: **Nabors B-22**
Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Mike McCammond**
Directional Drillers: **D Bohn/M Bader**
Survey Corrected To: **True North**
Vertical Section Direction: **90.29**
Survey Correction: **8.47**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
Tie in to Gyro Surveys									
Tie In	13238	88.90	90.40	224.00	10808.78	2659.99	54.60	2660.30	1.11
1	13302	88.50	90.70	219.00	10810.23	2723.97	53.99	2724.28	0.78
2	13333	88.80	90.00	219.00	10810.96	2754.96	53.80	2755.27	2.46
3	13365	87.00	88.60	221.00	10812.13	2786.94	54.19	2787.24	7.12
4	13396	87.40	87.40	221.00	10813.65	2817.87	55.27	2818.19	4.08
5	13427	87.50	87.10	221.00	10815.03	2848.80	56.75	2849.12	1.02
6	13458	89.30	87.40	222.00	10815.89	2879.74	58.24	2880.07	5.89
7	13491	89.70	88.10	222.00	10816.18	2912.71	59.54	2913.05	2.44
8	13521	89.60	88.40	222.00	10816.37	2942.69	60.45	2943.03	1.05
9	13552	90.60	89.20	224.00	10816.31	2973.68	61.10	2974.02	4.13
10	13583	90.60	88.90	224.00	10815.99	3004.67	61.61	3005.02	0.97
11	13615	90.60	88.80	226.00	10815.65	3036.66	62.26	3037.01	0.31
12	13646	90.40	89.20	226.00	10815.38	3067.65	62.80	3068.00	1.44
13	13677	90.20	88.50	226.00	10815.22	3098.64	63.42	3099.00	2.35
14	13708	90.20	88.90	226.00	10815.11	3129.62	64.12	3129.99	1.29
15	13739	90.60	89.10	228.00	10814.89	3160.62	64.66	3160.98	1.44
16	13771	91.20	89.20	224.00	10814.39	3192.60	65.14	3192.98	1.90
17	13802	91.20	89.10	226.00	10813.74	3223.59	65.60	3223.97	0.32
18	13833	91.00	90.40	228.00	10813.15	3254.58	65.73	3254.96	4.24
19	13864	91.30	90.50	226.00	10812.52	3285.58	65.49	3285.95	1.02
20	13896	91.20	90.60	228.00	10811.83	3317.57	65.18	3317.94	0.44
21	13927	90.40	91.30	228.00	10811.39	3348.56	64.67	3348.93	3.43
22	13958	90.90	91.40	228.00	10811.04	3379.56	63.94	3379.92	1.64
23	13989	90.20	91.30	228.00	10810.74	3410.55	63.21	3410.91	2.28
24	14021	89.70	90.80	230.00	10810.77	3442.55	62.62	3442.91	2.21
25	14052	88.70	91.30	230.00	10811.21	3473.54	62.05	3473.90	3.61
26	14083	88.90	91.30	228.00	10811.85	3504.53	61.35	3504.88	0.65
27	14114	90.90	92.10	230.00	10811.91	3535.52	60.43	3535.87	6.95
28	14146	92.10	91.80	230.00	10811.07	3567.49	59.34	3567.84	3.87
29	14177	91.70	91.70	228.00	10810.04	3598.47	58.40	3598.81	1.33
30	14208	91.20	90.90	231.00	10809.26	3629.45	57.69	3629.79	3.04
31	14240	89.70	90.90	231.00	10809.01	3661.45	57.19	3661.78	4.69
32	14271	90.00	91.10	231.00	10809.09	3692.44	56.65	3692.78	1.16
33	14302	90.10	91.30	230.00	10809.06	3723.44	56.00	3723.77	0.72
34	14334	90.60	91.70	231.00	10808.87	3755.43	55.16	3755.76	2.00
35	14365	90.20	91.50	230.00	10808.65	3786.42	54.30	3786.75	1.44
36	14396	90.00	91.70	231.00	10808.60	3817.42	53.43	3817.73	0.91
37	14427	89.30	91.30	231.00	10808.79	3848.41	52.62	3848.72	2.60
38	14459	89.30	91.40	233.00	10809.18	3880.40	51.87	3880.71	0.31
39	14490	89.10	91.20	233.00	10809.61	3911.39	51.16	3911.70	0.91
40	14521	89.00	90.70	233.00	10810.12	3942.39	50.65	3942.69	1.64
41	14552	88.70	90.80	233.00	10810.74	3973.38	50.24	3973.68	1.02
42	14584	89.00	90.90	233.00	10811.39	4005.37	49.77	4005.67	0.99
43	14615	89.40	90.90	233.00	10811.82	4036.37	49.28	4036.67	1.29
44	14646	90.20	90.90	235.00	10811.93	4067.36	48.80	4067.66	2.58
45	14677	90.00	90.50	235.00	10811.87	4098.36	48.42	4098.66	1.44
46	14709	90.50	90.40	235.00	10811.73	4130.36	48.17	4130.66	1.59
47	14740	90.90	90.50	235.00	10811.36	4161.36	47.92	4161.65	1.33
48	14771	91.20	90.40	235.00	10810.79	4192.35	47.68	4192.65	1.02
49	14803	90.90	90.20	233.00	10810.20	4224.35	47.51	4224.64	1.13
50	14834	91.10	89.90	235.00	10809.66	4255.34	47.48	4255.64	1.16
51	14865	89.60	90.30	235.00	10809.47	4286.34	47.43	4286.64	5.01
52	14896	88.90	89.90	237.00	10809.88	4317.34	47.38	4317.63	2.60
53	14927	88.40	89.30	237.00	10810.61	4348.33	47.59	4348.62	2.52
54	14959	88.20	89.00	237.00	10811.56	4380.31	48.07	4380.61	1.13
55	14990	88.20	88.70	235.00	10812.53	4411.28	48.69	4411.58	0.97
56	15021	88.30	88.70	237.00	10813.48	4442.26	49.39	4442.56	0.32
57	15052	89.10	89.30	237.00	10814.18	4473.24	49.93	4473.55	3.23
58	15084	88.50	89.50	237.00	10814.85	4505.23	50.27	4505.54	1.98
59	15115	89.50	89.80	237.00	10815.39	4536.22	50.46	4536.53	3.37
60	15146	90.10	89.90	237.00	10815.50	4567.22	50.54	4567.53	1.96
61	15177	90.80	90.20	239.00	10815.26	4598.22	50.51	4598.53	2.46
62	15209	90.30	90.30	239.00	10814.95	4630.22	50.37	4630.53	1.59
63	15240	90.60	89.80	239.00	10814.71	4661.22	50.34	4661.53	1.88
64	15271	90.40	89.60	239.00	10814.43	4692.21	50.51	4692.53	0.91
65	15302	90.30	89.20	239.00	10814.25	4723.21	50.83	4723.53	1.33

Report #: **2**
Date: **8-Sep-12**

 RYAN DIRECTIONAL SERVICES
A NABORS COMPANY
SURVEY REPORT

Ryan Job # **5451**
Kit # **9**

Customer: **Oasis Petroleum**
Well Name: **Ash Federal 5300 11-18T**
Block or Section: **Section 18 T-153N R-100W**
Rig #: **Nabors B-22**
Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Mike McCammond**
Directional Drillers: **D Bohn/M Bader**
Survey Corrected To: **True North**
Vertical Section Direction: **90.29**
Survey Correction: **8.47**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
66	15333	90.20	88.60	239.00	10814.11	4754.20	51.43	4754.52	1.96
67	15365	90.20	87.80	237.00	10814.00	4786.18	52.43	4786.50	2.50
68	15396	90.30	87.80	237.00	10813.86	4817.15	53.62	4817.48	0.32
69	15427	90.20	88.80	239.00	10813.73	4848.13	54.54	4848.47	3.24
70	15458	90.50	89.20	237.00	10813.54	4879.12	55.08	4879.46	1.61
71	15490	90.50	89.10	237.00	10813.26	4911.11	55.56	4911.46	0.31
72	15521	89.10	90.70	239.00	10813.37	4942.11	55.61	4942.45	6.86
73	15584	89.30	91.10	239.00	10814.25	5005.10	54.62	5005.44	0.71
74	15615	89.20	91.10	240.00	10814.65	5036.09	54.03	5036.43	0.32
75	15677	90.00	91.30	240.00	10815.09	5098.08	52.73	5098.42	1.33
76	15708	90.10	91.00	240.00	10815.06	5129.08	52.11	5129.41	1.02
77	15771	89.90	91.10	239.00	10815.06	5192.07	50.95	5192.40	0.35
78	15802	89.10	91.30	240.00	10815.33	5223.07	50.30	5223.39	2.66
79	15865	89.00	90.80	240.00	10816.37	5286.05	49.15	5286.37	0.81
80	15896	88.90	90.90	240.00	10816.94	5317.05	48.69	5317.36	0.46
81	15958	88.30	90.70	242.00	10818.46	5379.03	47.82	5379.34	1.02
82	15990	88.30	90.40	242.00	10819.41	5411.01	47.52	5411.32	0.94
83	16052	88.10	91.20	240.00	10821.35	5472.98	46.65	5473.28	1.33
84	16083	88.40	92.80	240.00	10822.30	5503.95	45.57	5504.25	5.25
85	16146	90.80	93.60	240.00	10822.74	5566.86	42.05	5567.15	4.02
86	16177	92.10	93.60	242.00	10821.96	5597.80	40.11	5598.07	4.19
87	16239	93.00	94.10	240.00	10819.20	5659.62	35.95	5659.87	1.66
88	16270	91.80	93.90	242.00	10817.90	5690.52	33.79	5690.77	3.92
89	16333	91.50	93.30	240.00	10816.09	5753.39	29.83	5753.62	1.06
90	16364	88.80	93.40	242.00	10816.00	5784.35	28.02	5784.56	8.72
91	16427	87.60	92.90	242.00	10817.98	5847.24	24.56	5847.43	2.06
92	16520	87.70	93.10	242.00	10821.80	5940.05	19.70	5940.23	0.24
93	16552	88.80	92.20	242.00	10822.77	5972.01	18.22	5972.18	4.44
94	16614	88.80	92.40	244.00	10824.07	6033.96	15.73	6034.12	0.32
95	16677	88.80	93.30	244.00	10825.39	6096.88	12.60	6097.02	1.43
96	16708	89.80	93.80	244.00	10825.77	6127.83	10.68	6127.96	3.61
97	16739	91.00	93.80	244.00	10825.55	6158.77	8.63	6158.89	3.87
98	16802	91.70	93.60	244.00	10824.07	6221.64	4.56	6221.74	1.16
99	16833	90.50	94.10	244.00	10823.47	6252.57	2.48	6252.67	4.19
100	16864	90.60	94.40	244.00	10823.18	6283.50	0.18	6283.58	1.02
101	16895	90.30	94.20	244.00	10822.93	6314.42	-2.14	6314.49	1.16
102	16958	88.90	94.30	246.00	10823.37	6377.27	-6.81	6377.31	2.23
103	16989	89.50	94.10	246.00	10823.81	6408.19	-9.08	6408.23	2.04
104	17020	89.70	93.60	246.00	10824.02	6439.13	-11.16	6439.16	1.74
105	17083	89.90	93.20	246.00	10824.24	6502.04	-14.90	6502.05	0.71
106	17114	90.00	93.20	246.00	10824.27	6533.00	-16.63	6533.00	0.32
107	17176	90.00	92.70	246.00	10824.27	6594.93	-19.82	6594.92	0.81
108	17208	88.40	92.30	246.00	10824.72	6626.90	-21.21	6626.88	5.15
109	17270	89.00	92.10	248.00	10826.12	6688.85	-23.59	6688.82	1.02
110	17302	89.70	92.30	248.00	10826.49	6720.83	-24.82	6720.79	2.28
111	17364	89.50	91.60	248.00	10826.92	6782.80	-26.93	6782.75	1.17
112	17395	88.90	91.60	248.00	10827.35	6813.79	-27.80	6813.74	1.94
113	17458	89.70	91.90	248.00	10828.12	6876.77	-29.72	6876.70	1.36
114	17489	89.80	92.00	248.00	10828.26	6907.75	-30.77	6907.69	0.46
115	17552	89.70	91.30	249.00	10828.53	6970.74	-32.59	6970.66	1.12
116	17583	89.40	91.00	249.00	10828.77	7001.73	-33.21	7001.65	1.37
117	17645	89.60	91.00	249.00	10829.32	7063.72	-34.29	7063.64	0.32
118	17707	91.00	91.20	249.00	10828.99	7125.71	-35.48	7125.63	2.28
119	17739	90.80	91.10	248.00	10828.49	7157.71	-36.12	7157.62	0.70
120	17801	89.20	90.40	249.00	10828.49	7219.70	-36.94	7219.61	2.82
121	17833	90.20	90.80	249.00	10828.66	7251.70	-37.27	7251.61	3.37
122	17864	90.50	90.70	249.00	10828.47	7282.70	-37.68	7282.60	1.02
123	17895	90.40	90.70	249.00	10828.22	7313.70	-38.06	7313.60	0.32
124	17926	91.00	90.70	249.00	10827.84	7344.69	-38.43	7344.59	1.94
125	18020	90.60	89.90	249.00	10826.53	7438.68	-38.93	7438.58	0.95
126	18051	90.10	89.90	249.00	10826.34	7469.68	-38.87	7469.58	1.61
127	18114	90.40	90.10	251.00	10826.07	7532.68	-38.87	7532.58	0.57
128	18177	91.20	89.40	251.00	10825.19	7595.67	-38.60	7595.57	1.69
129	18208	90.80	89.00	249.00	10824.65	7626.66	-38.17	7626.57	1.82
130	18270	89.00	88.50	249.00	10824.76	7688.64	-36.81	7688.55	3.01

Report #: **2**
Date: **8-Sep-12**

 **RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY
SURVEY REPORT

Ryan Job # **5451**
Kit # **9**

Customer: **Oasis Petroleum**
Well Name: **Ash Federal 5300 11-18T**
Block or Section: **Section 18 T-153N R-100W**
Rig #: **Nabors B-22**
Calculation Method: **Minimun Curvature Calculation**

MWD Operator: **Mike McCammond**
Directional Drillers: **D Bohn/M Bader**
Survey Corrected To: **True North**
Vertical Section Direction: **90.29**
Survey Correction: **8.47**
Temperature Forecasting Model (Chart Only): **Logarithmic**

Survey #	MD	Inc	Azm	Temp	TVD	VS	N/S	E/W	DLS
131	18302	90.00	88.60	251.00	10825.03	7720.62	-36.00	7720.54	3.14
132	18333	89.90	88.40	251.00	10825.06	7751.60	-35.19	7751.53	0.72
133	18395	90.40	87.60	251.00	10824.90	7813.55	-33.03	7813.49	1.52
134	18427	90.20	87.00	251.00	10824.73	7845.51	-31.52	7845.45	1.98
135	18489	90.10	86.90	251.00	10824.57	7907.40	-28.22	7907.36	0.23
136	18520	89.00	87.60	251.00	10824.81	7938.36	-26.73	7938.33	4.21
137	18583	88.70	87.90	251.00	10826.08	8001.28	-24.26	8001.26	0.67
138	18614	88.90	89.10	251.00	10826.73	8032.26	-23.45	8032.25	3.92
139	18676	88.60	89.10	253.00	10828.08	8094.23	-22.48	8094.22	0.48
140	18707	89.20	89.50	253.00	10828.67	8125.22	-22.10	8125.22	2.33
141	18770	88.60	89.50	253.00	10829.88	8188.21	-21.55	8188.20	0.95
142	18801	88.10	89.50	253.00	10830.78	8219.19	-21.28	8219.19	1.61
143	18863	88.60	89.20	253.00	10832.56	8281.16	-20.57	8281.16	0.94
144	18895	88.30	89.10	253.00	10833.43	8313.14	-20.10	8313.14	0.99
145	18926	88.10	89.00	253.00	10834.40	8344.11	-19.59	8344.12	0.72
146	18957	88.00	88.80	251.00	10835.46	8375.09	-18.99	8375.10	0.72
147	19020	88.00	89.40	253.00	10837.65	8438.04	-18.00	8438.05	0.95
148	19051	88.00	89.40	251.00	10838.74	8469.01	-17.68	8469.03	0.00
149	19113	88.40	90.00	253.00	10840.68	8530.98	-17.35	8531.00	1.16
150	19145	88.50	90.20	251.00	10841.55	8562.97	-17.41	8562.99	0.70
151	19207	89.10	90.30	253.00	10842.85	8624.95	-17.68	8624.97	0.98
152	19238	89.50	90.20	253.00	10843.23	8655.95	-17.82	8655.97	1.33
153	19332	90.20	90.20	257.00	10843.47	8749.95	-18.14	8749.97	0.74
154	19363	90.20	90.00	253.00	10843.36	8780.95	-18.20	8780.97	0.65
155	19425	90.60	90.00	255.00	10842.93	8842.95	-18.20	8842.97	0.65
156	19457	90.60	89.90	253.00	10842.60	8874.94	-18.17	8874.97	0.31
157	19519	91.10	90.20	255.00	10841.68	8936.94	-18.22	8936.96	0.94
158	19550	91.30	90.00	255.00	10841.03	8967.93	-18.28	8967.95	0.91
159	19613	91.30	89.80	253.00	10839.60	9030.91	-18.17	9030.94	0.32
160	19644	90.60	89.30	255.00	10839.08	9061.90	-17.92	9061.93	2.77
161	19706	90.80	88.60	255.00	10838.33	9123.88	-16.79	9123.91	1.17
162	19738	90.60	87.80	255.00	10837.94	9155.86	-15.78	9155.90	2.58
163	19800	90.10	87.20	257.00	10837.56	9217.78	-13.08	9217.84	1.26
164	19831	89.90	86.40	255.00	10837.56	9248.73	-11.35	9248.79	2.66
165	19890	89.60	86.00	251.00	10837.81	9307.57	-7.44	9307.66	0.85
166	19921	90.00	86.00	251.00	10837.92	9338.49	-5.28	9338.58	1.29
167	19984	89.90	86.10	253.00	10837.98	9401.32	-0.94	9401.43	0.22
168	20015	89.80	86.20	253.00	10838.06	9432.23	1.15	9432.36	0.46
169	20077	90.00	86.40	253.00	10838.17	9494.08	5.15	9494.23	0.46
170	20108	89.80	86.60	253.00	10838.22	9525.02	7.04	9525.17	0.91
171	20171	90.10	86.40	255.00	10838.28	9587.88	10.88	9588.06	0.57
172	20202	90.10	86.70	255.00	10838.22	9618.81	12.75	9619.00	0.97
173	20264	90.20	86.20	255.00	10838.06	9680.67	16.59	9680.88	0.82
174	20295	90.50	86.30	255.00	10837.87	9711.59	18.62	9711.81	1.02
175	20358	90.80	86.60	255.00	10837.16	9774.45	22.52	9774.69	0.67
176	20420	91.20	87.20	257.00	10836.07	9836.33	25.87	9836.59	1.16
Projection	20482	91.20	87.20		10834.77	9898.23	28.90	9898.50	0.00

WELL EVALUATION



**Figure 1. Nabors B22 drilling the Oasis Petroleum North America, LLC - Ash Federal 5300 11-18T during September, 2012 in Baker Field, McKenzie County, North Dakota.
(G. Wayne Peterson, Sunburst Consulting Geologist).**

INTRODUCTION

Oasis Petroleum North America, LLC Ash Federal 5300 11-18T [Lot 1 Sec. 18, T153N R100W] is located approximately 8 miles south and 2 miles east of the town of Williston in McKenzie County, North Dakota. The Ash Federal 5300 11-18T is a horizontal Three Forks development well in part of Oasis Petroleum's Baker prospect (Baker Field) within the Williston Basin. The vertical hole was planned to be drilled to approximately 10,328'. The curve will be built at 12 degrees per 100' to land within the upper Three Forks. This well is a two section lateral which originates in the northwest quarter of Section 18, then drills east to northeast quarter of Section 17 (Figure 2). Directional drilling technologies and geo-steering techniques will be used to land in the Three Forks reservoir and maintain exposure to the ideal target rock.

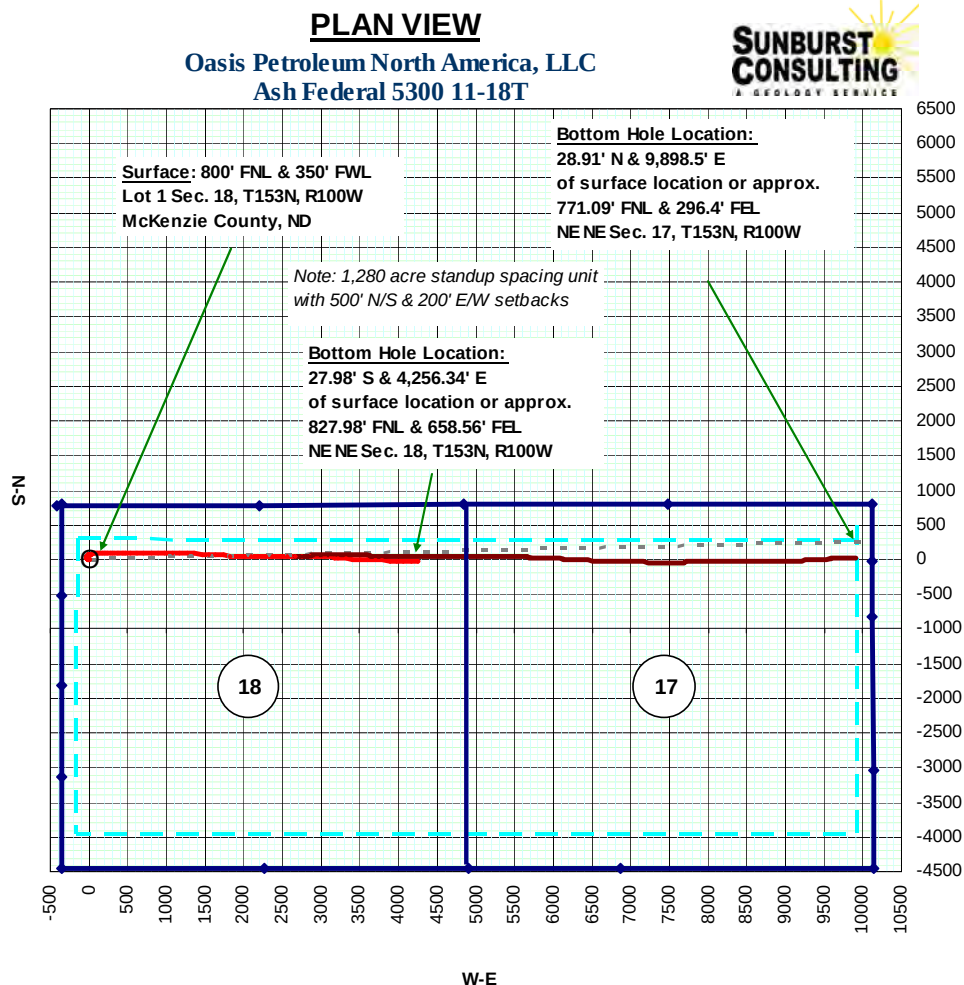


Figure 2. Plan view of Ash Federal 5300 11-18T spacing unit and well path.

OFFSET WELLS

The primary offset wells used for depth correlation during curve operations were the Gulf Oil Exploration and Production Lindvig 1-11HR and the Oasis Petroleum North America, LLC. Kline 5300 11-18H (Figure 3). The Gulf Oil Exploration and Production Lindvig 1-11HR [SE SE Section 11, T153N, R101W] is located approximately 1.5 miles west northwest of the Ash Federal 5300 11-18T. This well was completed in March of 1982 and reached a total depth of 13,800', true vertical depth (TVD). The Oasis Petroleum North America, LLC. Kline 5300 11-18H [Lot 1 Section 18 T153N, R100W] is located approximately 200' south of the Ash Federal 5300 11-18T. This well was completed in October of 2011 and reached a total depth of 20,978' MD. The formation thicknesses expressed by gamma signatures in these wells, and the Oasis Petroleum North America, LLC Larry 5301 44-12B [SE SE Section 12, T153N, R101W] were used to assist in landing the curve. This was accomplished by comparing gamma signatures from the offset wells to gamma data collected during drilling operations. The casing target landing true vertical depth (TVD) was periodically updated to ensure accurate landing of the curve. Data used in this evaluation are found in this report.

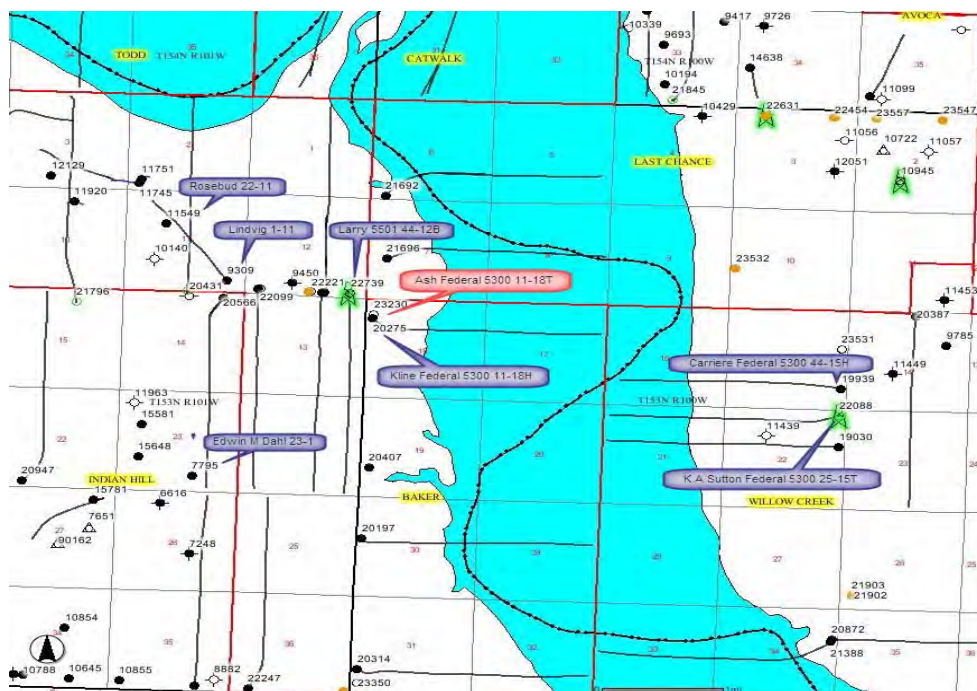


Figure 3. Base map of Ash Federal 5300 11-18T location with offset well locations and field boundaries.

ENGINEERING

Vertical Operations

The Ash Federal 5300 11-18T was spud on 23 August, 2012 by Nabors B22. A 13 ½” hole was drilled with fresh water to 2,187’ and isolated with 9 ⅝” J-55 casing and cemented to surface. After re-entry, the drilling fluid was diesel invert with target weight of 9.7-10.35 ppg for the vertical hole and 10.0-10.35 ppg for curve operations. The vertical hole was drilled from surface casing to the KOP of 10,335’ MD in 84.5 hours with bit #2, a Halliburton DX650, and a Baker 5/6 low speed motor.

Directional Operations

Ryan Energy Technologies provided personnel and equipment for MWD services in the vertical hole, the curve, and throughout the lateral. The directional drillers (RPM), MWD and Sunburst Consulting contractors worked closely together throughout the project to evaluate data and make steering decisions to maximize the amount of borehole in the targeted zones and increase rate of penetration (ROP) of the formation.

Curve Build

The curve was 781' and was drilled in approximately 26 hours by bit #3, a 8 3/4" Reed E1202-A1B PDC bit, attached to a 2.38 degree adjustable National 7/8 5 stage mud motor. The curve was successfully landed at 11,116' MD, approximately 30' below the base of the Lower Bakken Shale. Seven inch diameter 32# HCP-110 and 29# HCP-110 casing were set to 11,101' MD and cemented.

Lateral

After completion of curve operations, MWD tools and bit #4, a 6" Baker T406X, 6 blade PDC bit, attached to a 1.5° fixed Baker XL/LS 5/6 low speed motor were run in the hole. This assembly drilled to 14,938' MD, drilling 3,822' in 69 hours in the lateral before the decision was made to sidetrack due to excessive dog-legs. A new BHA consisting of MWD tools and bit #5 a Security FX64D attached to a new 1.5° fixed Baker XL/LS 5/6 low speed motor were run in the hole. This assembly drilled to 19,907' MD, drilling 6,572' in 149 hours before the MWD tool ceased pulsing. A new BHA consisting of new MWD tools and bit #6 a Varel VM613P2 attached to a new 1.5° fixed Baker XL/LS 5/6 low speed motor were run in the hole. At approximately 13,300' during a sidetrack check, the MWD tool ceased pulsing and was subsequently TOH. A new BHA with a new MWD tool and bit RR#6 and the previously run motor were assembled and TIH. This assembly drilled to TD, drilling 575' in 10.5 hours. Total depth of 20,482', with a final azimuth of 87.20° and inclination of 91.20° was achieved at 0416 hours CDT September 18, 2012. The resulting final vertical section was 9,898.23'.

The bottom hole location (BHL) is 28.91' north & 9,898.5' east of surface location or approximately 771.09' FNL & 296.4' FEL, NE NE Sec. 17, T153N, R100W. The hole was then circulated, and reamed for completion.

GEOLOGY

Methods

Geologic analysis for the Ash Federal 5300 11-18T was provided by Sunburst Consulting, Inc. Information was networked through Rig-watch's electronic data recorder system. This information network provided depth, drilling rate, pump strokes and total gas units to multiple areas throughout the well site. Gas data was fed from Sunburst's digital gas detector, a total gas chromatograph, through Rig-watch for dissemination. Hydrocarbon constituents (C₁ through C₄) were recorded in part-per-million concentrations. Gas sampling was pulled through ¼" poly tubing after agitation in Sunburst's gas trap at the shakers. Rig crews caught lagged samples at the direction of the well site geologists. Samples were collected at 30' intervals in the vertical/curve and 30' intervals in the lateral. Rock cuttings were analyzed in wet and dry conditions under a 10x45 power binocular microscope (for detailed lithologic descriptions see appendix). Cuttings were sent to North Dakota Geologic Survey. In addition to rock samples, rate of penetration and gamma ray data were also used in geologic analysis to aid geo-steering and dip calculations.

Lithology

Formation analysis began at 8,110' MD with a red orange siltstone and anhydrite characteristic of the Kibbey Formation [Mississippian Big Snowy Group]. This interval consisted of a blocky, red to orange, red brown, friable siltstone with calcareous cement moderately cemented with an earthy texture. Also present was a trace amount of anhydrite which was off white. It was soft, amorphous, with an earthy texture.

The Kibbey "Lime" came in at 8,347' MD 8,346' TVD (-6,268') subsea (SS). This marker is represented by an anhydrite which was off white. It was soft, amorphous, with an earthy texture. The anhydrite is often accompanied by slower penetration rates as the bit transitions out of the overlying siltstone. The rate of penetration (ROP) then increases as the bit exits the anhydrite into the underlying limestone mudstone. This carbonate was described as a light gray, light gray brown, off white, lime mudstone which was microcrystalline, friable, and dense, with an earthy texture. Samples in the lower section consisted of red to orange, red brown, light orange to off white siltstone that was friable and blocky with an earthy texture. This facies was moderately cemented with calcite and also was interbedded with soft, off white amorphous or microcrystalline anhydrite.

The Charles Formation [Mississippian Madison Group] consisted of salt, anhydrite, limestone and argillaceous lime mudstone. The first Charles salt was drilled at 8,492' MD 8,491' TVD (-6,413') SS. The Base of the Last Salt (BLS) was logged at 9,188' MD 9,187' TVD (-7,109') SS as indicated by slower penetration rates and increased weight on bit. The intervals of salt within the Charles Formation can be identified by an increase in penetration rates and lower gamma count. Samples of the salt intervals were described as clear frosted, crystalline, anhedral to rare subhedral with a trace euhedral, and hard, crystalline in texture with no visible porosity. Slower penetration rates are observed as the bit encounters sections of lime mudstone to wackestone, which was cream, light gray to dark gray, microcrystalline, firm to hard, and dense, with an earthy texture. This limestone was argillaceous in part and contained a trace of disseminated pyrite. In areas where the penetration rates were slowest anhydrite was observed. These samples were described as off white to white, soft with cryptocrystalline texture or amorphous. Within the Charles Formation at 9,090' MD slower penetration rates indicated the presence of anhydrite. This section was approximately 14' thick, its base, as indicated by the transition to faster penetration rates and higher gamma counts is indicative of the Upper Berentson which was drilled at 9,104' MD 9103 TVD (-7,025') SS.

The Ratcliffe interval [Charles Formation] was drilled at 9,221' MD 9,220' TVD (-7,142') SS. The top of this interval was observed as faster penetration rates were encountered, as the well bore transitioned from anhydrite and lime mudstone. This limestone was cream to light gray with a trace of dark brown to a trace of medium brown. It was microcrystalline, firm, and dense; with an earthy texture and a trace of disseminated pyrite. Also noted was anhydrite which was off white, cryptocrystalline or amorphous with no visible porosity.

The Mission Canyon Formation [Mississippian Madison Group] was logged at 9,415' MD 9,414' TVD (-7,336') SS. The Mission Canyon Formation consisted of lime mudstone. This limestone facies was described as off white, light gray to gray, light brown, off white, blackish gray and cream in color. The microcrystalline to crystalline structure was predominately friable to firm with an earthy texture. Certain samples were a trace siliceous with rare a trace of sparry calcite and a trace of disseminated pyrite. Fossil fragments, along with dark brown algal material were visible in some intervals throughout the Mission Canyon Formation. At connections and downtime, gas shows of 129 to 744 units were noted in this formation. A trace to rare *dark brown dead spotty oil stain* was occasionally observed while logging the Mission Canyon (Figure 4).

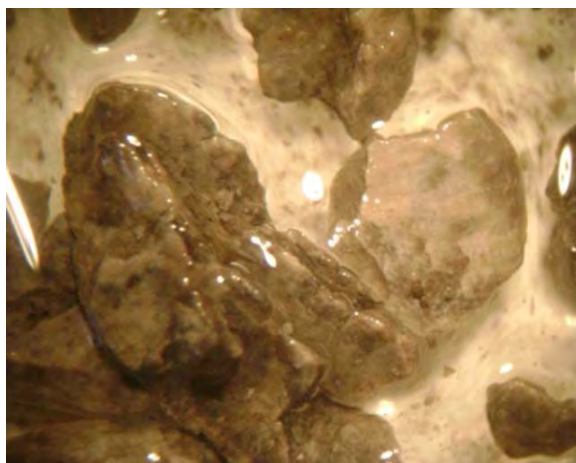


Figure 4. Limestone with spotty light brown & and dark brown dead oil staining in the Mission Canyon.

The Lodgepole Formation [Mississippian Madison Group] was encountered at 9,965' MD 9,964' TVD (-7,886') SS. This interval was characterized by light to medium gray, dark gray, argillaceous lime mudstone which was evidenced by the elevated gamma. The lithology was further characterized as being microcrystalline, firm to a trace hard, and dense, with an earthy to crystalline texture. Disseminated pyrite was also seen in the samples as a trace mineral. At connections and downtime, gas shows of 165 to 587 units were noted in this formation.

The “False Bakken” [Lodgepole Formation] was drilled at 10,749' MD 10,689' TVD (-8,611') SS, and the Scallion limestone at 10,751' MD 10,691' TVD (-8,613') SS. Samples from the False Bakken are typically brown, black, shale, friable with an earthy texture. This facies is calcareous, with a trace of disseminated pyrite. Due to the thin structure of the Scallion no samples were collected. Gas levels were slightly elevated through this interval with a maximum of 262 units likely due to fracture porosity.

The top of the Upper Bakken Shale Member [Mississippian] was drilled at 10,755' MD 10,695' TVD (-8,617') SS. Entry into this interval was characterized by high gamma, elevated background gas and increased rates of penetration. Unfortunately due to the thinness of this formation and the angle of penetration a sample of this formation was not collected. Drilling gas in this interval reached a maximum of 690 units.

The Bakken Middle Member [Mississippian-Devonian] was reached at 10,783' MD 10,709' TVD (-8,631') SS. This formation was predominantly siltstone and silty sandstone noted by the decreasing penetration rates, gamma units and recorded gas levels relative to the overlying source rock. The siltstone was tan gray, friable, subplaty to subblocky, with an earthy texture. It was moderately calcite cemented, with a trace of disseminated pyrite. Also present was silty sandstone which was light gray, very fine grained, friable, subangular to subrounded, well sorted, and contained possible intergranular porosity. It was moderately calcite cemented. Trace minerals included disseminated and nodular pyrite. Drilling gas in this interval reached a maximum of 503 units. Also observed was occasional *dark brown spotty oil stain*.

The Lower Bakken Shale Member [Devonian] reached at 10,880' MD 10,754' TVD (-8,676') SS. This was 13' to the Kline 5300 11-18H. Entry into this interval was characterized by high

gamma, elevated background gas and increased rates of penetration. The shale is black, friable, with an earthy texture and possible fracture porosity. Additionally the shale is described as carbonaceous and petroliferous (Figure 5). Trace minerals included disseminated pyrite. Drilling gas in this interval reached a maximum of 795 units with a connection gas of 894 units.

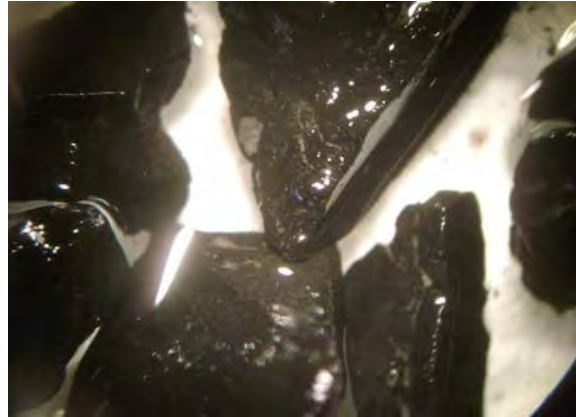


Figure 5. Sample of black, carbonaceous and petroliferous shale from the Lower Bakken Member.

The Pronghorn Member [Devonian-Bakken Formation] was reached at 10,913' MD 10,765' TVD (-8,687') SS. Entry into this interval was characterized by lower gamma, and slightly slower penetration rates. Samples from the Pronghorn were described as a siltstone which was medium gray, friable, and subplaty to sub-blocky, with an earthy texture. The siltstone contained dolomite cement moderately cemented and contained possible intergranular porosity. Trace minerals included disseminated and nodular pyrite. Drilling gas in this interval reached a maximum of 452 units with a connection gas of 509 units. A trace to rare *light brown spotty oil stain* was occasionally observed while logging the Pronghorn.

The Three Forks Formation [Devonian] was reached at 10,966' MD 10,779' TVD (-8,701') SS. The **target zone** of the Three Forks was to be drilled in the dolomitic facies in an eleven foot zone beginning 10 feet into the Three Forks Formation (Figure 6).

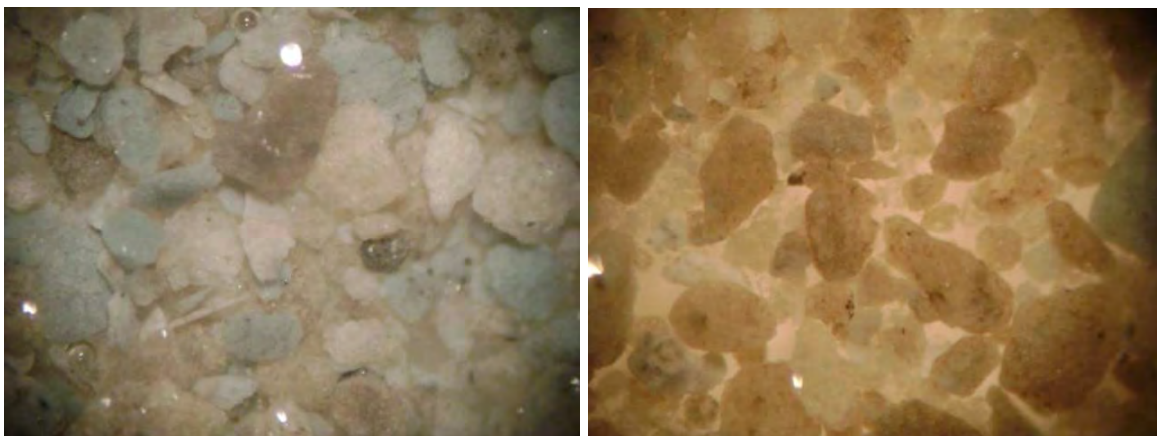


Figure 6. Sample of shale and dolomite from high in Three Forks target zone on left; sample from the low gamma low in zone dolomite and shale on right.

Samples in the Three Forks were generally dolomite which was tan, light brown, light pink to pink, and off white (see Figure 6). Additionally this facies was microcrystalline, friable, with an earthy to sucrosic texture; it also contained a trace of disseminated and nodular pyrite. Possible intercrystalline porosity was observed, as was a *common to a trace of light brown spotty oil stain*. Also present in varying amounts was a light green, light gray, shale. This shale was friable, sub-blocky to subplaty, with an earthy texture. Also a trace of nodular and disseminated pyrite was noted, and the shale contained no visible porosity.

Gas and Oil Shows

Gas monitoring and fluid gains provided evidence of a hydrocarbon saturated reservoir during the drilling of the Ash Federal 5300 11-18T. Oil and gas shows at the shakers and in samples were continuously monitored. In the closed mud system, hydrostatic conditions were maintained near balance. This allowed for gas and fluid gains from the well to be monitored. Gas on the Ash Federal 5300 11-18T varied according to penetration rates and stratigraphic position; with generally higher gas liberated from the dolomite-dominated intervals characterized by low gamma values (Figure 7). Observed concentrations ranged from 300 to 400 units of background gas, and connection peaks of 300 to 600 units in earlier portion of the lateral where shows were the best. In the later portion of the lateral flow was diverted through the gas buster, effectively muting recordable gas values. A trip gas of 1,334 units at 13,335' MD was observed, a later trip at 19,907' yielded a non- recorded 2,500 units of circulating gas. Chromatography of gas revealed typical concentrations of methane, ethane, propane and butane (Figure 8).

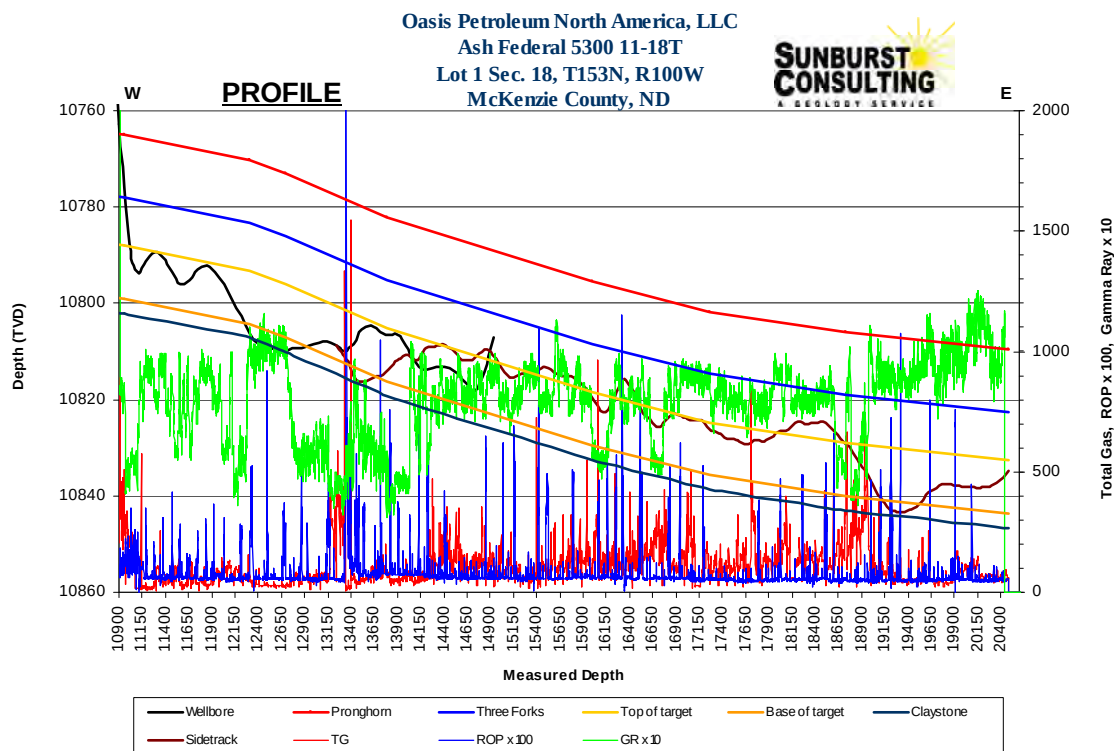


Figure 7. Profile displaying total gas, gamma ray and rate of penetration for the Ash Federal lateral.



Figure 8. Gas chromatography of 600 unit connection gas peaks and drilling gas. Note the measurable concentrations of C₄ at 4 minutes.

Geo-steering

Geologic structure maps in the vicinity of the Ash Federal 5300 11-18T estimated 0.28 degree down apparent dip for the first 4,000' of the lateral. This apparent dip would then flatten slightly for approximately 1,000' then resume the 0.28 degree down dip until total depth (TD) was reached. The Ash Federal 5300 11-18T preferred drilling interval consisted of an eleven foot zone located approximately ten feet below the Pronghorn Member. Stratigraphic location in the target zone was based on penetration rates, gas shows, gamma ray values and sample observations. The projected target landing was to be eight feet into the Three Forks and was successfully reached prior to casing operations. Using offsets provided by Oasis representatives, projected porosity zones were identified as the preferred drilling areas (Figure 9).

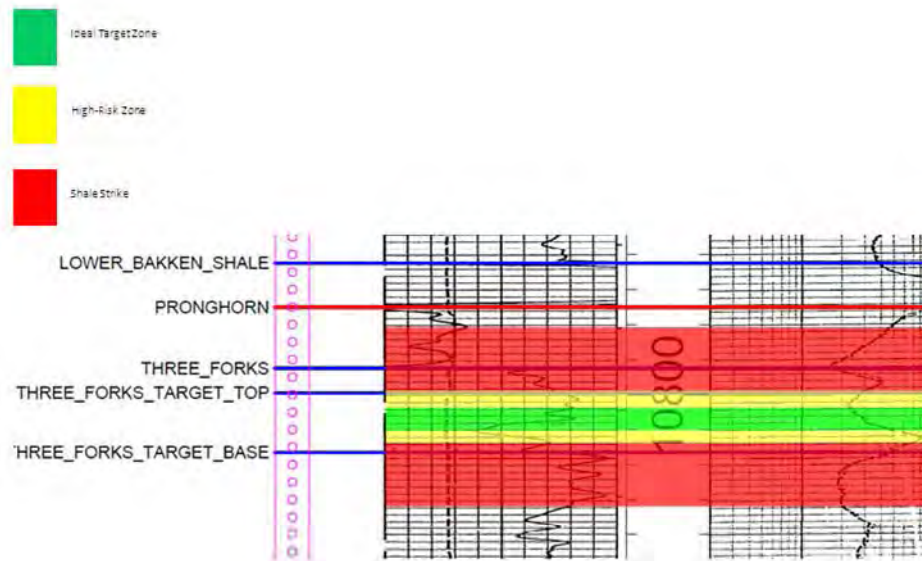


Figure 9. CND type log gamma profile from the Lindvig 1-11HR used for target zone gamma character.

As the well bore moved from the top to the bottom of the preferred drilling zone gamma counts, collected samples and gas were evaluated along with penetration rates to identify areas within the 11' preferred drilling zone that yielded the best prospect potential (refer to Figure 10).

It should be noted that due to the west to east direction of the well-bore the survey and gamma tools were positioned further back than normal. This positioning resulted in a delayed response to the deflections of the well-bore and also resulted in a greater potential for dog-legs. Drilling in the original well-bore was halted at 14,398' MD, and a sidetrack was drilled beginning at 13,335' MD as a result of severe dog-legs in the well-bore.

After beginning the lateral, Oasis representatives expressed a desire to remain in the upper portion of the preferred drilling zone. Due to problems during completions associated with hydraulic fracturing when the well-bore was positioned too close to the underlying claystone.

Drilling out of intermediate casing with the well-bore in the upper portion of the drilling zone, gamma values ranged from 100 to 75 api. When moving down section, gamma counts ranged from 50 to 60 api indicated that the well-bore had encountered a tight streak located in the middle of the drilling zone. The well-bore deflected away from this tight streak many times during the drilling of the Ash Federal 5300 11-18T. When in the top of the zone and moving down onsite personnel adopted a strategy to avoid dog-legs. This policy was not to steer after the gamma had dropped to between 50 and 60 api. Onsite staff would then monitor the well-bore and make steering decisions based on the direction of the well-bore after it encountered the tight streak. Gamma counts below the tight streak moving down would move first high to 90 to 100 api, then low to 30 to 80 api as the well-bore continued moving lower. If the well-bore entered the underlying claystone gamma counts were recorded from 90 to 120 api. A new gamma tool inserted towards the final portion of the lateral resulted in elevated gamma api values.

Samples tended to contain a greater quantity of the light pink light brown dolomite, where gamma values were lower. A greater quantity of the blue gray shale was observed in intervals with higher gamma. In the lower portion of the target zone there was observed a thin layer of dolomite that contained fair to good intercrystalline porosity and *common spotty to even spotty light to dark brown oil staining* was observed. As noted earlier in this report Oasis personnel decided not to drill near the claystone due to the problems during completions.

A TD of 20,482' MD was achieved at 0416 hours CDT September 18, 2012. The well site team worked well together maintaining the well bore in the desired target interval for 88% of the lateral, opening 9,518' of potentially productive reservoir rock. The bottom hole location (BHL) lies: 28.91' north & 9,898.5' east of surface location or approximately 771.09' FNL & 296.4' FEL, NE NE Sec. 17, T153N, R100W. The hole was then circulated and reamed for completion.

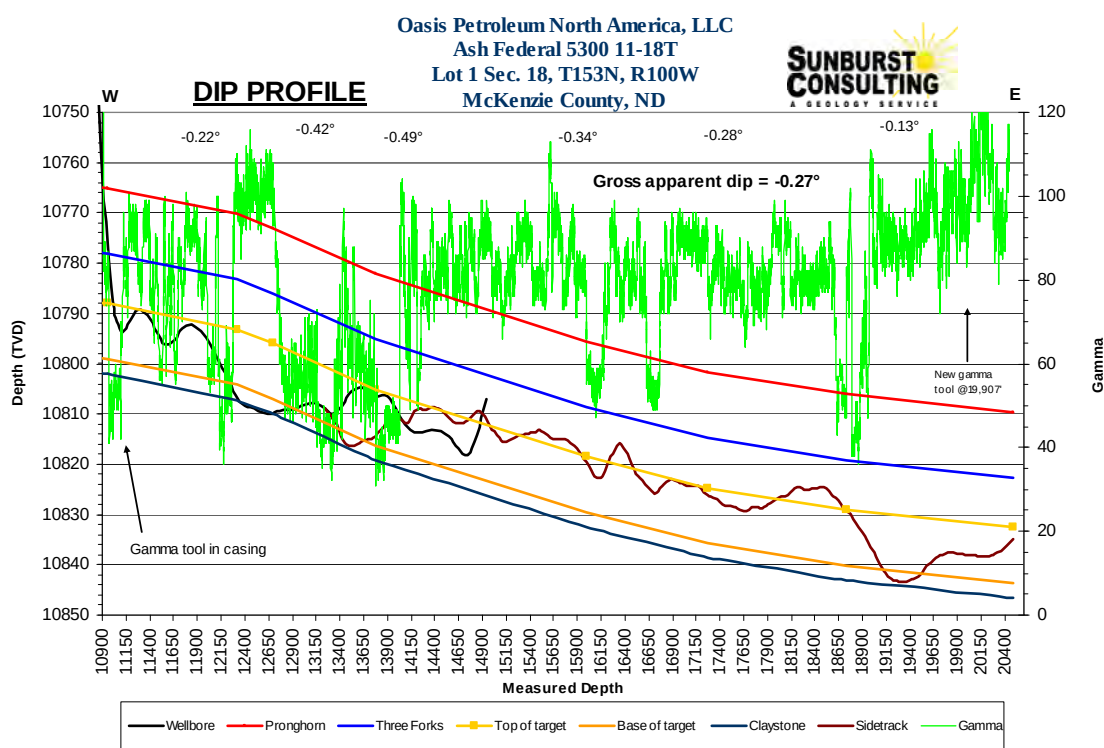


Figure 10. Cross-sectional interpretation of the Ash Federal 5300 11-18T borehole based upon lithology, MWD gamma ray.

SUMMARY

The Ash Federal 5300 11-18T is a successful well in Oasis Petroleum's horizontal Three Forks development program in Baker Field. The project was drilled from surface to TD in 26 days. The TD of 20,482' MD was achieved at 0416 hours CDT September 18, 2012. The well site team worked well together maintaining the well bore in the desired target interval for 88% of the lateral, opening 9,518' of potentially productive reservoir rock.

Samples in the Three Forks were generally dolomite which was tan, light brown, light pink to pink and off white. Additionally this facies was microcrystalline, friable, with an earthy to sucrosic texture. It also contained a trace of disseminated and nodular pyrite. Possible intercrystalline porosity was observed, as was a *common to a trace of light brown spotty oil stain*. Also present in varying amounts was light green and light gray shale. This shale was friable, sub-blocky to subplaty, with an earthy texture. Also a trace of nodular and disseminated pyrite was noted, and the shale contained no visible porosity.

Gas on the Ash Federal 5300 11-18T varied according to penetration rates and location of the well bore in the stratigraphic column. In general higher gas response was recorded from the dolomite-dominated intervals characterized by low gamma values. Observed concentrations ranged from 300 to 400 units of background gas, and connection peaks of 300 to 600 units in the earlier portion of the lateral where shows were the best. A trip gas of 1,334 units at 13,335' MD was observed, a later trip at 19,907' yielded an observed 2,500 units of circulating gas.

The Oasis Petroleum North America, LLC. Ash Federal 5300 11-18T awaits completion operations to determine its ultimate production potential.

Respectfully submitted,
G. Wayne Peterson
c/o Sunburst Consulting, Inc.
21 September, 2012

WELL DATA SUMMARY

<u>OPERATOR:</u>	Oasis Petroleum North America, LLC
<u>ADDRESS:</u>	1001 Fannin, Suite 1500 Houston, TX 77002
<u>WELL NAME:</u>	Ash Federal 5300 11-18T
<u>API #:</u>	33-053-04211
<u>SURFACE LOCATION:</u>	800' FNL & 350' FWL Lot 1 Section 18, T153N, R100W
<u>FIELD/ PROSPECT:</u>	Baker Field / Three Forks
<u>COUNTY, STATE</u>	McKenzie County, ND
<u>BASIN:</u>	Williston
<u>WELL TYPE:</u>	Three Forks Horizontal
<u>ELEVATION:</u>	GL: 2,053' KB: 2,078'
<u>SPUD/ RE-ENTRY DATE:</u>	August 23, 2012
<u>BOTTOM HOLE LOCATION1:</u>	27.98' south & 4,256.34' east of surface location or approx. 827.98' FNL & 658.56' FEL, NE NE Section 18, T153N, R100W
<u>CLOSURE COORDINATES:</u>	Closure Azimuth: 90.40° Closure Distance: 4357.17'
<u>TOTAL DEPTH / DATE:</u>	14,938' on September 7, 2012
<u>SIDETRACK BOTTOM HOLE LOCATION:</u>	28.91' north & 9,898.5' east of surface location or approx. 771.09' FNL & 296.4' FEL, NE NE Section 17, T153N, R100W
<u>CLOSURE COORDINATES:</u>	Closure Azimuth: 89.83° Closure Distance: 9,898.54'
<u>TOTAL DEPTH / DATE:</u>	20,482' on September 18, 2012

<u>TOTAL DRILLING DAYS:</u>	26 days
<u>CONTRACTOR:</u>	Nabors B22
<u>PUMPS:</u>	#1 & #2 - H&H Triplex (stroke length - 12")
<u>TOOLPUSHERS:</u>	Ron Cheney, Chase Erdman
<u>FIELD SUPERVISORS:</u>	Dominic Bohn, Mike Bader
<u>CHEMICAL COMPANY:</u>	NOV Fluid Control
<u>MUD ENGINEER:</u>	Mike McCall, Don Groetken
<u>MUD TYPE:</u>	Fresh water in surface hole Diesel invert in curve; Saltwater brine in lateral
<u>MUD LOSSES:</u>	Invert Mud: 797 bbls, Salt Water: 0 bbls
<u>PROSPECT GEOLOGIST:</u>	Brian Cornette
<u>WELLSITE GEOLOGISTS:</u>	G. Wayne Peterson, Eric Benjamin
<u>GEOSTEERING SYSTEM:</u>	Sunburst Digital Wellsite Geological System
<u>ROCK SAMPLING:</u>	30' from 8,110' - 10,960' & 11,110' - 20,482' (TD) 10' from 10,960' -11,110'
<u>SAMPLE EXAMINATION:</u>	Binocular microscope & fluoroscope
<u>SAMPLE CUTS:</u>	Trichloroethylene (Carbo-Sol)
<u>GAS DETECTION:</u>	MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph
<u>ELECTRIC LOGS:</u>	None
<u>DIRECTIONAL DRILLERS:</u>	RPM Inc. Dominic Bohn, Mike Bader, Rick Bansemer
<u>MWD:</u>	Ryan Mike McCammond, Edgar Maldonaldo

CASING:

Surface: 9 5/8" 36# J-55 set to 2,187'

Intermediate: 7" 3,654' 32# HCP-110, 7,447' 29# HCP-110
set to 11,101'

KEY OFFSET WELLS:

Oasis Petroleum North America, LLC

Larry 5301 44-12B

SESE Section 12 T153N R101W

McKenzie County, ND

Oasis Petroleum North America, LLC

Kline 5300 11-18H

Lot 1 Section 18 T153N R100W

McKenzie County, ND

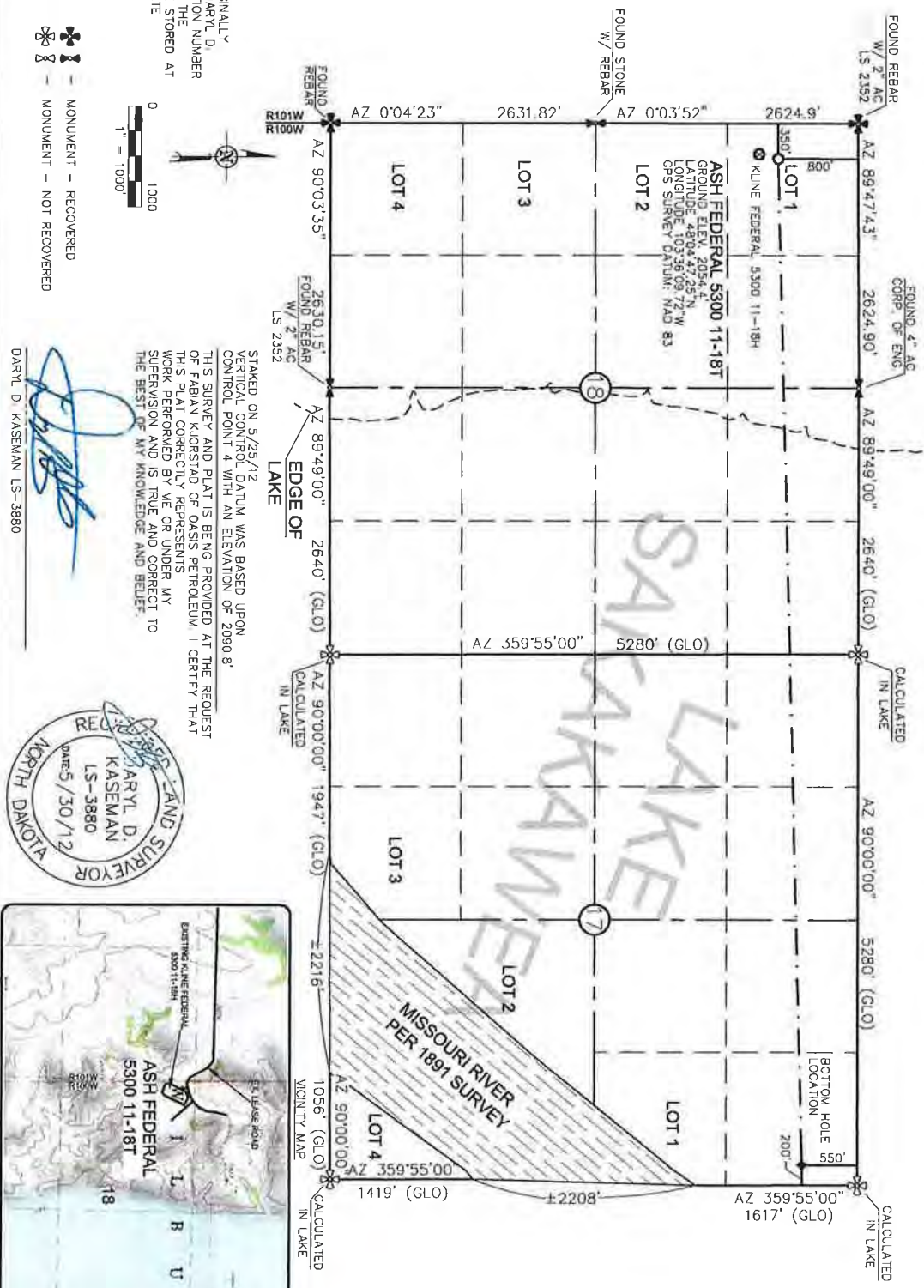
SM Energy Company

Lindvig 1-11HR

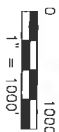
SESE Section 11, T153N, R101W

McKenzie County, ND

WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "ASH FEDERAL 5300 11-18T"
 800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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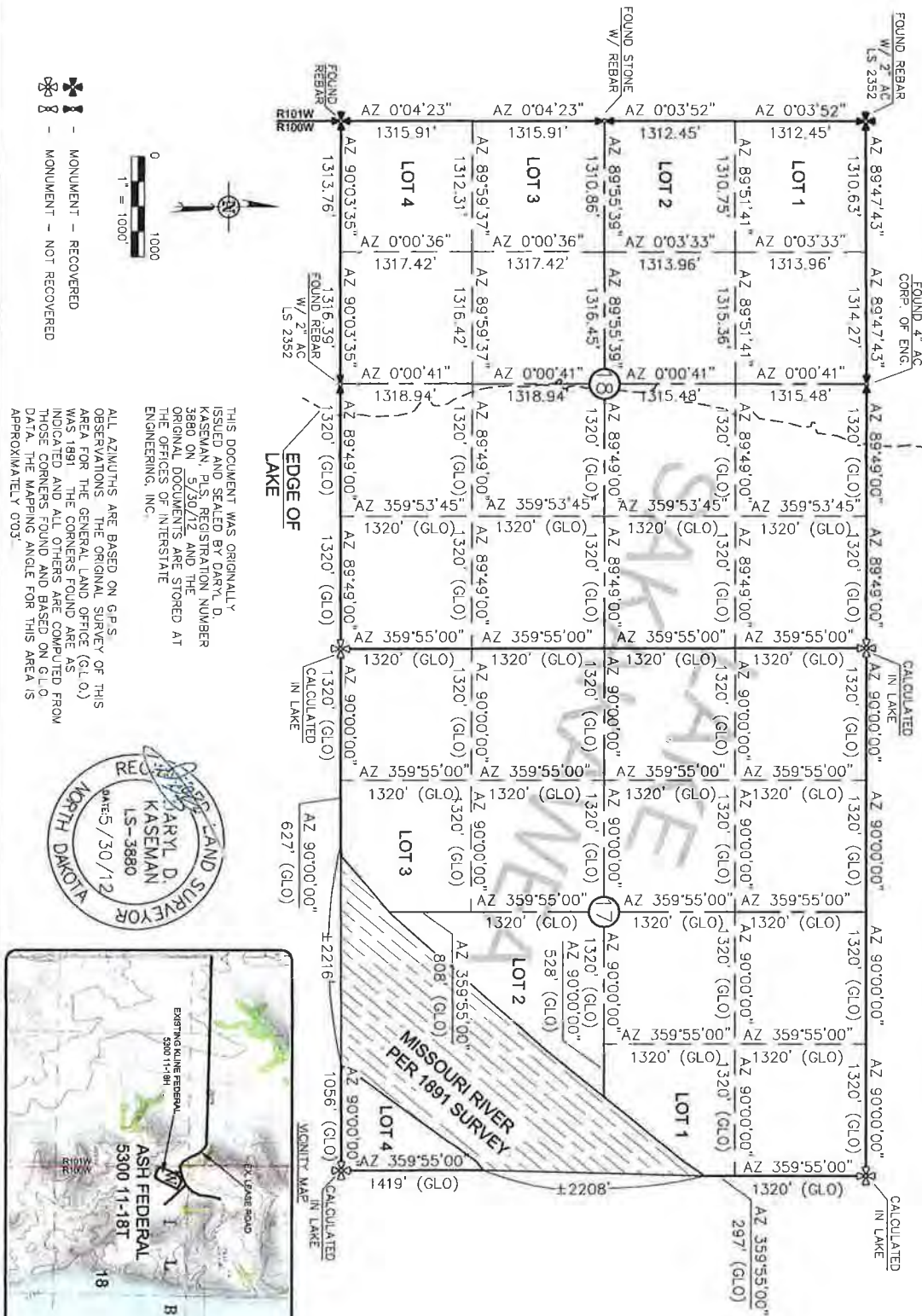
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 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: ALJH Project No.: S12-09-145
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Number	Date	By	Description

SECTION BREAKDOWN

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



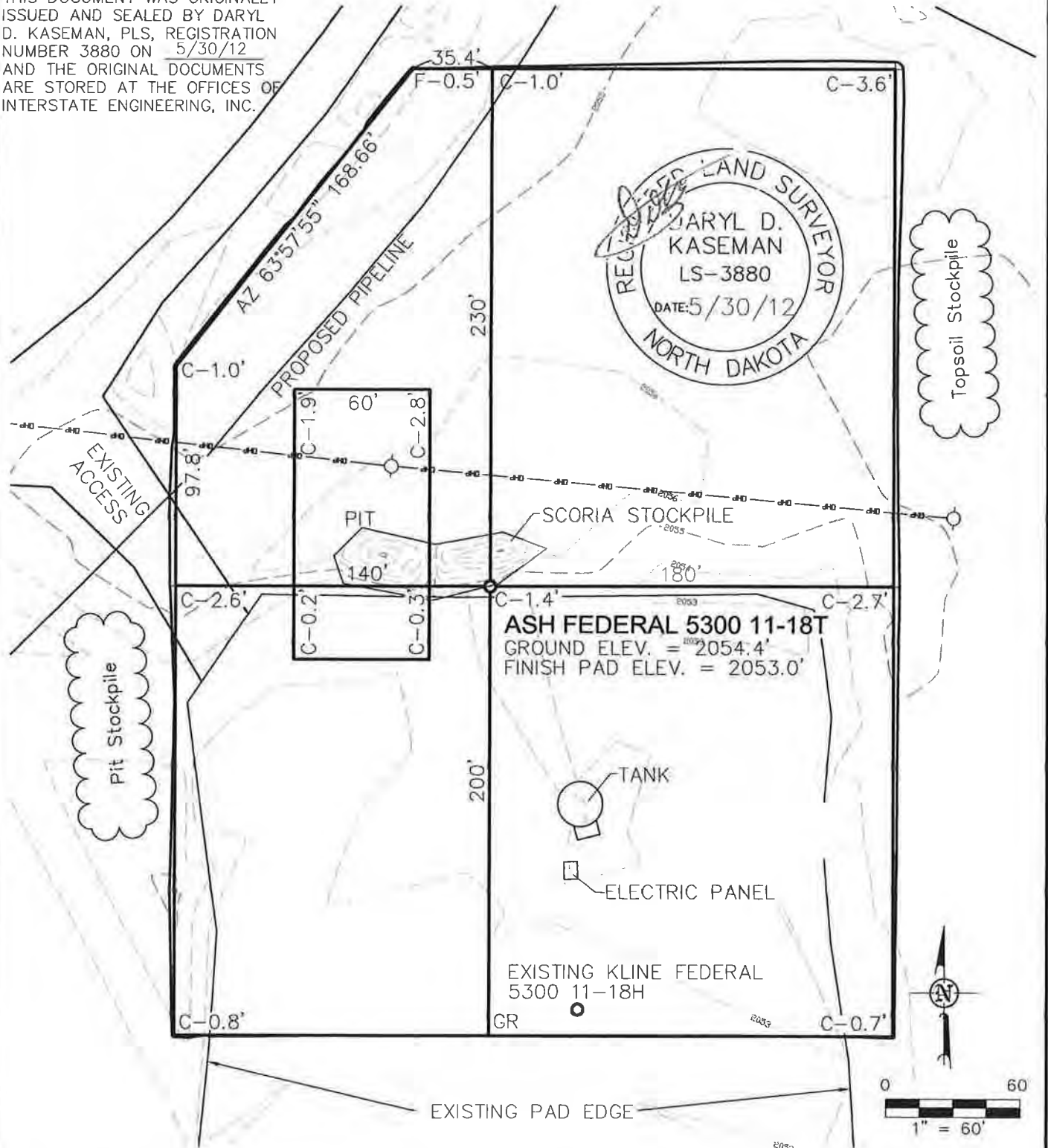
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PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

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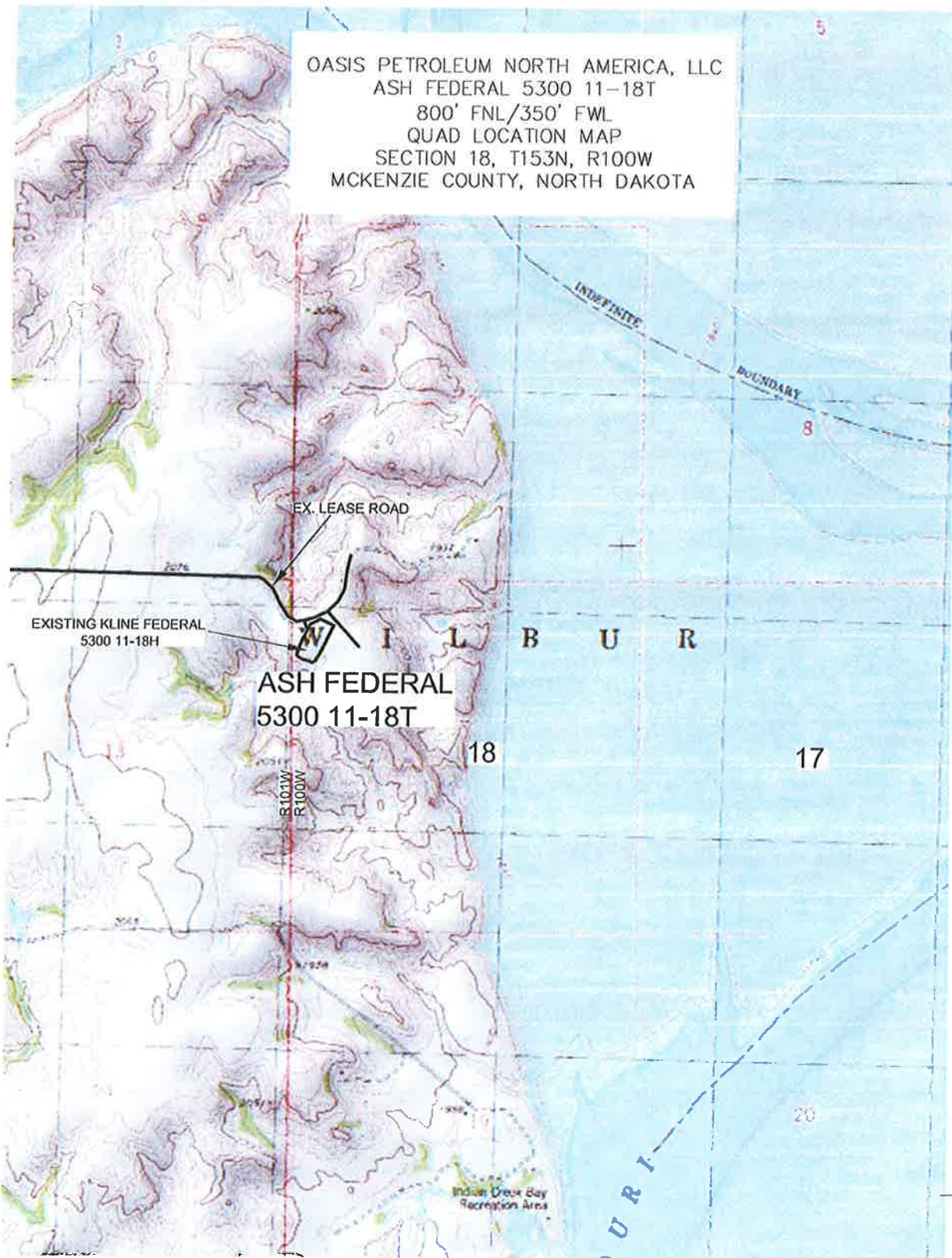
Other offices in Minnesota, North Dakota and South Dakota

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PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S12-09-145
Checked By: D.B.K. Date: MAY 2012

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
 ASH FEDERAL 5300 11-18T
 800' FNL/350' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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Operations in Montana, North Dakota and South Dakota

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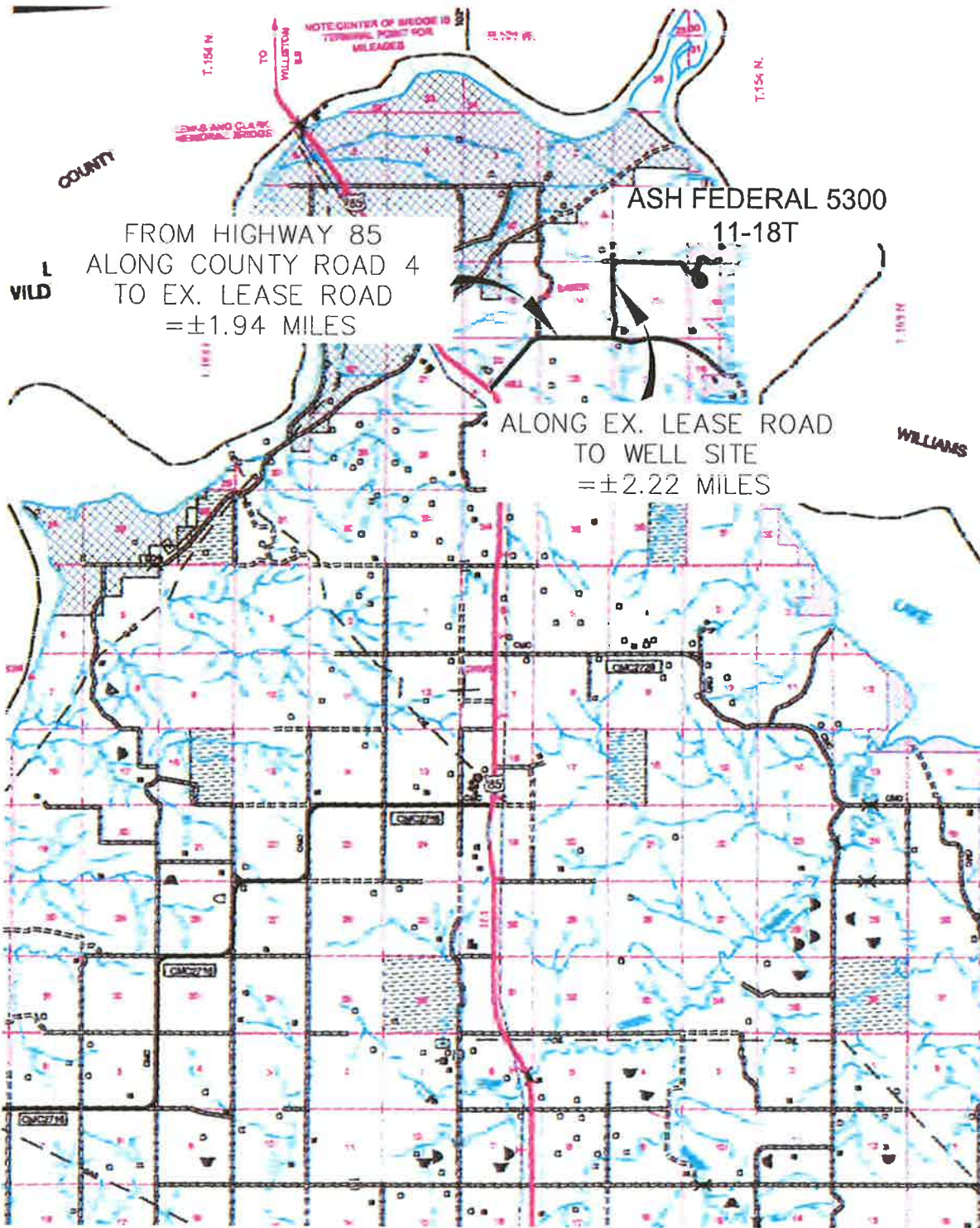
Revision	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

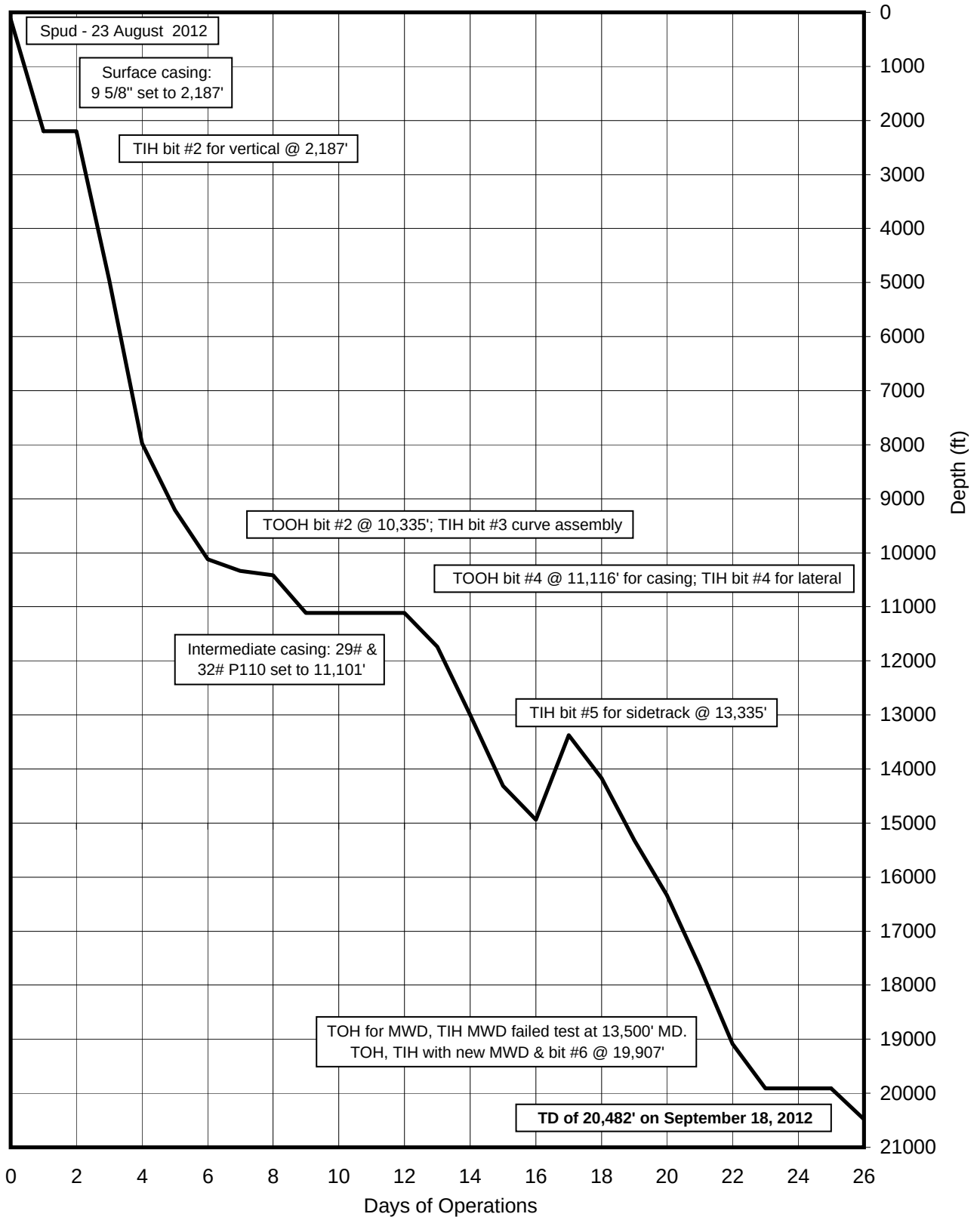
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Revision No.	Date	By	Description

TIME VS DEPTH

Oasis Petroleum North America, LLC

Ash Federal 5300 11-18T



DAILY DRILLING SUMMARY

Day	Date	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	WOB (Klbs) MM	RPM (RT)	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
0	8/23	126'	1,012'	-	-	-	-	-	-	-	-	-	Drill 126'-1012'	-
1	8/24	2,200'	2,200'	-	-	-	-	-	-	-	-	-	Drill 1012'-2200' TOH Lay down BHA Laydown 8" tools. Rig up to run casing.	-
2	8/25	2,200'	0	1	15	-	50	-	2900	110	0	541	Run Casing. Rig down casing. Rig up cementers. Cement. Rig down cementers. Install wellheads, prep wellhead. Stab BOP. Nipple up BOP. Test BOP, retest. Hook up super choke.	-
3	8/26	4,956'	2,756'	1	15	-	50	-	3200	0	110	541	Test BOP. Nipple up BOP. Pick up BHA TIH. Drill cement float and shoe track. Drill 2176'-2631'. Service top drive. Drill 2631'-4956'	-
4	8/27	7,970'	3,014'	2	20	-	60	118	3300	100	0	492	Drilling 4956'-5671'. Service top drive, Drilling 5671'-6790', Drilling 6790'-7970'	Kibbey
5	8/28	9,210'	1,240'	2	30	-	65	108	3300	50	50	451	Drilling 7941'-8470', Service rig, Drilling 8470'-8656', Drilling 8656'-8843', Service top drive-grease packing, Drilling 8843'-9210'	Base Last Salt
6	8/29	10,116'	906'	2	30	-	45	108	3500	50	50	451	Drilling 9200'-9497', Service rig, Drilling 9497'-9675', Circulate and condition add LCM to stop losses, Drilling 9675'-9714', Drilling 9714'-10116'	Lodgepole
7	8/30	10,335'	219'	2	34	-	45	89	3000	90	0	369	Drilling 10119'-10336', Circulate and condition, TOO H, Reaming back out, TOO H, Pump dry job and TOO H, Circulate and condition bottoms up. Pull rotating rubber and install trip nipple	Lodgepole
8	8/31	10,418'	83'	3	0	20	0	95	2300	0	80	394	L/D BHA, L/D mud motor and bit, P/U BHA, P/U new motor and bit, Service rig, TIH, Test MWD tool, TIH, Pull trip nipple and put rotating head in, TIH, TIH--reaming to bottom as needed, Drilling 10335'-10418'	Lodgepole
9	9/1	11,116'	698'	3	35	25	10	95	2395	0	80	394	Drilling 10414'-10725', Service top drive, Drilling 10725'-10819', Drilling 10819'-11116', TOO H	Three Forks
10	9/2	11,116'	0	-	-	-	-	-	-	-	-	-	Circulate and condition weight up system, L/D drill pipe, Circulate and condition, L/D drill pipe, L/D BHA	Three Forks
11	9/3	11,116'	0	-	-	-	-	-	-	-	-	-	L/D BHA, Rig up to run casing, Clean rig floor, Rig up casing, Run casing, Wash to bottom and land casing, Circulate and condition bottoms up, Primary cementing	Three Forks
12	9/4	11,116'	0	4	-	-	-	-	-	-	-	-	Primary cementing, Rig crew cementing, Lay landing joint out, Cut drilling line, P/U tools, Clean rig floor, Cut drilling line, P/U drill pipe, Change out quill, Strapping drill pipe, P/U BHA, Directional work/working on MWD tool, P/U drill pipe	Three Forks
13	9/5	11,738'	622'	4	15	20	20	115	2000	80	0	235	P/U drill pipe, Change rotating head rubber, Pull trip nipple and install rotating head, Ream and washing, Pressure test casing and shoe, Drillout of float and shoe, Formation integrity test, Drilling 11116'-11165', Service top drive, Drilling 11165'-11,738'	Three Forks

DAILY DRILLING SUMMARY

Day	Date 2012	Depth (0600 Hrs)	24 Hr Footage	Bit #	WOB (Klbs) RT	WOB (Klbs) MM	RPM (RT)	RPM (MM)	PP	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
14	9/6	13,000'	1,262	4	20	30	30	115	2600	80	0	235	Drilling 11735'-11752', Service top drive, Drilling 11752'-12347', Drilling 12347'-13000'	Three Forks
15	9/7	14,315'	1,315	4	20	36	30	253	2700	80	0	517	Drilling 13000'-13498', Service top drive, Drilling 13498'-13592', Drilling 13592'-14315'	Three Forks
16	9/8	14,938'	623	4	15	35	30	138	2600	0	80	282	Drilling 14325'-14809', Service rig, Drilling 14809'-14936', Circulate and condition, TOO H to BHA. Lay down BHA. Pick up BHA. TIH	Three Forks
17	9/9	13,375'	-1,563	5	-	30	-	138	2500	80	0	282	TIH, Directional work, Trough for time drilling, Directional work, Time drilling, Drilling 13,335'-13,375'	Three Forks
18	9/10	14,170'	795	5	20	13	30	115	2500	80	0	235	Directional work, Drilling 13375'-13497', Service rig, Drilling 13497'-13717', Drilling 13717'-14170'	Three Forks
19	9/11	15,320'	1,150	5	25	17	30	138	3200	0	80	282	Drilling 14165'-14530', Service rig, Drilling 14530'-14748', Drilling 14748'-15279', Service rig, Drilling 15279'-15320'	Three Forks
20	9/12	16,330'	1,010	5	25	10	30	138	3200	0	80	282	Drilling 15311'-15561', Service rig, Drilling 15561'-15842', Drilling 15842'-16330'	Three Forks
21	9/13	17,660'	1,330	5	25	10	30	138	3200	0	80	282	Drilling 16341'-16591', Service rig, Drilling 16591'-16873', Drilling 16873'-17660'	Three Forks
22	9/14	19,090'	1,430	5	25	19	30	115	3400	80	0	235	Drilling 17654'-17997', Service rig, Drilling 17997'-18404', Drilling 18404'-19090'	Three Forks
23	9/15	19,907'	817	5	25	28	30	138	3400	0	80	282	Drilling 19091'-19309', Service rig, Drilling 19309'-19528', Drilling 19528'-19907', Directional work--trouble shoot MWD tool, Circulate bottoms up, Pump pill, TOO H	Three Forks
24	9/16	19,907'	0	5	-	-	-	-	-	-	-	-	TOOH, Service rig, TOO H, L/D BHA, P/U BHA, TIH, Cut drilling line, TIH, Change saver sub, install rotating head, Ream and washing	Three Forks
25	9/17	19,907'	0	6	-	-	-	-	-	-	-	-	TIH, Reaming, TOO H due to MWD failure, TOO H, Circulate bottoms up, Pump dry job, TOO H, L/D BHA, P/U BHA, Make up bit, TIH, Directional work--test MWD tool, TIH, Directional work--test MWD tool, TIH	Three Forks
26	9/18	20,482'	575	6	17	48	30	138	3710	0	80	282	TIH, Service rig, TIH, Drilling 19907'-20482', TOO H	Three Forks

DAILY MUD SUMMARY

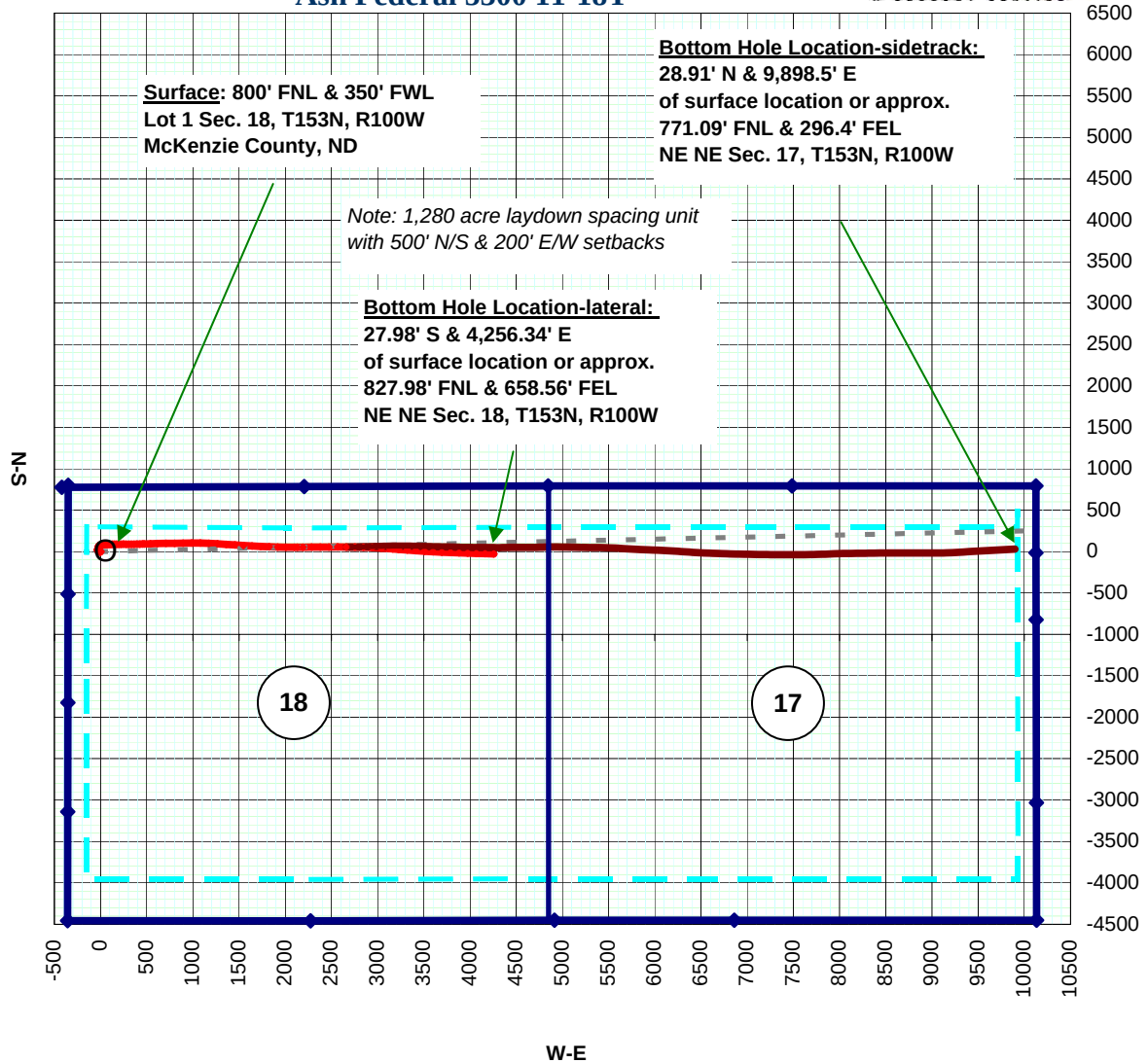
Day	Date 2012	Mud Depth	Mud WT (ppg)	VIS (sec/qt)	PV (cP)	YP (lbs/100 ft2)	Gels (lbs/100 ft ³)	600/300	NAP/ H ₂ O (ratio)	API/HTHP Filtrate (ml)	Cake (API/HTHP)	Cor. Solids (%)	Oil/ H ₂ O (%)	Alk	pH	Excess Lime (lb/bbl)	Cl ⁻ (mg/L)	LGS/ HGS (%)	Salinity (ppm)	Electrical Stability	Gain/ Loss (bbls)
0	08/23	pit	8.4	27	1	-	1/1-	2/1	-	-	-	0	0/100		9	-	500	0.1/0	-	-	-
1	08/24	1,151'	8.6	30	1	1	1/1/-	3/2	-	-	-	0	0.5/99.5		8.5	-	500	0.3/0	-	-	-
2	08/25	2,200'	9	34	2	5	4/5/-	7/9	-	-	-	2	0/98		8.5	-	300	1.5/0.5	-	-	-
3	08/26	2,200'	10.4	90	27	15	10/18/-	69/42	76.7/23.3	4.2	2	11.3	66/20	2.8	-	3.6	79k	2.3/9	264,320	320	-
4	08/27	5,489'	10.35	73	22	16	13/21/-	60/38	75.6/24.4	4.4	2	11.2	65/21	4.2	-	5.4	67k	3.0/8.3	264,320	580	-/88
5	08/28	8,119'	10	56	19	11	9/13/-	49/30	79.3/20.7	4.2	2	10.6	69/18	2.2	-	2.8	57k	3.5/7.0	264,320	620	-/176
6	08/29	9,290'	9.85	56	22	16	12/20/-	60/38	79.3/20.7	4.4	2	10.6	69/18	2.9	-	3.8	64k	5.8/4.8	264,320	727	-/69
7	08/30	10,204'	9.7	44	16	8	5/8/-	40/24	84.1/15.9	5	2	10.1	74/14	1.3	-	1.7	50k	4.0/6.1	264,320	760	-/331
8	08/31	10,335'	10.25	58	15	8	7/11/-	38/23	82.6/17.4	5.8	2	12	71/15	1.8	-	2.3	52k	3.7/8.3	264,320	750	-/131
9	09/01	10,479'	10	44	12	7	6/10/-	31/19	80.5/19.5	6.4	2	10.7	70/17	1.8	-	2.3	55k	3.5/3.5	264,320	590	-
10	09/02	11,116'	10.35	48	16	7	6/10/-	39/23	80.5/19.5	5.2	2	10.7	70/17	2.1	-	2.7	55k	1.8/9.0	264,320	625	-
11	09/03																				
Change over from diesel invert to saltwater brine drilling fluid																					
12	09/04	11,116'	10.2	29	2	1	1/1/-	5/3	-	-	-	-	1/89	-	8	-	167k	1.4/0.5	-	-	-
13	09/05	11,738'	10.2	29	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
14	09/06	12,117'	9.95	28	1	1	1/1/-	3/2	-	-	-	-	1/90	-	9	-	156k	1.7/1.6	-	-	-
15	09/07	-	9.85	48	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16	09/08	14,905'	9.8	28	1	1	1/1/-	3/2	-	-	-	-	0.5/91.5	-	9.5	-	156k	0.7/0.7	-	-	-
17	09/09	-	9.8	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
18	09/10	13,677'	9.9	28	1	1	1/1/-	3/2	-	-	-	-	0.5/91.5	-	9.5	-	168k	0.7/0.7	-	-	-
19	09/11	-	9.9	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
20	09/12	15,835'	9.9	28	1	1	1/1/-	3/2	-	-	-	-	2/90	-	8.5	-	164k	1.2/1.2	-	-	-
21	09/13	-	9.9	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
22	09/14	17,996'	9.9	28	1	1	1/1/-	3/2	-	-	-	-	0/92	-	9	-	171k	0.5/0.4	-	-	-
23	09/15	-	9.9	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
24	09/16	19,907'	9.9	27	1	1	1/1/-	3/2	-	-	-	-	0/92	-	8.5	-	169k	0.5/0.6	-	-	-
25	09/17	-	9.9	27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
26	09/18	19,907'	9.9	28	1	1	1/1/-	3/2	-	-	-	-	1/90	-	8.5	-	176k	0.4/0.5	-	-	-

BIT RECORD

Bit #	Size	Type	Make	Model	Serial #	Jets	Depth In	Depth Out	Footage	Hours	Accum. Hours	Vert. Dev.
1	13 1/2"	PDC	Hughs	MX-1	5184113	3x18 1x22	106'	2,187'	2,081'	20	20.00	Surface
2	8 3/4"	PDC	Security	DX650	12060802	6x12	2,187'	10,335'	8,148'	84.5	104.50	Vertical
3	8 3/4"	PDC	Reed	E1202-A1B	E156190	5x18	10,335'	11,116'	781'	26	130.50	Curve
4	6"	PDC	HTC	T406X	156190	3x22 3x13	11,116'	14,938'	3,822'	69	199.50	Lateral
5	6"	PDC	Security	FX64D	12026004	6x16	13,335'	19,907'	6,572'	149	348.50	Lateral
6	6"	PDC	Varel	VM613P2	4004836	6x20	19,907'	20,482'	575'	10.5	359.00	Lateral

PLAN VIEW

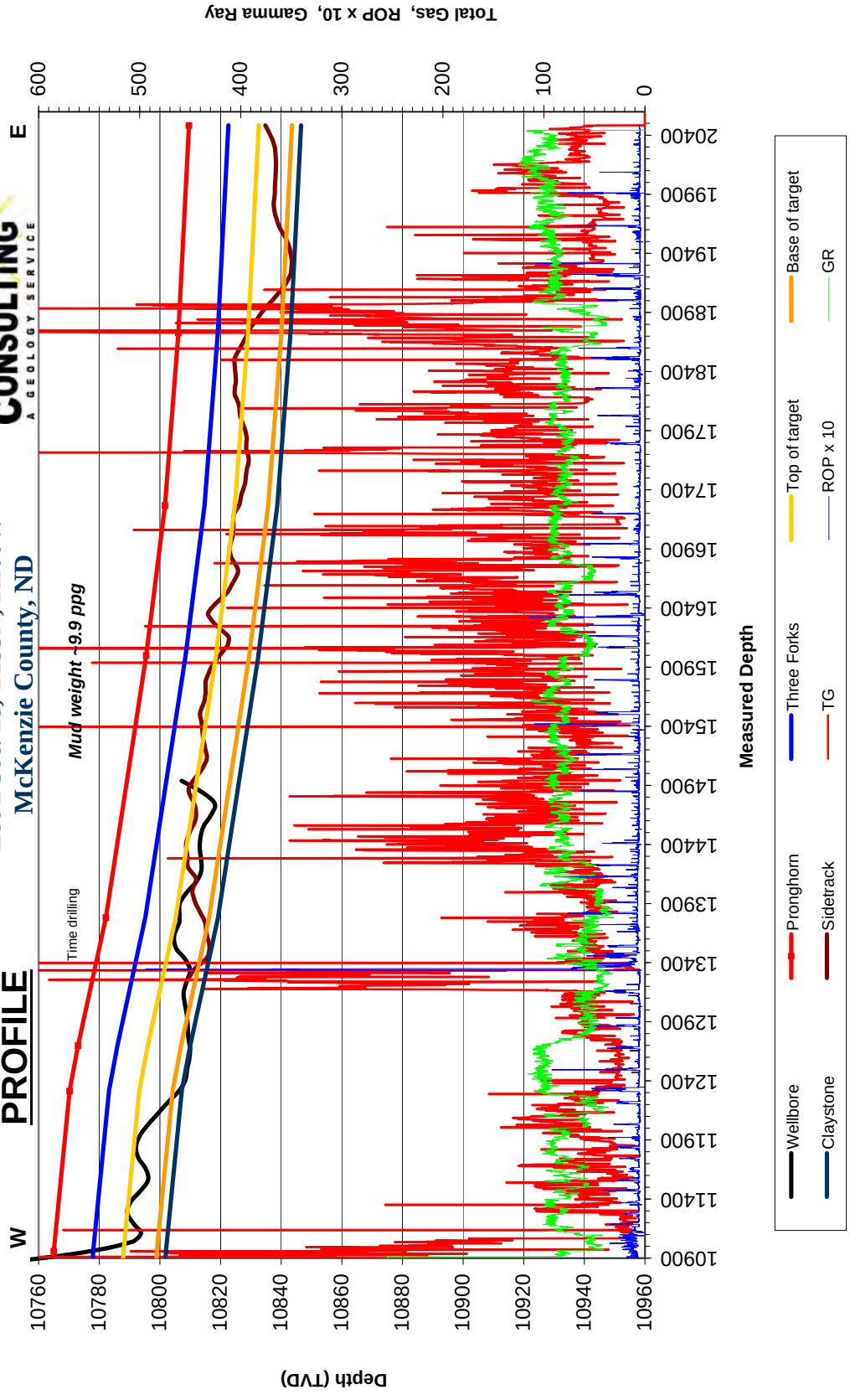
Oasis Petroleum North America, LLC
Ash Federal 5300 11-18T



Oasis Petroleum North America, LLC
Ash Federal 5300 11-18T
Lot 1 Sec. 18, T153N, R100W
McKenzie County, ND



PROFILE



FORMATION MARKERS & DIP ESTIMATES

Oasis Petroleum North America, LLC - Ash Federal 5300 11-18T

Dip Change Points	MD	TVD	TVD diff.	MD diff.	Dip	Dipping up/down	Type of Marker
Marker							
Initial Target Contact	10,960'	10,788.00					Gamma
Claystone	12,314'	10,793.20	5.20	1354.00	-0.22	Down	Gamma
Claystone	12,698'	10,796.00	2.80	384.00	-0.42	Down	Gamma
Low gamma	13,781'	10,805.20	9.20	1083.00	-0.49	Down	Gamma
Tight streak, low gamma	16,000'	10,818.50	13.30	2219.00	-0.34	Down	Gamma
Top of zone	17,267'	10,824.70	6.20	1267.00	-0.28	Down	Gamma
Tight streak, low gamma	18,729'	10,829.10	4.40	1462.00	-0.17	Down	Gamma
TD	20,482'	10,832.60	3.50	1753.00	-0.11		
Gross Dip							
Initial Target Contact	10,960'	10,788.00					
Projected Final Target Contact	20,482'	10,832.60	44.60	9522.00	-0.27	Down	Projection

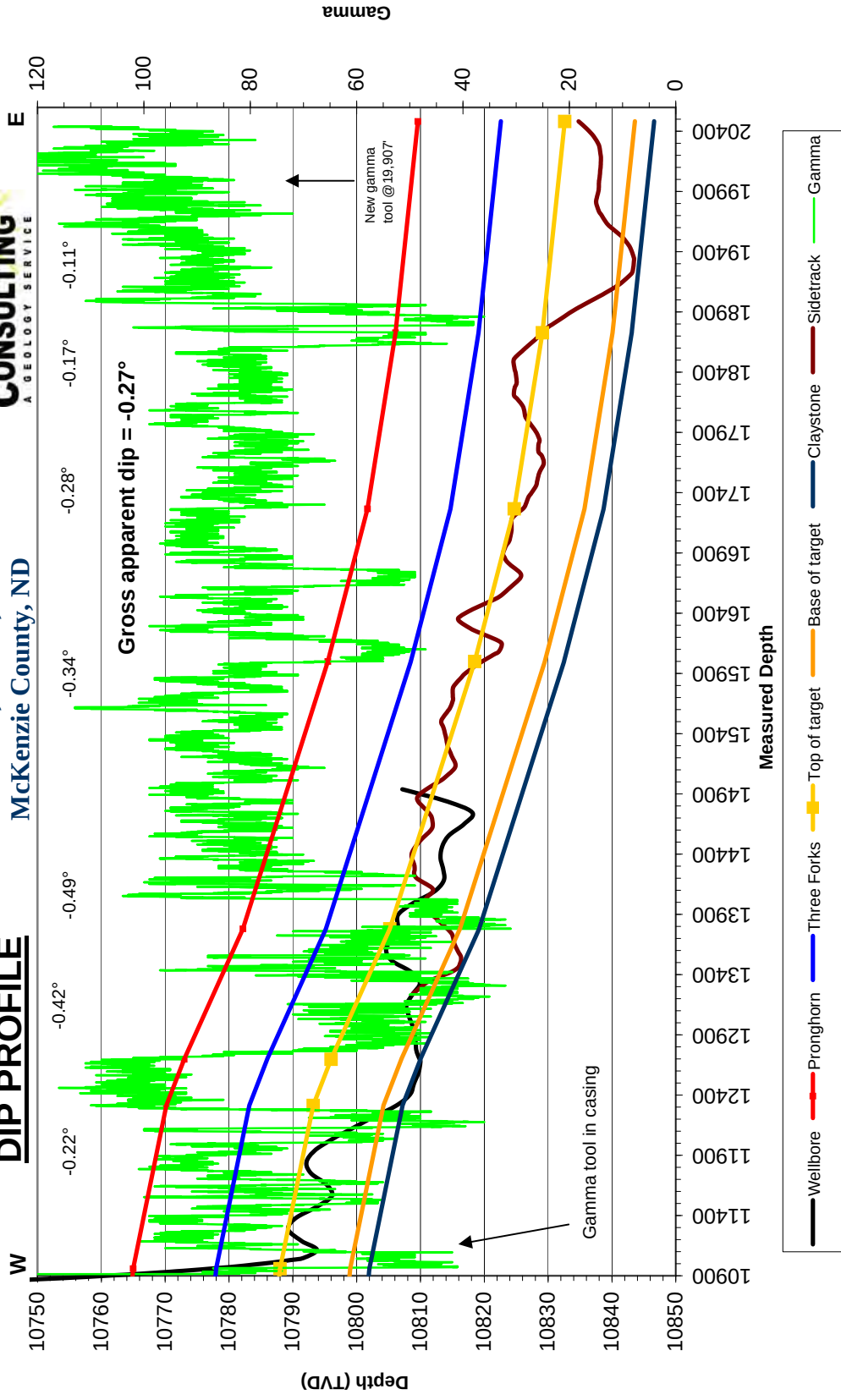
Oasis Petroleum North America, LLC

Ash Federal 5300 11-18T

Lot 1 Sec. 18, T153N, R100W

McKenzie County, ND

DIP PROFILE



<

SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	8/31/2012
Finish:	9/8/2012
Directional Supervision:	RPM Inc.

Date: 9/26/2012
Time: 13:34

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 90.29

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
Tie	2183.00	0.00	0.00		2183.00	0.00	0.00	0.00	
1	2261.00	1.20	173.10		2260.99	-0.81	0.10	0.10	1.54
2	2354.00	0.90	183.40		2353.98	-2.51	0.17	0.18	0.38
3	2448.00	0.70	203.60		2447.97	-3.77	-0.10	-0.08	0.36
4	2541.00	0.40	249.50		2540.97	-4.40	-0.63	-0.61	0.55
5	2634.00	0.80	271.30		2633.96	-4.50	-1.59	-1.56	0.49
6	2727.00	1.00	274.10		2726.95	-4.43	-3.05	-3.02	0.22
7	2821.00	1.10	288.60		2820.93	-4.08	-4.72	-4.70	0.30
8	2914.00	1.20	294.00		2913.91	-3.40	-6.45	-6.44	0.16
9	3007.00	1.50	299.60		3006.89	-2.41	-8.40	-8.39	0.35
10	3101.00	1.40	305.10		3100.86	-1.14	-10.41	-10.41	0.18
11	3194.00	1.30	305.50		3193.83	0.13	-12.20	-12.20	0.11
12	3287.00	1.20	304.30		3286.81	1.29	-13.86	-13.87	0.11
13	3381.00	1.20	313.70		3380.79	2.52	-15.39	-15.40	0.21
14	3474.00	1.20	308.00		3473.77	3.80	-16.86	-16.88	0.13
15	3568.00	1.10	310.30		3567.75	4.99	-18.32	-18.35	0.12
16	3661.00	1.00	303.00		3660.74	6.01	-19.68	-19.71	0.18
17	3754.00	1.00	310.00		3753.72	6.97	-20.99	-21.02	0.13
18	3848.00	1.00	311.40		3847.71	8.04	-22.23	-22.27	0.03
19	3941.00	0.80	310.70		3940.70	9.00	-23.33	-23.38	0.22
20	4034.00	0.80	314.00		4033.69	9.87	-24.29	-24.34	0.05
21	4127.00	0.80	316.20		4126.68	10.79	-25.21	-25.26	0.03
22	4221.00	0.80	320.90		4220.67	11.78	-26.07	-26.13	0.07
23	4314.00	0.50	322.80		4313.66	12.60	-26.73	-26.79	0.32
24	4407.00	0.40	321.20		4406.66	13.18	-27.18	-27.24	0.11
25	4500.00	0.40	324.30		4499.66	13.70	-27.57	-27.64	0.02
26	4594.00	0.50	331.60		4593.65	14.32	-27.96	-28.03	0.12
27	4687.00	0.80	321.50		4686.65	15.19	-28.56	-28.63	0.34
28	4780.00	0.80	340.50		4779.64	16.31	-29.18	-29.26	0.28
29	4873.00	0.60	1.50		4872.63	17.41	-29.38	-29.47	0.35
30	4966.00	0.60	12.50		4965.63	18.37	-29.26	-29.35	0.12
31	5059.00	0.60	38.30		5058.62	19.23	-28.85	-28.95	0.29
32	5153.00	0.40	59.70		5152.62	19.78	-28.27	-28.37	0.29
33	5246.00	0.30	183.10		5245.62	19.70	-28.00	-28.10	0.66
34	5339.00	0.70	141.20		5338.61	19.01	-27.66	-27.75	0.56
35	5432.00	0.20	152.80		5431.61	18.43	-27.23	-27.32	0.54
36	5526.00	0.10	56.20		5525.61	18.33	-27.08	-27.18	0.25
37	5619.00	0.30	339.60		5618.61	18.60	-27.10	-27.19	0.32
38	5712.00	0.30	264.60		5711.61	18.80	-27.43	-27.52	0.39
39	5806.00	0.60	296.80		5805.61	19.00	-28.11	-28.21	0.41
40	5899.00	0.50	310.60		5898.60	19.49	-28.85	-28.95	0.18

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC		
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Minimum Curvature Method (SPE-3362)

Proposed dir: **90.29**

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
41	5992.00	0.50	341.80		5991.60	20.14	-29.29	-29.39	0.29
42	6086.00	0.80	344.20		6085.59	21.16	-29.60	-29.70	0.32
43	6179.00	1.20	1.20		6178.58	22.76	-29.75	-29.87	0.53
44	6273.00	1.60	3.90		6272.55	25.05	-29.64	-29.77	0.43
45	6366.00	1.50	10.20		6365.52	27.54	-29.34	-29.48	0.21
46	6458.00	1.90	20.20		6457.48	30.16	-28.60	-28.75	0.54
47	6552.00	2.20	21.60		6551.42	33.30	-27.40	-27.57	0.32
48	6645.00	2.70	22.70		6644.33	36.98	-25.89	-26.08	0.54
49	6738.00	1.30	359.40		6737.27	40.06	-25.06	-25.26	1.71
50	6832.00	1.20	345.30		6831.25	42.07	-25.32	-25.53	0.34
51	6925.00	1.60	358.30		6924.22	44.31	-25.61	-25.83	0.55
52	7018.00	1.50	4.30		7017.19	46.82	-25.55	-25.79	0.20
53	7111.00	1.50	1.50		7110.16	49.26	-25.43	-25.68	0.08
54	7205.00	1.40	7.40		7204.13	51.62	-25.25	-25.51	0.19
55	7298.00	1.50	7.70		7297.10	53.96	-24.94	-25.21	0.11
56	7391.00	1.60	21.90		7390.06	56.37	-24.29	-24.58	0.43
57	7484.00	1.60	32.50		7483.03	58.67	-23.11	-23.41	0.32
58	7578.00	1.60	32.10		7576.99	60.89	-21.71	-22.02	0.01
59	7671.00	1.50	39.50		7669.96	62.93	-20.25	-20.56	0.24
60	7764.00	1.70	38.80		7762.92	64.94	-18.61	-18.94	0.22
61	7857.00	1.00	37.00		7855.89	66.66	-17.25	-17.59	0.75
62	7951.00	0.80	48.50		7949.88	67.75	-16.27	-16.61	0.29
63	8044.00	0.80	58.20		8042.87	68.53	-15.23	-15.58	0.15
64	8137.00	0.80	53.90		8135.86	69.25	-14.15	-14.50	0.06
65	8231.00	0.60	46.40		8229.86	69.98	-13.27	-13.62	0.23
66	8324.00	0.50	50.80		8322.85	70.57	-12.60	-12.96	0.12
67	8417.00	0.50	46.40		8415.85	71.10	-11.99	-12.35	0.04
68	8511.00	0.60	51.60		8509.85	71.69	-11.31	-11.67	0.12
69	8604.00	0.80	68.00		8602.84	72.24	-10.33	-10.69	0.30
70	8697.00	0.90	59.40		8695.83	72.85	-9.10	-9.46	0.17
71	8791.00	1.00	69.20		8789.82	73.52	-7.69	-8.07	0.20
72	8884.00	0.80	66.70		8882.80	74.07	-6.34	-6.71	0.22
73	8977.00	0.80	70.50		8975.79	74.54	-5.13	-5.51	0.06
74	9071.00	0.90	79.80		9069.78	74.89	-3.78	-4.16	0.18
75	9164.00	0.80	79.40		9162.77	75.14	-2.43	-2.81	0.11
76	9257.00	0.80	75.00		9255.77	75.43	-1.16	-1.54	0.07
77	9351.00	0.80	79.80		9349.76	75.71	0.12	-0.27	0.07
78	9444.00	0.70	92.20		9442.75	75.80	1.32	0.94	0.20
79	9538.00	0.70	79.20		9536.74	75.89	2.46	2.08	0.17

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	8/31/2012
Finish:	9/8/2012
Directional Supervision:	RPM Inc.

Date: 9/26/2012
Time: 13:34

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: 90.29

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		TVD	N-S	E-W	SECT	DLS/
			AZM						100
80	9631.00	0.60	92.70		9629.74	75.97	3.51	3.12	0.20
81	9724.00	0.70	83.50		9722.73	76.01	4.56	4.17	0.16
82	9818.00	0.70	78.00		9816.72	76.20	5.69	5.30	0.07
83	9911.00	0.50	78.00		9909.72	76.40	6.64	6.25	0.22
84	10004.00	0.50	50.20		10002.71	76.75	7.35	6.96	0.26
85	10098.00	0.60	45.70		10096.71	77.35	8.02	7.63	0.12
86	10191.00	0.70	50.70		10189.70	78.05	8.81	8.41	0.12
87	10283.00	0.70	30.90		10281.70	78.89	9.53	9.13	0.26
88	10315.00	0.80	42.50		10313.69	79.22	9.78	9.38	0.57
89	10346.00	3.20	87.70		10344.67	79.42	10.79	10.39	8.70
90	10377.00	8.10	88.40		10375.51	79.51	13.84	13.44	15.81
91	10408.00	13.90	85.90		10405.93	79.84	19.74	19.34	18.77
92	10440.00	17.90	85.90		10436.70	80.47	28.49	28.08	12.50
93	10471.00	18.50	85.20		10466.15	81.22	38.14	37.73	2.06
94	10502.00	21.50	82.70		10495.28	82.35	48.68	48.26	10.06
95	10533.00	25.40	85.50		10523.71	83.60	60.94	60.52	13.08
96	10564.00	28.50	86.20		10551.34	84.61	74.96	74.53	10.05
97	10595.00	33.90	87.50		10577.85	85.48	90.98	90.55	17.55
98	10626.00	38.10	87.60		10602.92	86.25	109.19	108.75	13.55
99	10657.00	40.10	87.50		10626.98	87.09	128.72	128.27	6.45
100	10688.00	45.50	88.40		10649.72	87.84	149.76	149.31	17.53
101	10720.00	50.00	89.60		10671.23	88.24	173.44	172.99	14.33
102	10751.00	51.20	89.60		10690.90	88.41	197.39	196.94	3.87
103	10782.00	56.70	88.90		10709.14	88.74	222.44	221.99	17.84
104	10813.00	61.10	90.40		10725.15	88.89	248.98	248.52	14.79
105	10844.00	64.40	91.30		10739.34	88.48	276.53	276.08	10.95
106	10875.00	67.80	90.60		10751.90	88.01	304.86	304.41	11.16
107	10906.00	71.40	89.30		10762.70	88.04	333.91	333.46	12.26
108	10937.00	74.70	88.50		10771.74	88.61	363.56	363.10	10.93
109	10969.00	77.90	88.10		10779.32	89.54	394.63	394.17	10.07
110	11000.00	80.20	88.40		10785.21	90.47	425.05	424.58	7.48
111	11031.00	82.60	88.40		10789.84	91.32	455.68	455.22	7.74
112	11048.00	84.80	88.10		10791.71	91.84	472.57	472.10	13.06
113	11116.00	91.70	87.60		10793.78	94.39	540.45	539.97	10.17
114	11190.00	92.10	88.60		10791.33	96.84	614.37	613.87	1.45
115	11284.00	90.40	88.80		10789.28	98.97	708.32	707.81	1.82
116	11378.00	88.20	90.00		10790.43	99.96	802.30	801.78	2.67
117	11470.00	87.80	89.20		10793.64	100.60	894.24	893.72	0.97
118	11562.00	89.10	88.80		10796.13	102.20	986.19	985.66	1.48
119	11656.00	91.90	90.90		10795.31	102.45	1080.17	1079.64	3.72

SUNBURST CONSULTING, INC.

Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	8/31/2012
Finish:	9/8/2012
Directional Supervision:	RPM Inc.

Date: 9/26/2012
Time: 13:34

F9 to re-calculate

Proposed dir: **90.29**

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
120	11749.00	91.10	92.80	10792.87	99.45	1173.09	1172.57	2.22
121	11842.00	89.80	94.30	10792.14	93.69	1265.90	1265.41	2.13
122	11937.00	88.70	94.40	10793.38	86.49	1360.62	1360.16	1.16
123	12031.00	88.00	94.00	10796.09	79.60	1454.33	1453.90	0.86
124	12124.00	88.00	94.20	10799.34	72.96	1547.03	1546.64	0.21
125	12218.00	88.00	93.60	10802.62	66.57	1640.76	1640.40	0.64
126	12310.00	87.60	93.30	10806.15	61.04	1732.52	1732.19	0.54
127	12403.00	89.70	91.80	10808.34	56.90	1825.40	1825.08	2.77
128	12495.00	89.50	91.60	10808.98	54.17	1917.35	1917.05	0.31
129	12585.00	89.60	90.60	10809.69	52.44	2007.33	2007.04	1.12
130	12676.00	90.00	89.90	10810.01	52.05	2098.33	2098.04	0.89
131	12708.00	90.80	89.70	10809.78	52.16	2130.33	2130.04	2.58
132	12769.00	90.00	89.40	10809.36	52.64	2191.32	2191.03	1.40
133	12863.00	90.20	89.70	10809.19	53.38	2285.32	2285.02	0.38
134	12956.00	90.20	89.90	10808.87	53.70	2378.32	2378.02	0.22
135	13050.00	90.60	89.20	10808.21	54.44	2472.31	2472.01	0.86
136	13144.00	89.90	90.10	10807.80	55.01	2566.31	2566.00	1.21
137	13238.00	88.90	90.40	10808.79	54.60	2660.30	2659.99	1.11
138	13331.00	89.70	90.90	10809.92	53.55	2753.29	2752.98	1.01
139	13425.00	92.10	91.40	10808.45	51.66	2847.25	2846.95	2.61
140	13519.00	91.60	93.20	10805.41	47.89	2941.12	2940.84	1.99
141	13618.00	89.30	94.20	10804.63	41.50	3039.90	3039.66	2.53
142	13711.00	89.10	94.30	10805.93	34.61	3132.64	3132.42	0.24
143	13742.00	89.20	94.90	10806.39	32.13	3163.54	3163.33	1.96
144	13805.00	90.70	95.10	10806.45	26.64	3226.30	3226.12	2.40
145	13899.00	89.20	94.90	10806.53	18.44	3319.93	3319.80	1.61
146	13992.00	88.00	94.90	10808.80	10.50	3412.57	3412.47	1.29
147	14086.00	88.00	94.30	10812.08	2.97	3506.21	3506.15	0.64
148	14180.00	90.10	93.20	10813.64	-3.18	3599.98	3599.95	2.52
149	14274.00	90.00	92.30	10813.56	-7.69	3693.87	3693.87	0.96
150	14368.00	90.50	92.40	10813.15	-11.54	3787.79	3787.80	0.54
151	14462.00	89.20	92.00	10813.39	-15.15	3881.72	3881.75	1.45
152	14555.00	89.50	92.80	10814.45	-19.05	3974.63	3974.68	0.92
153	14649.00	87.60	91.70	10816.83	-22.73	4068.53	4068.59	2.34
154	14743.00	90.80	91.70	10818.14	-25.52	4162.46	4162.54	3.40
155	14837.00	94.00	91.30	10814.20	-27.98	4256.34	4256.42	3.43
156	14938.00	94.00	91.30	10807.16	-30.27	4357.06	4357.16	0.00

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T Sidetrack #1		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	9/8/2012
Finish:	9/18/2012
Directional Supervision:	RPM Inc

Date: 9/26/2012
Time: 13:34

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: **90.29**

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
Tie	13238.00	88.90	90.40	10808.78	54.60	2660.30	2659.99	1.11
1	13300.00	88.50	90.70	10810.19	54.01	2722.28	2721.98	0.81
2	13333.00	88.80	90.00	10810.96	53.81	2755.27	2754.97	2.31
3	13365.00	87.00	88.60	10812.14	54.20	2787.25	2786.94	7.12
4	13396.00	87.40	87.40	10813.65	55.28	2818.19	2817.88	4.08
5	13427.00	87.50	87.10	10815.03	56.76	2849.12	2848.80	1.02
6	13458.00	89.30	87.40	10815.90	58.25	2880.08	2879.74	5.89
7	13491.00	89.70	88.10	10816.18	59.55	2913.05	2912.71	2.44
8	13521.00	89.60	88.40	10816.37	60.46	2943.03	2942.69	1.05
9	13552.00	90.60	89.20	10816.31	61.11	2974.03	2973.68	4.13
10	13583.00	90.60	88.90	10815.99	61.62	3005.02	3004.67	0.97
11	13615.00	90.60	88.80	10815.65	62.27	3037.01	3036.66	0.31
12	13646.00	90.40	89.20	10815.38	62.81	3068.01	3067.65	1.44
13	13677.00	90.20	88.50	10815.22	63.43	3099.00	3098.64	2.35
14	13708.00	90.20	88.90	10815.11	64.13	3129.99	3129.63	1.29
15	13739.00	90.60	89.10	10814.90	64.67	3160.99	3160.62	1.44
16	13771.00	91.20	89.20	10814.39	65.15	3192.98	3192.61	1.90
17	13802.00	91.20	89.10	10813.74	65.61	3223.97	3223.59	0.32
18	13833.00	91.00	90.40	10813.15	65.74	3254.96	3254.59	4.24
19	13864.00	91.30	90.50	10812.53	65.50	3285.95	3285.58	1.02
20	13896.00	91.20	90.60	10811.83	65.19	3317.94	3317.57	0.44
21	13927.00	90.40	91.30	10811.40	64.68	3348.94	3348.57	3.43
22	13958.00	90.90	91.40	10811.04	63.95	3379.93	3379.56	1.64
23	13989.00	90.20	91.30	10810.75	63.22	3410.92	3410.55	2.28
24	14021.00	89.70	90.80	10810.77	62.63	3442.91	3442.55	2.21
25	14052.00	88.70	91.30	10811.21	62.06	3473.90	3473.54	3.61
26	14083.00	88.90	91.30	10811.86	61.36	3504.89	3504.53	0.65
27	14114.00	90.90	92.10	10811.91	60.44	3535.87	3535.52	6.95
28	14146.00	92.10	91.80	10811.07	59.35	3567.84	3567.50	3.87
29	14177.00	91.70	91.70	10810.05	58.41	3598.81	3598.47	1.33
30	14208.00	91.20	90.90	10809.26	57.70	3629.79	3629.45	3.04
31	14240.00	89.70	90.90	10809.01	57.20	3661.79	3661.45	4.69
32	14271.00	90.00	91.10	10809.09	56.66	3692.78	3692.45	1.16
33	14302.00	90.10	91.30	10809.06	56.01	3723.77	3723.44	0.72
34	14334.00	90.60	91.70	10808.87	55.17	3755.76	3755.44	2.00
35	14365.00	90.20	91.50	10808.65	54.31	3786.75	3786.43	1.44
36	14396.00	90.00	91.70	10808.60	53.44	3817.74	3817.42	0.91
37	14427.00	89.30	91.30	10808.79	52.63	3848.73	3848.41	2.60
38	14459.00	89.30	91.40	10809.18	51.88	3880.71	3880.40	0.31
39	14490.00	89.10	91.20	10809.61	51.17	3911.70	3911.39	0.91

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T Sidetrack #1		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	9/8/2012
Finish:	9/18/2012
Directional Supervision:	
	RPM Inc

Date: 9/26/2012
Time: 13:34

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir:

90.29

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/ 100
			AZM	TVD				
38	14521.00	89.00	90.70	10810.12	50.66	3942.69	3942.39	1.64
40	14552.00	88.70	90.80	10810.75	50.25	3973.69	3973.38	1.02
41	14584.00	89.00	90.90	10811.39	49.78	4005.68	4005.37	0.99
42	14615.00	89.40	90.90	10811.82	49.29	4036.67	4036.37	1.29
43	14646.00	90.20	90.90	10811.93	48.81	4067.66	4067.37	2.58
44	14677.00	90.00	90.50	10811.88	48.43	4098.66	4098.36	1.44
45	14709.00	90.50	90.40	10811.74	48.18	4130.66	4130.36	1.59
46	14740.00	90.90	90.50	10811.36	47.93	4161.66	4161.36	1.33
47	14771.00	91.20	90.40	10810.79	47.69	4192.65	4192.36	1.02
48	14803.00	90.90	90.20	10810.20	47.52	4224.65	4224.35	1.13
49	14834.00	91.10	89.90	10809.66	47.49	4255.64	4255.35	1.16
50	14865.00	89.60	90.30	10809.47	47.44	4286.64	4286.34	5.01
51	14896.00	88.90	89.90	10809.88	47.39	4317.64	4317.34	2.60
52	14927.00	88.40	89.30	10810.61	47.60	4348.63	4348.33	2.52
53	14959.00	88.20	89.00	10811.56	48.08	4380.61	4380.31	1.13
54	14990.00	88.20	88.70	10812.53	48.70	4411.59	4411.28	0.97
55	15021.00	88.30	88.70	10813.48	49.40	4442.57	4442.26	0.32
56	15052.00	89.10	89.30	10814.18	49.94	4473.55	4473.24	3.23
57	15084.00	88.50	89.50	10814.85	50.28	4505.54	4505.23	1.98
58	15115.00	89.50	89.80	10815.39	50.47	4536.54	4536.22	3.37
59	15146.00	90.10	89.90	10815.50	50.55	4567.54	4567.22	1.96
60	15177.00	90.80	90.20	10815.26	50.52	4598.54	4598.22	2.46
61	15209.00	90.30	90.30	10814.95	50.38	4630.53	4630.22	1.59
62	15240.00	90.60	89.80	10814.71	50.35	4661.53	4661.22	1.88
63	15271.00	90.40	89.60	10814.44	50.52	4692.53	4692.22	0.91
64	15302.00	90.30	89.20	10814.25	50.84	4723.53	4723.21	1.33
65	15333.00	90.20	88.60	10814.11	51.44	4754.52	4754.20	1.96
66	15365.00	90.20	87.80	10814.00	52.44	4786.51	4786.18	2.50
67	15396.00	90.30	87.80	10813.87	53.63	4817.48	4817.15	0.32
68	15427.00	90.20	88.80	10813.73	54.55	4848.47	4848.13	3.24
69	15458.00	90.50	89.20	10813.54	55.09	4879.46	4879.12	1.61
70	15490.00	90.50	89.10	10813.26	55.57	4911.46	4911.11	0.31
71	15521.00	89.10	90.70	10813.37	55.62	4942.46	4942.11	6.86
72	15584.00	89.30	91.10	10814.25	54.63	5005.44	5005.10	0.71
73	15615.00	89.20	91.10	10814.65	54.04	5036.43	5036.10	0.32
74	15677.00	90.00	91.30	10815.09	52.74	5098.42	5098.09	1.33
75	15708.00	90.10	91.00	10815.06	52.12	5129.41	5129.08	1.02
76	15771.00	89.90	91.10	10815.06	50.96	5192.40	5192.08	0.35
77	15802.00	89.10	91.30	10815.33	50.31	5223.39	5223.07	2.66
78	15865.00	89.00	90.80	10816.38	49.16	5286.37	5286.06	0.81

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SUNBURST CONSULTING, INC.

>

Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T Sidetrack #1		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	9/8/2012
Finish:	9/18/2012
Directional Supervision:	RPM Inc

Date: 9/26/2012
Time: 13:34

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir: **90.29**

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
79	15896.00	88.90	90.90	10816.94	48.70	5317.37	5317.05	0.46
80	15958.00	88.30	90.70	10818.46	47.83	5379.34	5379.03	1.02
81	15990.00	88.30	90.40	10819.41	47.53	5411.32	5411.02	0.94
82	16052.00	88.10	91.20	10821.36	46.66	5473.29	5472.98	1.33
83	16083.00	88.40	92.80	10822.30	45.58	5504.25	5503.95	5.25
84	16146.00	90.80	93.60	10822.74	42.06	5567.15	5566.86	4.02
85	16177.00	92.10	93.60	10821.96	40.12	5598.08	5597.80	4.19
86	16239.00	93.00	94.10	10819.20	35.96	5659.87	5659.62	1.66
87	16270.00	91.80	93.90	10817.90	33.80	5690.77	5690.53	3.92
88	16333.00	91.50	93.30	10816.09	29.84	5753.62	5753.40	1.06
89	16364.00	88.80	93.40	10816.01	28.03	5784.56	5784.35	8.72
90	16427.00	87.60	92.90	10817.98	24.57	5847.44	5847.24	2.06
91	16520.00	87.70	93.10	10821.80	19.71	5940.23	5940.06	0.24
92	16552.00	88.80	92.20	10822.78	18.23	5972.18	5972.01	4.44
93	16614.00	88.80	92.40	10824.07	15.74	6034.12	6033.96	0.32
94	16677.00	88.80	93.30	10825.39	12.61	6097.03	6096.88	1.43
95	16708.00	89.80	93.80	10825.77	10.69	6127.96	6127.83	3.61
96	16739.00	91.00	93.80	10825.56	8.64	6158.89	6158.77	3.87
97	16802.00	91.70	93.60	10824.07	4.57	6221.74	6221.64	1.16
98	16833.00	90.50	94.10	10823.48	2.49	6252.67	6252.58	4.19
99	16864.00	90.60	94.40	10823.18	0.19	6283.58	6283.50	1.02
100	16895.00	90.30	94.20	10822.93	-2.13	6314.49	6314.42	1.16
101	16958.00	88.90	94.30	10823.37	-6.80	6377.32	6377.27	2.23
102	16989.00	89.50	94.10	10823.81	-9.07	6408.23	6408.19	2.04
103	17020.00	89.70	93.60	10824.02	-11.15	6439.16	6439.13	1.74
104	17083.00	89.90	93.20	10824.24	-14.89	6502.05	6502.04	0.71
105	17114.00	90.00	93.20	10824.27	-16.62	6533.00	6533.00	0.32
106	17176.00	90.00	92.70	10824.27	-19.81	6594.92	6594.93	0.81
107	17208.00	88.40	92.30	10824.72	-21.20	6626.88	6626.91	5.15
108	17270.00	89.00	92.10	10826.12	-23.58	6688.82	6688.85	1.02
109	17302.00	89.70	92.30	10826.49	-24.81	6720.80	6720.83	2.28
110	17364.00	89.50	91.60	10826.92	-26.92	6782.76	6782.81	1.17
111	17395.00	88.90	91.60	10827.35	-27.79	6813.74	6813.80	1.94
112	17458.00	89.70	91.90	10828.12	-29.71	6876.71	6876.77	1.36
113	17489.00	89.80	92.00	10828.26	-30.76	6907.69	6907.76	0.46
114	17552.00	89.70	91.30	10828.53	-32.58	6970.66	6970.74	1.12
115	17583.00	89.40	91.00	10828.78	-33.20	7001.65	7001.73	1.37
116	17645.00	89.60	91.00	10829.32	-34.28	7063.64	7063.73	0.32
117	17707.00	91.00	91.20	10828.99	-35.47	7125.63	7125.72	2.28
118	17739.00	90.80	91.10	10828.49	-36.11	7157.62	7157.71	0.70

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T Sidetrack #1		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	9/8/2012
Finish:	9/18/2012
Directional Supervision:	RPM Inc

Date: 9/26/2012
Time: 13:34

F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir:

90.29

[North and East are positive and South and West are negative, relative to surface location]

No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
119	17801.00	89.20	90.40	10828.49	-36.93	7219.61	7219.71	2.82
120	17833.00	90.20	90.80	10828.66	-37.26	7251.61	7251.70	3.37
121	17864.00	90.50	90.70	10828.47	-37.67	7282.61	7282.70	1.02
122	17895.00	90.40	90.70	10828.22	-38.05	7313.60	7313.70	0.32
123	17926.00	91.00	90.70	10827.85	-38.42	7344.60	7344.70	1.94
124	18020.00	90.60	89.90	10826.53	-38.92	7438.59	7438.69	0.95
125	18051.00	90.10	89.90	10826.34	-38.86	7469.58	7469.69	1.61
126	18114.00	90.40	90.10	10826.07	-38.86	7532.58	7532.68	0.57
127	18177.00	91.20	89.40	10825.19	-38.59	7595.58	7595.67	1.69
128	18208.00	90.80	89.00	10824.65	-38.16	7626.57	7626.66	1.82
129	18270.00	89.00	88.50	10824.76	-36.80	7688.55	7688.64	3.01
130	18302.00	90.00	88.60	10825.04	-35.99	7720.54	7720.62	3.14
131	18333.00	89.90	88.40	10825.06	-35.18	7751.53	7751.61	0.72
132	18395.00	90.40	87.60	10824.90	-33.02	7813.49	7813.56	1.52
133	18427.00	90.20	87.00	10824.73	-31.51	7845.45	7845.51	1.98
134	18489.00	90.10	86.90	10824.57	-28.21	7907.37	7907.41	0.23
135	18520.00	89.00	87.60	10824.81	-26.72	7938.33	7938.36	4.21
136	18583.00	88.70	87.90	10826.08	-24.25	8001.27	8001.29	0.67
137	18614.00	88.90	89.10	10826.73	-23.44	8032.25	8032.26	3.92
138	18676.00	88.60	89.10	10828.08	-22.47	8094.23	8094.24	0.48
139	18707.00	89.20	89.50	10828.68	-22.09	8125.22	8125.23	2.33
140	18770.00	88.60	89.50	10829.89	-21.54	8188.20	8188.21	0.95
141	18801.00	88.10	89.50	10830.78	-21.27	8219.19	8219.19	1.61
142	18863.00	88.60	89.20	10832.56	-20.56	8281.16	8281.16	0.94
143	18895.00	88.30	89.10	10833.43	-20.09	8313.14	8313.14	0.99
144	18926.00	88.10	89.00	10834.40	-19.58	8344.13	8344.12	0.72
145	18957.00	88.00	88.80	10835.46	-18.98	8375.10	8375.09	0.72
146	19020.00	88.00	89.40	10837.66	-17.99	8438.05	8438.04	0.95
147	19051.00	88.00	89.40	10838.74	-17.67	8469.03	8469.02	0.00
148	19113.00	88.40	90.00	10840.69	-17.34	8531.00	8530.98	1.16
149	19145.00	88.50	90.20	10841.55	-17.40	8562.99	8562.97	0.70
150	19207.00	89.10	90.30	10842.85	-17.67	8624.98	8624.96	0.98
151	19238.00	89.50	90.20	10843.23	-17.81	8655.97	8655.95	1.33
152	19332.00	90.20	90.20	10843.47	-18.13	8749.97	8749.95	0.74
153	19363.00	90.20	90.00	10843.37	-18.19	8780.97	8780.95	0.65
154	19425.00	90.60	90.00	10842.93	-18.19	8842.97	8842.95	0.65
155	19457.00	90.60	89.90	10842.60	-18.16	8874.97	8874.95	0.31
156	19519.00	91.10	90.20	10841.68	-18.21	8936.96	8936.94	0.94
157	19550.00	91.30	90.00	10841.03	-18.27	8967.95	8967.93	0.91
158	19613.00	91.30	89.80	10839.60	-18.16	9030.94	9030.91	0.32

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SUNBURST CONSULTING, INC.

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Operator:	Oasis Petroleum North America, LLC		
Well :	Ash Federal 5300 11-18T Sidetrack #1		
County:	McKenzie	State:	ND
QQ:	Lot 1	Section:	18
Township:	153	N/S:	N
Range:	100	E/W:	W
Footages:	800	FN/SL:	N
	350	FE/WL:	W

Kick-off:	9/8/2012
Finish:	9/18/2012
Directional Supervision:	RPM Inc

Date: 9/26/2012
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F9 to re-calculate

Minimum Curvature Method (SPE-3362)

Proposed dir:

90.29

[North and East are positive and South and West are negative, relative to surface location]

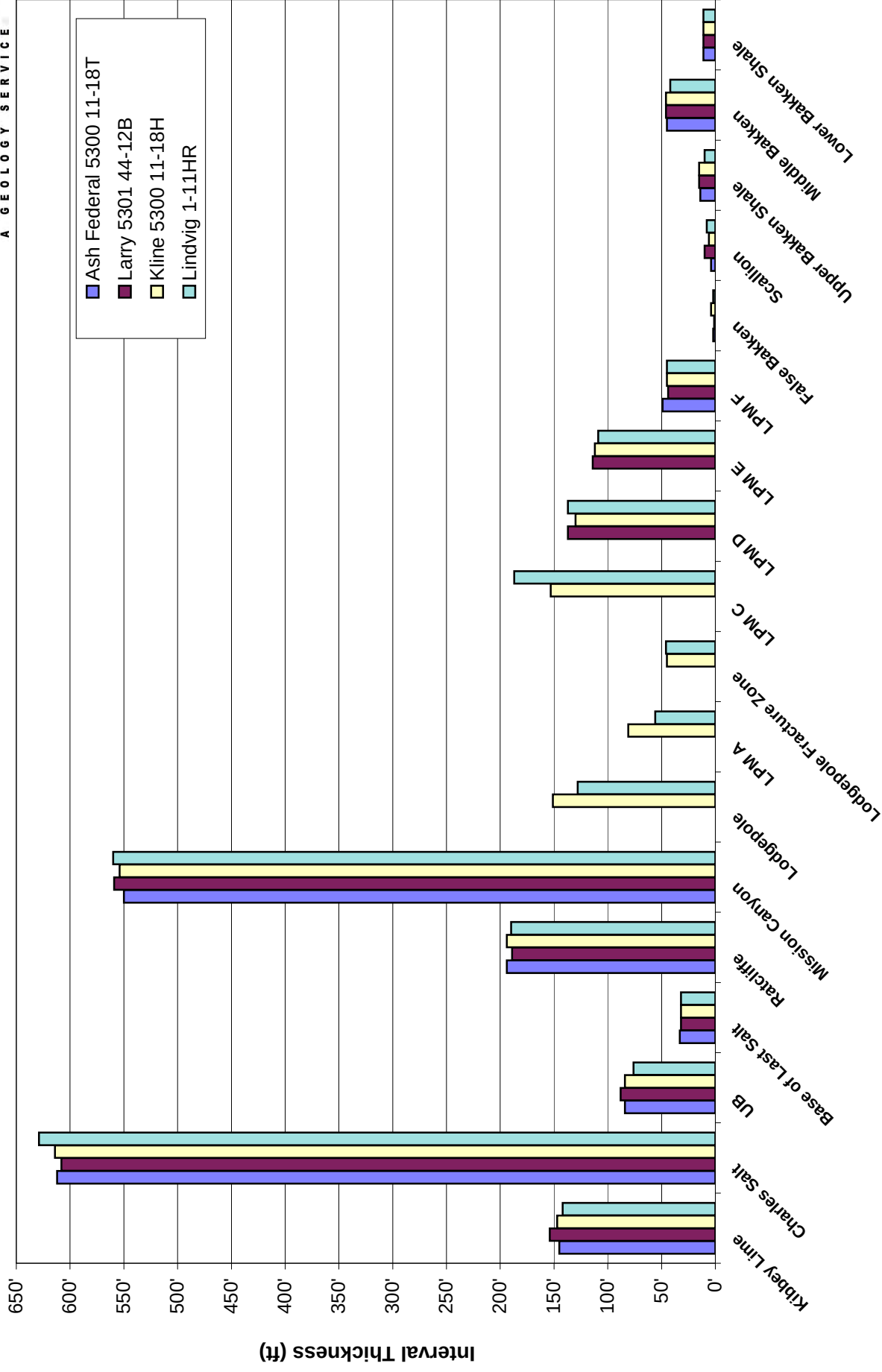
No.	MD	INC	TRUE		N-S	E-W	SECT	DLS/
			AZM	TVD				100
159	19644.00	90.60	89.30	10839.09	-17.91	9061.93	9061.91	2.77
160	19706.00	90.80	88.60	10838.33	-16.78	9123.92	9123.89	1.17
161	19738.00	90.60	87.80	10837.94	-15.77	9155.90	9155.86	2.58
162	19800.00	90.10	87.20	10837.56	-13.07	9217.84	9217.79	1.26
163	19831.00	89.90	86.40	10837.56	-11.34	9248.79	9248.73	2.66
164	19890.00	89.60	86.00	10837.82	-7.43	9307.66	9307.58	0.85
165	19921.00	90.00	86.00	10837.92	-5.27	9338.58	9338.49	1.29
166	19984.00	89.90	86.10	10837.98	-0.93	9401.43	9401.32	0.22
167	20015.00	89.80	86.20	10838.06	1.16	9432.36	9432.24	0.46
168	20077.00	90.00	86.40	10838.17	5.16	9494.23	9494.09	0.46
169	20108.00	89.80	86.60	10838.22	7.05	9525.18	9525.02	0.91
170	20171.00	90.10	86.40	10838.28	10.89	9588.06	9587.88	0.57
171	20202.00	90.10	86.70	10838.22	12.76	9619.00	9618.81	0.97
172	20264.00	90.20	86.20	10838.06	16.60	9680.88	9680.67	0.82
173	20295.00	90.50	86.30	10837.87	18.63	9711.82	9711.60	1.02
174	20358.00	90.80	86.60	10837.16	22.53	9774.69	9774.45	0.67
175	20420.00	91.20	87.20	10836.07	25.88	9836.59	9836.33	1.16
176	20482.00	91.20	87.20	10834.78	28.91	9898.50	9898.23	0.00

CONTROL DATA

Operator: Well Name: Location: Elevation:	Oasis Petroleum North America, LLC Larry 5301 44-12B SESE Section 12 T153N R101W McKenzie County, ND .25 mile NW of the Ash Federal 5300 11-18T KB: 2,083'					Oasis Petroleum North America, LLC Kline 5300 11-18H Lot 1 Section 18 T153N R100W McKenzie County, ND 200' south of the Ash Federal 5300 11-18T KB: 2,079'					SM Energy Company Lindvig 1-11HR SESE Section 11, T153N, R101W McKenzie County, ND 1.5 miles NW of the Ash Federal 5300 11-18T KB: 2,105'					
Formation/ Zone	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target	E-Log Top	Datum (MSL)	Interval Thickness	Thickness to Target
Kibbey Lime	8,348'	-6,265'	154'	2,454'	8,355'	-6,276'	147'	2,451'	8,363'	-6,258'	142'	2,436'	8,363'	-6,258'	142'	2,436'
Charles Salt	8,502'	-6,419'	608'	2,300'	8,502'	-6,423'	614'	2,304'	8,505'	-6,400'	629'	2,294'	8,505'	-6,400'	629'	2,294'
UB	9,110'	-7,027'	88'	1,692'	9,116'	-7,037'	84'	1,690'	9,134'	-7,029'	76'	1,665'	9,134'	-7,029'	76'	1,665'
Base of Last Salt	9,198'	-7,115'	32'	1,604'	9,200'	-7,121'	32'	1,606'	9,210'	-7,105'	32'	1,589'	9,210'	-7,105'	32'	1,589'
Ratcliffe	9,230'	-7,147'	189'	1,572'	9,232'	-7,153'	194'	1,574'	9,242'	-7,137'	190'	1,557'	9,242'	-7,137'	190'	1,557'
Mission Canyon	9,419'	-7,336'	559'	1,383'	9,426'	-7,347'	554'	1,380'	9,432'	-7,327'	560'	1,367'	9,432'	-7,327'	560'	1,367'
Lodgepole	9,978'	-7,895'	-	824'	9,980'	-7,901'	151'	826'	9,992'	-7,887'	128'	807'	9,992'	-7,887'	128'	807'
LPM A	-	-	-	-	10,131'	-8,052'	81'	675'	10,120'	-8,015'	56'	679'	10,120'	-8,015'	56'	679'
Lodgepole Fracture Zone	-	-	-	-	10,212'	-8,133'	45'	594'	10,176'	-8,071'	46'	623'	10,176'	-8,071'	46'	623'
LPM C	-	-	-	-	10,257'	-8,178'	153'	549'	10,222'	-8,117'	187'	577'	10,222'	-8,117'	187'	577'
LPM D	10,397'	-8,314'	137'	405'	10,410'	-8,331'	130'	396'	10,409'	-8,304'	137'	390'	10,409'	-8,304'	137'	390'
LPM E	10,534'	-8,451'	114'	268'	10,540'	-8,461'	112'	266'	10,546'	-8,441'	109'	253'	10,546'	-8,441'	109'	253'
LPM F	10,648'	-8,565'	44'	154'	10,652'	-8,573'	45'	154'	10,655'	-8,550'	45'	144'	10,655'	-8,550'	45'	144'
False Bakken	10,692'	-8,609'	1'	110'	10,697'	-8,618'	4'	109'	10,700'	-8,595'	2'	99'	10,700'	-8,595'	2'	99'
Scallion	10,693'	-8,610'	10'	109'	10,701'	-8,622'	6'	105'	10,702'	-8,597'	8'	97'	10,702'	-8,597'	8'	97'
Upper Bakken Shale	10,703'	-8,620'	15'	99'	10,707'	-8,628'	15'	99'	10,710'	-8,605'	10'	89'	10,710'	-8,605'	10'	89'
Middle Bakken	10,718'	-8,635'	46'	84'	10,722'	-8,643'	46'	84'	10,720'	-8,615'	42'	79'	10,720'	-8,615'	42'	79'
Lower Bakken Shale	10,764'	-8,681'	11'	38'	10,768'	-8,689'	11'	38'	10,762'	-8,657'	11'	37'	10,762'	-8,657'	11'	37'
Pronghorn	10,775'	-8,692'	19'	27'	10,779'	-8,700'	19'	27'	10,773'	-8,668'	19'	26'	10,773'	-8,668'	19'	26'
Three Forks	10,794'	-8,711'	8'	8'	10,798'	-8,719'	8'	8'	10,792'	-8,687'	7'	7'	10,792'	-8,687'	7'	7'
Three Forks Landing	10,802'	-8,719'	19'	0'	10,806'	-8,727'	19'	0'	10,799'	-8,694'	19'	0'	10,799'	-8,694'	19'	0'
Base of Target	10,821'	-8,738'	-	-	10,825'	-	-	-	10,818'	-8,713'	-	-	10,818'	-8,713'	-	-

INTERVAL THICKNESS

Oasis Petroleum North America, LLC - Ash Federal 5300 11-18T



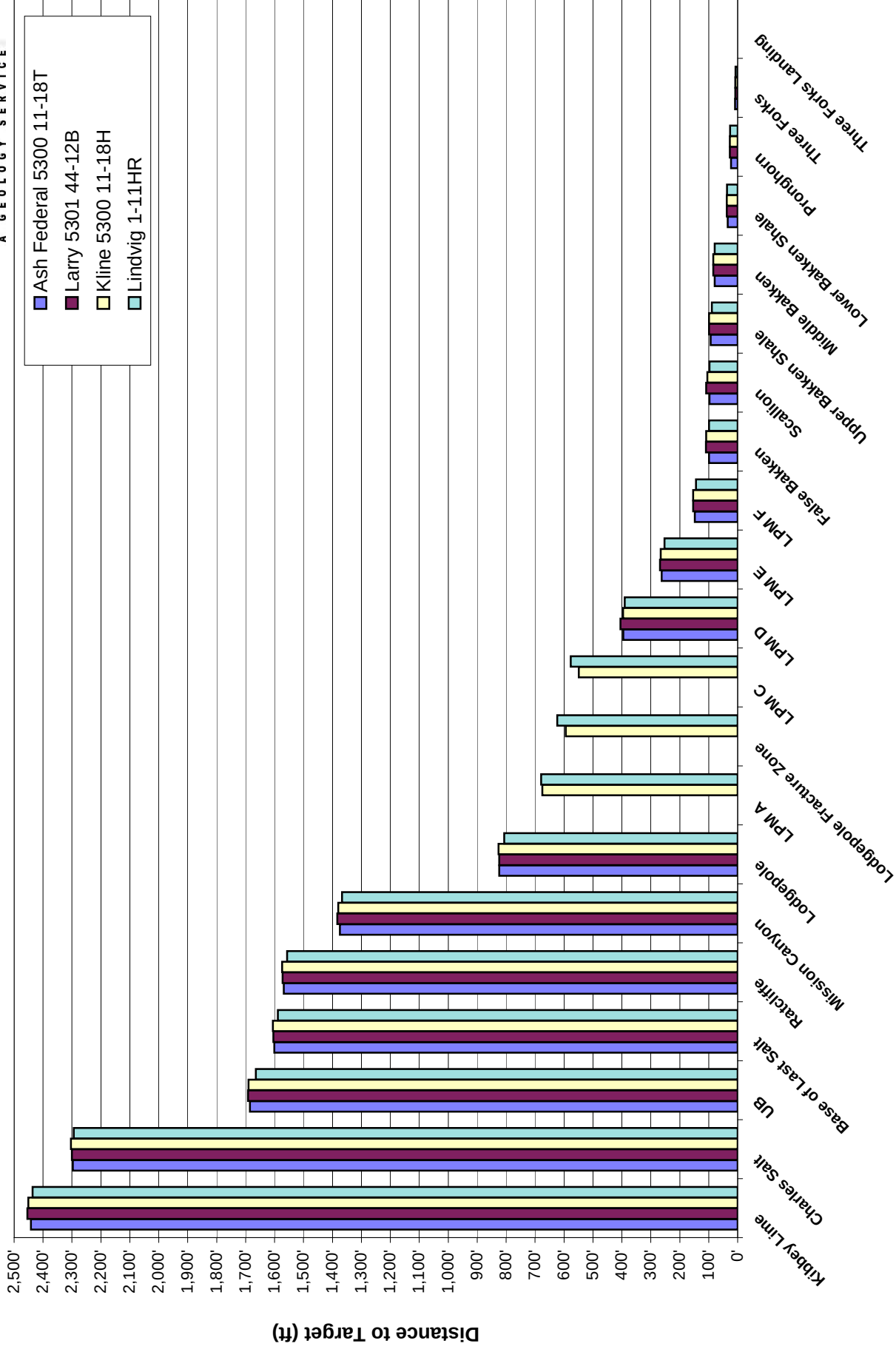
TARGET PROXIMATION

Formation/ Zone:	Proposed Landing Target From:		
	Larry 5301 44-12B	Kline 5300 11-18H	Lindvig 1-11HR
			Average of Offset Wells
Kibbey Lime	10,800'	10,797'	10,782'
Charles Salt	10,791'	10,795'	10,785'
UB	10,795'	10,793'	10,768'
Base of Last Salt	10,791'	10,793'	10,776'
Ratcliffe	10,792'	10,794'	10,777'
Mission Canyon	10,797'	10,794'	10,781'
Lodgepole	10,788'	10,790'	10,771'
LPM A	-	-	-
Lodgepole Fracture Zone	-	-	-
LPM C	-	-	-
LPM D	10,798'	10,789'	10,783'
LPM E	10,793'	10,791'	10,778'
LPM F	10,794'	10,794'	10,784'
False Bakken	10,799'	10,798'	10,788'
Scallion	10,800'	10,796'	10,788'
Upper Bakken Shale	10,794'	10,794'	10,784'
Middle Bakken	10,793'	10,793'	10,788'
Lower Bakken Shale	10,792'	10,792'	10,791'
Pronghorn	10,792'	10,792'	10,791'
Three Forks	10,787'	10,787'	10,786'
Three Forks Landing	10,788'	10,788'	10,788'

Landing Target (26' below the base of the LBS): **10,788'**

ISOPACH TO TARGET

Oasis Petroleum North America, LLC - Ash Federal 5300 11-18T



LITHOLOGY

Oasis Petroleum North America, LLC Ash Federal 11-18T

Rig crews caught lagged samples, under the supervision of Sunburst geologists, in 30' intervals from 8,140' to 10,960', 10' intervals from 10,960' to 11,110', and then 30' intervals to the TD at 20,482' MD. Sample or gamma ray marker tops have been inserted in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope. The drilling fluid was diesel-based invert from surface casing to the TD of the curve at 11,110', and salt water from 11,110' to 20,482' MD.

Drilling in Kibbey Formation

8140-8170 SILTSTONE: off white, light orange, light brown, friable, blocky, calcareous cement, moderately cemented

8170-8200 SILTSTONE: red to orange, red brown, friable, blocky, calcareous cement moderately cemented; rare ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

8200-8230 SILTSTONE: red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented

8230-8260 SILTSTONE: red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented; trace ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

8260-8290 red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented; trace ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

8290-8320 red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented; trace ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

Kibbey "Lime": 8,347' MD 8,346' TVD (-6,268')

8320-8350 ANHYDRITE: off white, microcrystalline, soft, massive, amorphous; SILTSTONE: red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented

8350-8380 LIMESTONE: mudstone to wackestone, light gray light gray brown, off white, microcrystalline, friable, dense, earthy

8380-8410 SILTSTONE: red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented; occasional LIMESTONE: mudstone to wackestone, light gray light gray brown, off white, microcrystalline, friable, dense, earthy; trace ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

8410-8440 SILTSTONE: red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented; trace ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

8440-8470 SILTSTONE: red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented; trace ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

Charles Salt:**8,492' MD 8,491' TVD (-6,413')**

8470-8500 SILTSTONE: red to orange, red brown, light orange to off white, trace light brown, friable, blocky, calcareous cement moderately cemented; common SALT: clear to milky, crystalline, hard, euhedral crystalline texture; trace ANHYDRITE: off white, microcrystalline, soft, massive, amorphous

8500-8530 SALT: frosted to trace clear, crystalline, hard, anhedral to rare subhedral to trace euhedral, crystalline texture, no visible porosity; rare SILTSTONE: red orange, firm, sub-blocky to subplaty, calcareous cement, moderately cemented, no visible porosity

8530-8560 SALT: clear to rare frosted, crystalline, hard, anhedral to rare subhedral, crystalline texture, no visible porosity; rare SILTSTONE: red orange, friable to trace firm, sub-blocky to subplaty, calcareous cement, moderately cemented, no visible porosity

8560-8590 SALT: frosted to rare clear, crystalline, hard, anhedral to rare subhedral to trace euhedral, crystalline texture, no visible porosity; trace SILTSTONE: red orange, friable, sub-blocky to subplaty, calcareous cement, moderately cemented, no visible porosity

8590-8620 No Sample

8620-8650 SALT: frosted to rare clear to trace milky, crystalline, hard, anhedral to rare subhedral, crystalline texture, no visible porosity; rare SILTSTONE: red orange, firm to trace friable, sub-blocky to subplaty, calcareous cement, moderately cemented, no visible porosity; trace ANHYDRITE: off white, soft, amorphous, no visible porosity

8650-8680 SALT: frosted to rare clear, crystalline, hard, anhedral to rare subhedral, crystalline texture, no visible porosity; rare SILTSTONE: medium gray to trace red orange, firm to trace friable, sub-blocky to subplaty, calcareous cement, moderately cemented, no visible porosity; trace ANHYDRITE: off white, soft, amorphous, no visible porosity

8680-8710 No Sample

8710-8740 SILTSTONE: red orange to trace medium gray, firm, sub-blocky to subplaty, calcareous cement, well cemented, no visible porosity; rare ANHYDRITE: off white, soft, amorphous, no visible porosity; rare SALT: frosted to trace clear, crystalline, hard, anhedral to trace subhedral, crystalline texture, no visible porosity

8740-8770 No Sample

8770-8800 SILTSTONE: medium gray, firm, sub-blocky to subplaty, calcareous cement, well cemented, no visible porosity; occasional ANHYDRITE: off white, soft, amorphous, no visible porosity

8800-8830 SALT: frosted to occasional clear, anhedral to rare subhedral to rare euhedral, crystalline texture, no visible porosity; trace SILTSTONE: medium gray, firm, sub-blocky to subplaty, calcareous cement, well cemented, no visible porosity

8830-8860 LIMESTONE: mudstone, cream, light gray to trace medium brown, microcrystalline, firm to trace brittle, earthy texture, no visible porosity; rare ANHYDRITE: off white, soft, amorphous, no visible porosity; rare SALT: clear to trace frosted, anhedral, crystalline texture, no visible porosity

8860-8890 LIMESTONE: mudstone, cream to rare light gray to trace medium gray, firm to trace hard, earthy to trace crystalline texture, trace siliceous, no visible porosity

8890-8920 LIMESTONE: mudstone, cream, microcrystalline, firm, earthy texture, possible intergranular porosity, trace dark light brown dead spotty oil stain; trace SALT: clear, anhedral, crystalline texture, no visible porosity

8920-8950 ANHYDRITE: off white, soft, amorphous, no visible porosity

8950-8980 SALT: frosted to occasional clear, crystalline, hard, anhedral to occasional subhedral to trace euhedral, crystalline texture, no visible porosity; trace ANHYDRITE: off white, soft, amorphous, no visible porosity

8980-9010 ANHYDRITE: off white, soft, amorphous, no visible porosity; rare SILTSTONE: medium gray, firm, sub-blocky to subplaty, calcareous cement, well cemented, no visible porosity; rare SALT: clear to trace frosted, crystalline, hard, anhedral to trace subhedral, crystalline texture, no visible porosity

9010-9040 ANHYDRITE: off white, soft, amorphous, no visible porosity; trace LIMESTONE: mudstone, light gray to trace dark gray, microcrystalline, firm to rare hard, earthy to rare crystalline texture, siliceous in part, no visible porosity; trace SALT: clear, crystalline, hard, anhedral to trace subhedral, crystalline texture, no visible porosity

9040-9070 ARGILLACEOUS LIMESTONE: mudstone, dark gray, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity; rare ANHYDRITE: off white, soft, amorphous, no visible porosity; trace LIMESTONE: mudstone, cream to trace dark gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace siliceous, no visible porosity

9070-9100 ARGILLACEOUS LIMESTONE: mudstone, dark gray, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity; occasional ANHYDRITE: off white, soft, amorphous, no visible porosity

Upper Berentson: **9,104' MD 9103 TVD (-7,025')**

9100-9130 No sample

9130-9160 SALT: clear to trace frosted, crystalline, hard, anhedral to trace subhedral, crystalline texture, no visible porosity; ARGILLACEOUS LIMESTONE: mudstone, light gray to gray, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity

9160-9190 ARGILLACEOUS LIMESTONE: mudstone, light gray, light gray brown, off white, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity; ANHYDRITE: off white, soft, amorphous; trace SALT: clear to trace frosted, crystalline, hard, anhedral to trace subhedral, crystalline texture

Base Last Salt: **9,188' MD 9,187' TVD (-7,109')**

9190-9220 ARGILLACEOUS LIMESTONE: mudstone, light gray to gray, light gray brown, off white, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity; ANHYDRITE: off white to white, soft, amorphous

Ratcliffe: **9,221' MD 9,220' TVD (-7,142')**

9250-9280 ANHYDRITE: off white, soft, amorphous, no visible porosity; common LIMESTONE: mudstone, creamish light gray to trace dark brown to trace medium brown, microcrystalline, firm, earthy to trace crystalline texture, trace siliceous, no visible porosity

Mission Canyon: **9,415' MD 9,414' TVD (-7,336')**

9220-9250 LIMESTONE: mudstone, light gray to gray, light brown, off white, trace dark gray, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity; ANHYDRITE: off white to white, soft, amorphous

9250-9280 LIMESTONE: mudstone, light brown, off white, trace dark gray, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity trace fossil fragment; ANHYDRITE: off white to white, soft, amorphous

9280-9310 LIMESTONE: mudstone, light brown, off white, light brown gray, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity; ANHYDRITE: off white to white, soft, amorphous

9310-9340 LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity; trace ANHYDRITE: off white to white, soft, amorphous

9340-9370 LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity trace fossil fragment

9370-9400 LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity

9400-9430 LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity

9430-9460 LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity, trace fossil fragments

9460-9490 ARGILLACEOUS LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity, trace fossil fragments

9490-9520 ARGILLACEOUS LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity, trace fossil fragments

9520-9550 ARGILLACEOUS LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity, trace fossil fragments, trace alga material

9550-9580 ARGILLACEOUS LIMESTONE: mudstone, light gray to gray, light brown, off white, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity, trace fossil fragments, trace alga material

9580-9610 ARGILLACEOUS LIMESTONE: mudstone, off white, light brown, light gray, microcrystalline, firm to friable, earthy texture, trace disseminated pyrite, no visible porosity, trace spar calcite, trace alga material

9610-9640 LIMESTONE: mudstone, cream, microcrystalline, firm, earthy texture, possible intergranular porosity, trace dark brown dead spotty oil stain

9640-9670 LIMESTONE: mudstone, cream to trace dark to light gray, microcrystalline, firm to trace hard, earthy texture, trace argillaceous, no visible porosity

9670-9700 LIMESTONE: mudstone, creamish light gray to occasional tan to trace dark brown, microcrystalline, firm to trace hard, earthy to trace crystalline texture, siliceous in part, no visible porosity

9700-9730 LIMESTONE: mudstone, light gray, microcrystalline, firm, earthy texture, no visible porosity; occasional ANHYDRITE: off white, soft, amorphous, no visible porosity

9730-9760 No Sample

9760-9790 LIMESTONE: mudstone, cream to trace light to dark gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace siliceous, trace spar calcareous, possible intergranular porosity, trace dark brown dead spotty oil stain

9790-9820 LIMESTONE: mudstone, cream to rare off white, microcrystalline, hard to trace firm, earthy texture, rare spar calcareous, possible intergranular porosity, trace dark brown dead spotty oil stain

9820-9850 LIMESTONE: mudstone, tan to trace dark gray, microcrystalline, firm, earthy texture, trace argillaceous, possible intergranular porosity, rare dark brown dead spotty oil stain

9850-9880 ARGILLACEOUS LIMESTONE: mudstone, blackish gray to trace cream, microcrystalline, firm to trace friable, earthy texture, trace disseminated pyrite, trace spar calcareous, trace intercrystalline porosity; trace LIMESTONE: mudstone, cream to trace light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace siliceous, trace spar calcareous, no visible porosity

9880-9910 LIMESTONE: mudstone, cream to trace light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, no visible porosity; trace ARGILLACEOUS LIMESTONE: mudstone, blackish gray, microcrystalline, friable, earthy texture, trace disseminated pyrite, no visible porosity

9910-9940 No Sample

9940-9970 LIMESTONE: mudstone, light gray, microcrystalline, hard, earthy to occasional crystalline texture, siliceous in part, no visible porosity; rare ARGILLACEOUS LIMESTONE: mudstone, blackish gray, firm, earthy texture, trace disseminated pyrite, no visible porosity

Lodgepole: 9,965' MD 9,964' TVD (-7,886')

9970-10000 ARGILLACEOUS LIMESTONE: mudstone, medium gray to trace dark gray, microcrystalline, firm, earthy texture, trace disseminated pyrite, no visible porosity

10000-10030 ARGILLACEOUS LIMESTONE: mudstone, medium gray to rare light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace siliceous, trace disseminated pyrite, no visible porosity

10030-10060 ARGILLACEOUS LIMESTONE: mudstone, medium gray to rare light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace siliceous, trace disseminated pyrite, no visible porosity

10060-10090 ARGILLACEOUS LIMESTONE: mudstone, medium gray to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10090-10120 No sample

10120-10150 ARGILLACEOUS LIMESTONE: mudstone, light to dark gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10150-10180 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10180-10210 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10210-10240 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10240-10270 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10270-10300 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10300-10330 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10335-10360 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10360-10390 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10390-10420 No sample

10420-10450 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10450-10480 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10480-10510 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, crystalline to earthy texture, trace disseminated pyrite, no visible porosity

10510-10540 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10540-10570 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10570-10600 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10600-10630 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10630-10660 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10660-10690 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10690-10720 ARGILLACEOUS LIMESTONE: mudstone, medium to light gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace disseminated pyrite, no visible porosity

10720-10750 LIMESTONE: mudstone, tan light gray to trace medium gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace siliceous, no visible porosity; trace SHALE: black, friable, earthy texture, trace disseminated pyrite, petroliferous, carbonaceous

False Bakken: **10,749' MD 10,689' TVD (-8,611')**

10720-10750 LIMESTONE: mudstone, tan light gray to trace medium gray, microcrystalline, firm to trace hard, earthy to trace crystalline texture, trace siliceous, no visible porosity; trace SHALE: black, friable, earthy texture, trace disseminated pyrite

Scallion: **10,751' MD 10,691' TVD (-8,613')**

10750-10780 No Sample

Upper Bakken Shale: **10,755' MD 10,695' TVD (-8,617')**

10750-10780 No Sample

Middle Bakken: **10,783' MD 10,709' TVD (-8,631')**

10780-10810 SILTSTONE: tan gray, friable, subplaty to sub-blocky, earthy texture, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, occasional dark brown spotty oil stain; trace SHALE: black, friable, earthy texture, trace disseminated pyrite, petroliferous, carbonaceous, fracture porosity

10810-10840 SILTSTONE: light gray, friable, subplaty to sub-blocky, earthy texture, calcareous cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, occasional dark brown spotty oil stain; trace SLITY SANDSTONE: light gray, very fine gray, friable, subangular to subrounded, well sorted, calcareous cement, moderately cemented, trace disseminated and nodular pyrite, possible intergranular porosity, occasional dark brown spotty oil stain

Lower Bakken Shale: **10,880' MD 10,754' TVD (-8,676')**

10840-10870 SHALE: black, friable, earthy texture, trace disseminated pyrite, petroliferous, carbonaceous, fracture porosity; occasional SLITY SANDSTONE: medium gray to tan, very fine gray, firm to trace friable, subangular to subrounded, well sorted, calcareous cement, well to trace moderately cemented, possible intergranular porosity, trace dark brown spotty oil stain

10870-10900 SHALE: black, friable, earthy texture, trace disseminated pyrite, petroliferous, carbonaceous, fracture porosity; trace SLITY SANDSTONE: tan gray, very fine gray, firm to trace friable, subangular to subrounded, well sorted, calcareous cement, well to trace moderately cemented, possible intergranular porosity, trace dark brown spotty oil stain

10900-10913 SHALE: black, friable, earthy texture, trace disseminated pyrite, petroliferous, carbonaceous, fracture porosity

Pronghorn: **10,913' MD 10,765' TVD (-8,687')**

10913-10930 SHALE: black, friable, earthy texture, trace disseminated pyrite, petroliferous, carbonaceous, fracture porosity

10930-10960 SHALE: dark brown black, friable, earthy texture, trace disseminated pyrite, petroliferous, carbonaceous, fracture porosity; trace SILTSTONE: medium gray, friable, subplaty to sub-blocky, earthy texture, dolomite cement, moderately cemented, trace disseminated and nodular pyrite, possible intergranular porosity, trace light brown spotty oil stain

Three Forks:**10,966' MD 10,779' TVD (-8,701')**

10960-10970 No Sample

10970-10980 SHALE: light green to rare dark black brown, friable, earthy texture, trace disseminated pyrite, no visible porosity; rare SILTSTONE: off white, friable, subplaty to sub-blocky, earthy texture, dolomite cement, moderately cemented, trace disseminated and nodular pyrite, possible intergranular porosity, trace light brown spotty oil stain

10980-10990 SHALE: light green to rare dark black brown, friable, earthy texture, trace disseminated pyrite, no visible porosity; rare SILTSTONE: off white, friable, subplaty to sub-blocky, earthy texture, dolomite cement, moderately cemented, trace disseminated and nodular pyrite, possible intergranular porosity, trace light brown spotty oil stain; trace DOLOMITE: mudstone, cream, microcrystalline, firm, crystalline texture, no visible porosity

10990-11000 SHALE: light green to trace dark black brown, friable, earthy texture, trace disseminated pyrite, no visible porosity; common DOLOMITE: cream, cream, microcrystalline, firm, crystalline texture, no visible porosity; trace SILTSTONE: off white, friable, subplaty to sub-blocky, earthy texture, dolomite cement, moderately cemented, trace disseminated and nodular pyrite, possible intergranular porosity, occasional light brown spotty oil stain

11000-11010 SHALE: light green, friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity; occasional DOLOMITE: mudstone, pink tan, microcrystalline, hard, crystalline texture, no visible porosity

11010-11020 No Sample

11020-11030 SHALE: light green, friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity; rare SILTSTONE: off white, friable, subplaty to sub-blocky, earthy texture, dolomite cement, moderately cemented, trace disseminated and nodular pyrite, possible intergranular porosity, rare light brown spotty oil stain; rare DOLOMITE: mudstone, pink tan, microcrystalline, hard, crystalline texture, no visible porosity

11030-11040 SHALE: light green, firm, earthy texture, trace disseminated pyrite, no visible porosity; occasional SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain; trace DOLOMITE: mudstone, pink tan, microcrystalline, hard, crystalline texture, no visible porosity

11040-11050 SHALE: light green, firm to trace friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity; rare SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain; trace DOLOMITE: mudstone, pink tan, microcrystalline, hard, crystalline texture, no visible porosity

11050-11060 SHALE: light green, firm to trace friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity; rare SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain

11060-11070 SHALE: light green, firm to trace friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity; rare SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain

11070-11080 SHALE: light green, friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity; occasional SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain

11080-11090 SHALE: light green, friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity; occasional SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain

11090-11100 SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain; SHALE: light green, friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity

11100-11110 SILTSTONE: off white, friable, subplaty to sub-blocky, well sorted, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity, trace light brown spotty oil stain; SHALE: light green, friable, earthy texture, trace disseminated and nodular pyrite, no visible porosity

11110-11140 DOLOMITE: mudstone, tan pink, friable, earthy texture, possible intergranular porosity, trace light brown spotty oil stain; occasional SHALE: light green, firm, sub-blocky to subplaty, earthy texture, no visible porosity

11140-11170 DOLOMITE: mudstone, tan pink, friable to trace firm, earthy texture, possible intergranular porosity, trace light brown spotty oil stain; rare SHALE: light green, firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11170-11200 SHALE: light green, firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, rare medium to light brown spotty oil stain

11200-11230 SHALE: light green, firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, rare medium to light brown spotty oil stain

11230-11260 No Sample

11260-11290 SHALE: light green, firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, possible intergranular porosity, trace medium brown laminated oil stain; abundant DOLOMITE: tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, common medium to light brown spotty to even oil stain

11290-11320 SHALE: light green, firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, possible intergranular porosity, trace medium brown laminated oil stain; common DOLOMITE: tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, common medium to light brown spotty to even oil stain

11320-11350 SHALE: light green, firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, possible intergranular porosity, trace medium brown laminated oil stain; common DOLOMITE: tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

11350-11380 SHALE: light green, firm to rare friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

11380-11410 DOLOMITE: mudstone, tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, firm to rare friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11410-11440 DOLOMITE: mudstone, tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, firm to rare friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11440-11470 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; abundant DOLOMITE: mudstone, tan pink to rare off white, friable to trace firm, earthy texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

11470-11500 DOLOMITE: mudstone, tan pink to rare off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, occasional medium to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11500-11530 DOLOMITE: mudstone, tan pink to rare off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, occasional medium to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11530-11560 SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

11560-11590 SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain

11590-11620 DOLOMITE: mudstone, tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11620-11650 DOLOMITE: mudstone, tan, light pink to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11650-11680 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

11680-11710 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

2010-12040 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12040-12070 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12070-12100 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12100-12130 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12130-12160 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; trace SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12160-12190 DOLOMITE: mudstone, tan, light brown to off white, light pink, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; trace SHALE: light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12190-12220 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; rare SHALE: light gray, light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12220-12250 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; rare SHALE: light gray, light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12250-12280 DOLOMITE: mudstone, tan, light brown to off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, trace dark to light brown spotty to even oil stain; rare SHALE: light gray, light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12280-12310 CLAYSTONE: light gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

12310-12340 No Sample

12340-12370 CLAYSTONE: medium to light gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

12370-12400 DOLOMITE: mudstone, tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain; occasional CLAYSTONE: light to medium gray to rare light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12400-12430 CLAYSTONE: light gray to trace medium gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; rare DOLOMITE: tan pink to rare off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain

12430-12460 CLAYSTONE: medium gray to trace light gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, possible intergranular porosity, trace medium brown laminated oil stain; trace DOLOMITE: tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain

12460-12490 CLAYSTONE: dark to medium gray to trace light green, firm to occasional friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; trace DOLOMITE: tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12490-12520 CLAYSTONE: medium to rare light gray to trace green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; rare DOLOMITE: off white to rare tan pink, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12520-12550 CLAYSTONE: light gray to occasional medium gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; trace DOLOMITE: off white to trace tan pink, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12550-12580 CLAYSTONE: medium gray to rare light gray to rare light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: off white to rare tan pink, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12580-12610 CLAYSTONE: light gray to occasional medium gray to rare dark gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; rare DOLOMITE: tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12610-12640 CLAYSTONE: light gray to common medium gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; rare DOLOMITE: off white to trace tan pink, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12640-12670 CLAYSTONE: light gray to occasional medium gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; trace DOLOMITE: tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12670-12700 CLAYSTONE: light gray to common medium gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; rare DOLOMITE: off white to trace tan pink, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12700-12730 CLAYSTONE: medium gray to occasional light gray to trace light green, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: off white to rare tan pink, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain

12730-12760 DOLOMITE: tan pink to trace off white, friable to trace firm, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green to common dark gray, friable to trace firm, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12760-12790 DOLOMITE: tan pink to trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12790-12820 DOLOMITE: tan pink to trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12820-12850 DOLOMITE: tan pink to trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12850-12880 DOLOMITE: tan pink to trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12880-12910 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12910-12940 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12940-12970 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

12970-13000 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13000-13030 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13030-13060 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13060-13090 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13090-13120 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13120-13150 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13150-13180 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13180-13210 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13210-13240 Sample heavily contaminated well lube; DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13240-13270 Sample heavily contaminated well lube; DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13270-13300 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13300-13330 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13330-13360 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13360-13390 DOLOMITE: tan, light pink, trace off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13390-13420 DOLOMITE: tan, light pink, trace off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13420-13450 No sample

13450-13480 DOLOMITE: tan, light pink, trace off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, occasional dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13480-13510 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13510-13540 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13540-13570 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13570-13600 No Sample

13600-13630 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13630-13660 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; rare DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty oil stain

13660-13690 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; rare DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty oil stain

13690-13720 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional medium to light brown spotty to even oil stain

13720-13750 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional medium to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13750-13780 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, rare nodular pyrite, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain

13780-13810 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional medium to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

13810-13840 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity; rare DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain

13840-13870 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

13870-13900 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace medium to light brown spotty oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

13900-13930 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

13930-13960 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

13960-13990 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional medium to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated, rare nodular pyrite, no visible porosity

13990-14020 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace light brown spotty oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

14020-14050 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace light brown spotty oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

14050-14080 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace light brown spotty oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

14080-14110 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated pyrite, rare nodular pyrite, no visible porosity

14110-14140 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

14140-14170 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14170-14200 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14200-14230 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14230-14260 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14260-14290 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14290-14320 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14320-14350 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14350-14380 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14380-14410 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14410-14440 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14440-14470 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14470-14500 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14500-14530 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14530-14560 Sample heavily contaminated well lube; DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14560-14590 Sample heavily contaminated well lube; DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14590-14620 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; common SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14620-14650 SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain

14650-14680 SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain

14680-14710 SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain

14710-14740 SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain

14740-14770 SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain

14770-14800 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14800-14830 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14830-14860 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14860-14890 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14890-14920 DOLOMITE: tan, light pink, off white, trace light brown, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; SHALE: light gray, light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

Sidetrack @ 13,335' MD:

13335-13360 DOLOMITE: off white, tan, light brown, light pink, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13360-13390 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13390-13420 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13420-13450 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13450-13480 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13480-13510 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13510-13540 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13540-13570 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13570-13600 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13600-13630 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13630-13660 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13660-13690 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13690-13720 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13720-13750 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; rare SHALE: light green to occasional dark to medium green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13750-13780 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; rare SHALE: light green to occasional dark to medium green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13780-13810 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green to rare medium green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13810-13840 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; rare SHALE: light green to trace medium green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13840-13870 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13870-13900 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13900-13930 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13930-13960 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13960-13990 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

13990-14020 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14020-14050 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14050-14080 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14080-14110 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14110-14140 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14140-14170 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14170-14200 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14200-14230 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14230-14260 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14260-14290 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14290-14320 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14320-14350 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14350-14380 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14380-14410 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14410-14440 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14440-14470 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14470-14500 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14500-14530 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14530-14560 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14560-14590 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14590-14620 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

14620-14650 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty oil stain; occasional SHALE: light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14650-14680 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty oil stain; occasional SHALE: light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

14680-14710 SHALE: light gray to rare light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

14710-14740 SHALE: light gray to rare light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

14740-14770 SHALE: light gray to occasional light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

14770-14800 SHALE: light gray to occasional light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

14800-14830 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

14830-14860 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

14860-14890 SHALE: light gray to common light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

14890-14920 SHALE: light gray to common light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

14920-14950 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, rare nodular pyrite, trace disseminated pyrite, no visible porosity; rare DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

14950-14980 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, rare nodular pyrite, trace disseminated pyrite, no visible porosity; rare DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

14980-15010 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

15010-15040 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

15040-15070 No sample

15070-15100 No sample

15100-15130 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15130-15160 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15160-15190 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15190-15220 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15220-15250 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15250-15280 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15280-15310 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15310-15340 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15340-15370 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15370-15400 SHALE: medium to light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, possible intergranular porosity, trace medium brown laminated oil stain; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15400-15430 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15430-15460 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15460-15490 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15490-15520 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15520-15550 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15550-15580 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15580-15610 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15610-15640 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15640-15670 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15670-15700 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; trace SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

15700-15730 No Sample

15730-15760 No Sample

15760-15790 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated pyrite, rare nodular pyrite, possible intergranular porosity, trace dark laminated oil stain

15790-15820 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

15820-15850 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

15850-15880 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

15880-15910 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated pyrite, rare nodular pyrite, no visible porosity

15910-15940 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated pyrite, rare nodular pyrite, no visible porosity

15940-15960 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

15960-16000 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16000-16030 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16030-16060 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16060-16090 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16090-16120 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16120-16150 No Sample

16150-16180 No Sample

16180-16210 DOLOMITE: mudstone, off white to common tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16210-16240 DOLOMITE: mudstone, off white to common tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16240-16270 DOLOMITE: mudstone, off white to common tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16270-16300 DOLOMITE: mudstone, off white to common tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace disseminated and nodular pyrite, no visible porosity

16300-16330 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16330-16360 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16360-16390 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16390-16420 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16420-16450 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16450-16480 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16480-16510 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16510-16540 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16540-16570 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16570-16600 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16600-16640 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16640-16660 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16660-16690 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16690-16720 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16720-16750 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16750-16780 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

16780-16810 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain

16810-16840 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain

16840-16870 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

16870-16900 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

16900-16930 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

16930-16960 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

16960-16990 SHALE: light green to trace light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

16990-17020 SHALE: light green to trace light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

17020-17050 SHALE: light green to trace light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain

17050-17080 SHALE: light green to trace light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty to even oil stain

17080-17110 SHALE: light gray to trace medium gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty oil stain

17110-17140 SHALE: light gray to trace medium gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark to light brown spotty oil stain

17140-17170 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

17170-17200 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain

17200-17230 SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

17230-17260 SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

17260-17290 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark brown spotty to even oil stain

17290-17320 SHALE: light gray to trace light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark brown spotty to even oil stain

17320-17350 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark brown spotty to even oil stain; occasional SHALE: light gray to rare light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17350-17380 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark brown spotty to even oil stain; occasional SHALE: light gray to rare light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17380-17410 DOLOMITE: mudstone, off white to tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark brown spotty to even oil stain; occasional SHALE: light green to occasional light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17410-17440 DOLOMITE: mudstone, off white to tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, trace dark brown spotty to even oil stain; occasional SHALE: light green to occasional light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17440-17470 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17470-17500 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17500-17530 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17530-17560 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17890-17920 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17920-17950 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17950-17980 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

17980-18010 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18010-18040 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18040-18070 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18070-18100 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18100-18130 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18130-18160 DOLOMITE: tan, light brown, light pink, off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18160-18190 SHALE: light green to trace light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark-light brown spotty to even oil stain

18190-18220 SHALE: light green to trace light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark-light brown spotty to even oil stain

18220-18250 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18250-18280 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18280-18310 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18310-18340 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18340-18370 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18370-18400 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain; common SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18400-18430 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

18430-18460 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; occasional DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common dark to light brown spotty to even oil stain

18460-18490 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18490-18520 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18520-18550 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

18550-18580 SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity; common DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain

18580-18610 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common medium to light brown spotty to even oil stain; rare SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18610-18640 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, common medium to light brown spotty to even oil stain; rare SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18640-18670 No Sample

18670-18700 No Sample

18700-18730 DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18730-18760 DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18760-18790 DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; rare SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18790-18820 DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; rare SHALE: light green to rare light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18820-18850 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18850-18880 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18880-18910 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18910-18940 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18940-18970 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

18970-19000 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19000-19030 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19030-19060 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19060-19090 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19090-19120 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19120-19150 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19150-19180 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19180-19210 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19210-19240 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19240-19270 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19270-19300 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19300-19330 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19330-19360 DOLOMITE: tan, light brown, light pink, trace off white, friable, earthy to sucrosic texture, possible intergranular porosity, trace very fine sand grains, common dark to light brown spotty to even oil stain; occasional SHALE: light green, light gray, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19360-19390 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19390-19420 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19420-19450 DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19450-19480 DOLOMITE: mudstone, off white to occasional tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19480-19510 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19510-19540 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19540-19570 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19570-19600 DOLOMITE: mudstone, tan pink to occasional off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19600-19630 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19630-19660 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19660-19690 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19690-19720 DOLOMITE: mudstone, tan pink to rare off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19720-19750 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19750-19780 DOLOMITE: mudstone, tan pink to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, occasional dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19780-19810 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19810-19840 DOLOMITE: mudstone, off white to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19840-19870 DOLOMITE: mudstone, off white to occasional medium gray to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19870-19900 DOLOMITE: mudstone, off white to occasional medium gray to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19900-19930 DOLOMITE: mudstone, off white to rare light gray to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19930-19960 DOLOMITE: mudstone, off white to rare light gray to rare tan pink, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19960-19990 DOLOMITE: mudstone, tan pink to rare off white to rare medium gray, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

19990-20020 DOLOMITE: mudstone, tan pink to rare off white to rare medium gray, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; occasional SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20020-20050 DOLOMITE: mudstone, off white to rare tan pink to trace medium gray, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20050-20080 DOLOMITE: mudstone, off white to rare tan pink to trace medium gray, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20080-20110 DOLOMITE: mudstone, tan pink to rare medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20110-20140 DOLOMITE: mudstone, tan pink to rare medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare dark to light brown spotty to even oil stain; rare SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20140-20170 DOLOMITE: mudstone, tan pink to rare medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20170-20200 DOLOMITE: mudstone, tan pink to rare medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20200-20230 DOLOMITE: mudstone, tan pink to occasional medium gray to trace medium gray, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20230-20260 DOLOMITE: mudstone, tan pink to occasional medium gray to trace medium gray, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20260-20290 DOLOMITE: mudstone, tan pink to occasional medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20290-20320 DOLOMITE: mudstone, tan pink to occasional medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20320-20350 DOLOMITE: mudstone, tan pink to occasional medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20350-20380 DOLOMITE: mudstone, tan pink to occasional medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

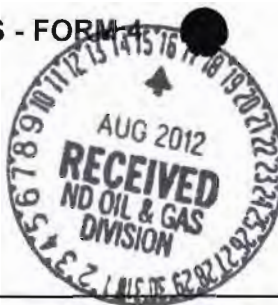
20380-20410 DOLOMITE: mudstone, tan pink to occasional medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20410-20440 DOLOMITE: mudstone, tan pink to occasional medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

20440-20482 DOLOMITE: mudstone, tan pink to occasional medium gray to trace off white, microcrystalline, friable, earthy to sucrosic texture, possible intergranular porosity, rare medium to light brown spotty to even oil stain; trace SHALE: light green, friable, sub-blocky to subplaty, earthy texture, trace nodular and disseminated pyrite, no visible porosity

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
23230

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date August 15, 2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Revise BHL and Casing Point

Well Name and Number Ash Federal 5300 11-18T					
Footages		Qtr-Qtr	Section	Township	Range
800 F N L 350 F W L		LOT1	18	153 N	100 W
Field	Pool	County			
Baker	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests to make the following changes to the well plan for the subject well:

Intermediate casing point changed to: 798' FNL & 840'FWL Sec. 18 T153N R100W
Bottom hole changed to: 850'FNL & 200'FEL Sec. 17 T153N R100W

Attached is revised plats, drill plan, directional plan and plot

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Brandi Terry		
Title Regulatory Specialist	Date August 15, 2012		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date August 20, 2012	
By 	
Title Petroleum Resource Specialist	

DRILLING PLAN									
OPERATOR		COUNTY/STATE		MA Pad 100W/S18					
WELL NAME		RIG		Kibby 30-32					
WELL TYPE									
LOCATION		Surface Location (survey plat): 200 100 200 100							
EST. T.D.		GROUND ELEV:		Finished Pad Elev.		Sub Height:			
TOTAL LATERS		9,504'		KB ELEV:					
PROGNOSIS:				LOGS:					
MARKER DEPTH (Surf Loc) DATUM (Surf Loc)				Type Interval OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota CBL/GR: Above top of cement/GR to base of casing MWD GR: KOP to lateral TD					
Pierre NDIC MAP 2,028 50' Greenhorn 4,615 -2,537' Mowry 5,025 -2,947' Dakota 5,429 -3,351' Rierdon 6,449 -4,371' Dunham Salt 6,784 -4,706' Dunham Salt Base 6,901 -4,823' Spearfish 6,985 -4,907' Pine Salt 7,250 -5,172' Pine Salt Base 7,297 -5,219' Opeche Salt 7,341 -5,263' Opeche Salt Base 7,372 -5,294' Broom Creek (Top of Minnelusa Gp.) 7,576 -5,498' Amsden 7,653 -5,575' Tyler 7,823 -5,745' Otter (Base of Minnelusa Gp.) 7,994 -5,916' Kibby Lime 8,368 -6,290' Charles Salt 8,516 -6,438' UB 9,140 -7,062' Base Last Salt 9,216 -7,138' Ratcliffe 9,262 -7,184' Mission Cyn 9,440 -7,362' Lodgepole 9,993 -7,915' Lodgepole Fracture Zone 10,209 -8,131' False Bakken 10,697 -8,619' Upper Bakken 10,706 -8,628' Middle Bakken 10,720 -8,642' Lower Bakken 10,766 -8,688' Three Forks 10,785 -8,707' Three Forks Target 10,805 -8,727' Base Target/Claystone 10,817 -8,739'				DEVIATION: Surf: 3 deg. max., 1 deg / 100'; srvy every 500' Prod: 5 deg. max., 1 deg / 100'; srvy every 100'					
DST'S:				None planned					
CORES:				None planned					
MUDLOGGING:				Two-Man: 8,316' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land					
BOP:				11" 5000 psi blind, pipe & annular					
Dip Rate:				12 deg / 100'					
Max. Anticipated BHP:				Surface Formation: Glacial till					
MUD:				Interval Type WT Vis WL Remarks					
Surface:				0' - 2,130' FW/Gel - Lime Sweeps 8.4-9.0 28-32 NC Circ Mud Tanks					
Intermediate:				2,130' - 11,090' Inven 9.5-10.4 40-50 30+HtHp Circ Mud Tanks					
Lateral:				11,090' - 20,594' Salt Water 9.8-10.2 28-32 NC Circ Mud Tanks					
CASING:				Size Wt ppf Hole Depth Cement WOC Remarks					
Surface:				9-5/8" 36# 13-1/2" 2,130' To Surface 12 100' into Pierre					
Intermediate:				7" 29/32# 8-3/4" 11,090' 3,929' 24 1500' above Dakota					
Production Liner:				4.5" 11.6# 6" 20,594' TOL @ 10,278' 50' above KOP					
PROBABLE PLUGS, IF REQ'D:									
OTHER:									
		MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI		
Surface:		2,130	2,130					Survey Company:	
KOP:		10,328	10,328					Build Rate: 12 deg / 100'	
EOC:		11,075	10,805					90.3	
Casing Point:		11,090	10,805					90.3	
Three Forks Lateral TD:		20,594	10,851					90.3	
Comments:									
Request Sundry for Open Hole Log Waiver									
Exception well: Kline 5300 11-18H (33-053-03426); 153N/100W/S18; located 190' S and 45' W of Ash Surface location									
36 Stage Plug and Perf									
A frac string will be utilized on this well									
									
Geology: 3 - 3000 2010/12				Engineering: M. Brown 8/15/2012					

**Oasis Petroleum
Well Summary
Ash Federal 5300 11-18T
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 2,130'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,130'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.02	3520 / 3.53	453 / 2.72	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.0 ppg fluid on backside (2150' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2150' setting depth).
- c) Based on string weight in 9.0 ppg fluid at 2125' TVD plus 100k# overpull. (Buoyed weight equals 66k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 55% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **413 sks** (234 bbls) Conventional system with 94 lb/sk cement, 4% extender, 2% expanding agent, 2% CaCl₂ and 0.25 lb/sk lost circulation control agent

Tail Slurry: **225 sks** (60 bbls) Conventional system with 94 lb/sk cement, 3% NaCl, and .25 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Ash Federal 5300 11-18T
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,600'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,600' – 10,328'	32	HCP-110	LTC	6.094"	6.000"	6,730	8,970	9,870
7"	10,328' – 11,090'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

**Special Drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,600'	6,600'	7", 29#, P-110, LTC, 8rd	8530 / 2.41*	11220 / 1.19	797 / 2.08
6,600' – 10,328'	3,728'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.20*	12460 / 1.29	
6,600' – 10,328'	3,728'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.06**	12460 / 1.29	
10,328' – 11,090'	762'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.51*	11220 / 1.15	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9,000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9.0 ppg fluid on backside—to 10,805' TVD.
- c. Based on string weight in 10 ppg fluid, (281k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess. TOC targeted to 1,500' above Dakota (3,929').

Pre-flush (Spacer): **125 bbls** Saltwater
20 sks Pozmix A
10 bbls Fresh Water

Lead Slurry: **246 sks** (95 bbls) Conventional system with 47 lb/sk cement, 10% NaCl, 34 lb/sk extender, 10% D020 extender, 1% D079 extender, 1% anti-settling agent, 1% fluid loss agent, 0.2% anti-foam agent, 0.6% retarder, and 0.4% dispersant

Tail Slurry: **666 sks** (196 bbls) Conventional system with 94 lb/sk cement, 10% NaCl, 35% Silica, 0.2% fluid loss agent, 0.8% dispersant and 0.3% retarder

**Oasis Petroleum
Well Summary
Ash Federal 5300 11-18T
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10,278' to 20,594'	13.5	P-110	BTC	4.000"	3.875"	2,270	3,020	3,780

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
10,278' to 20,594'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10670 / 1.99	12410 / 1.28	443 / 2.02	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,851' TVD.
- b) Burst pressure based on 9,000 psi treating pressure with 10.2 ppg internal fluid gradient and 9.0 ppg external fluid gradient @ 10,851' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 119k lbs.) plus 100k lbs overpull.



Azimuths to True North
Magnetic North: 8.47°

Magnetic Field
Strength: 56673.3snT
Dip Angle: 73.07°
Date: 6/6/2012
Model: IGRF200510

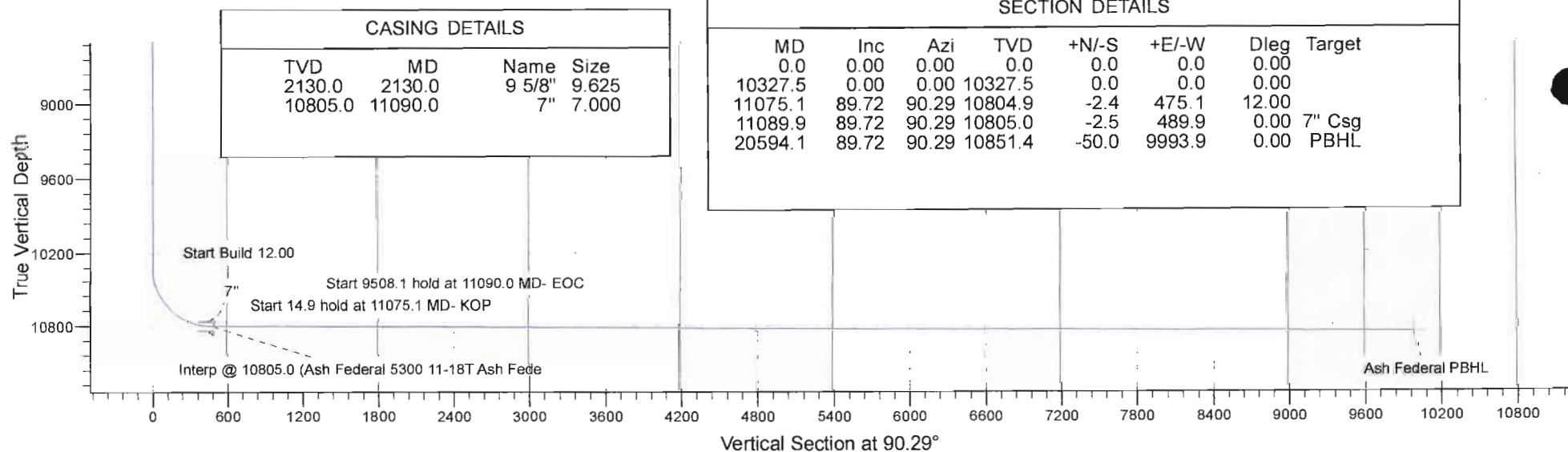
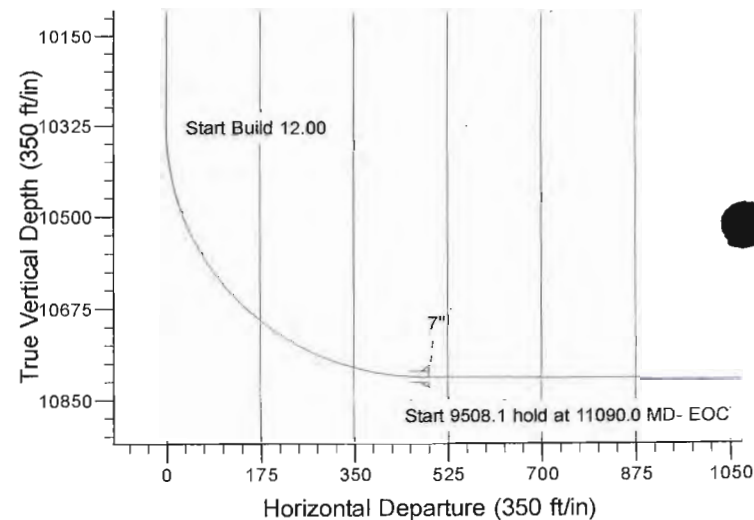
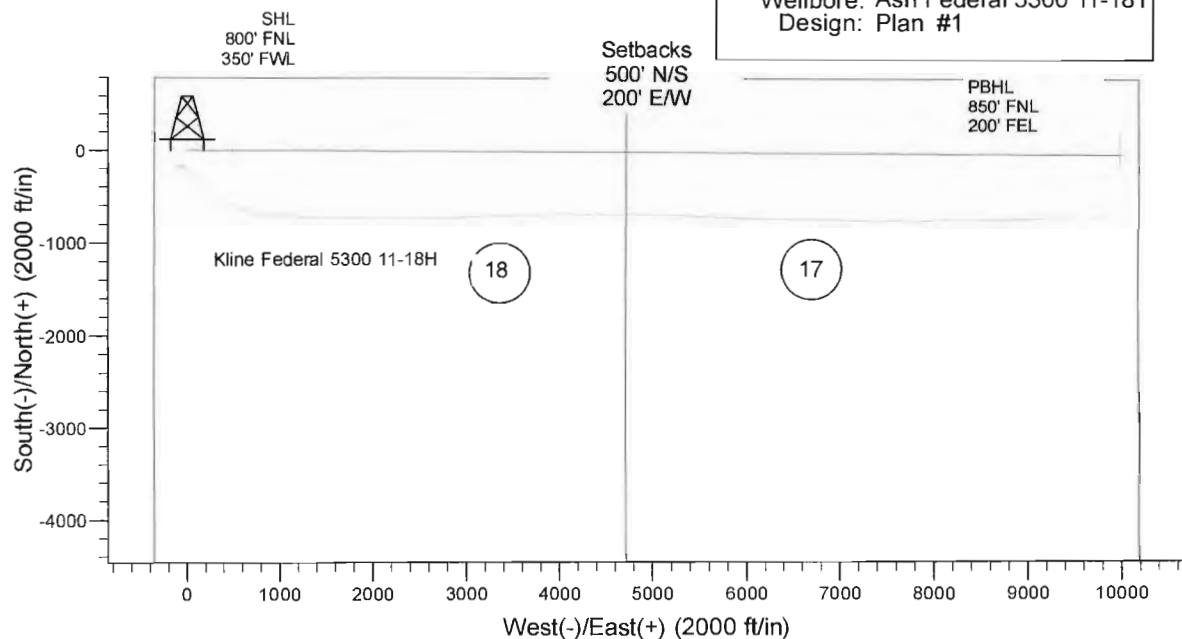


Project: Indian Hills
Site: 153N-100W-18/17
Well: Ash Federal 5300 11-18T
Wellbore: Ash Federal 5300 11-18T
Design: Plan #1

SITE DETAILS: 153N-100W-18/17

Site Centre Latitude: 48° 4' 47.250 N
Longitude: 103° 36' 9.720 W

Positional Uncertainty: 0.0
Convergence: -2.31
Local North: True



Oasis

Indian Hills

153N-100W-18/17

Ash Federal 5300 11-18T

Ash Federal 5300 11-18T

Plan: Plan #1

Standard Planning Report

15 August, 2012

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-18/17
Well: Ash Federal 5300 11-18T
Wellbore: Ash Federal 5300 11-18T
Design: Plan #1

Local Co-ordinate Reference: Well Ash Federal 5300 11-18T
TVD Reference: WELL @ 2078.0ft (Original Well Elev)
MD Reference: WELL @ 2078.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-18/17		
Site Position:		Northing:	124,709.16 m
From:	Lat/Long	Easting:	368,894.57 m
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in
		Latitude:	48° 4' 47.250 N
		Longitude:	103° 36' 9.720 W
		Grid Convergence:	-2.31 °

Well	Ash Federal 5300 11-18T		
Well Position	+N/-S	0.0 ft	Northing: 124,709.16 m
	+E/-W	0.0 ft	Easting: 368,894.57 m
Position Uncertainty	0.0 ft	Wellhead Elevation:	Latitude: 48° 4' 47.250 N
			Longitude: 103° 36' 9.720 W
			Ground Level: 2,053.0 ft

Wellbore	Ash Federal 5300 11-18T				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	6/6/2012	8.47	73.07	56,673

Design	Plan #1				
Audit Notes:					
Version:	Phase:	PROTOTYPE		Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)	
	0.0	0.0	0.0	90.29	

Plan Sections										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	

Planning Report

Database: OpenWellsCompass - EDM Prod
 Company: Oasis
 Project: Indian Hills
 Site: 153N-100W-18/17
 Well: Ash Federal 5300 11-18T
 Wellbore: Ash Federal 5300 11-18T
 Design: Plan #1

Local Co-ordinate Reference: Well Ash Federal 5300 11-18T
 TVD Reference: WELL @ 2078.0ft (Original Well Elev)
 MD Reference: WELL @ 2078.0ft (Original Well Elev)
 North Reference: True
 Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
2,028.0	0.00	0.00	2,028.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,130.0	0.00	0.00	2,130.0	0.0	0.0	0.0	0.00	0.00	0.00
9 5/8"									
4,615.0	0.00	0.00	4,615.0	0.0	0.0	0.0	0.00	0.00	0.00
Greenhorn									
5,025.0	0.00	0.00	5,025.0	0.0	0.0	0.0	0.00	0.00	0.00
Mowry									
5,429.0	0.00	0.00	5,429.0	0.0	0.0	0.0	0.00	0.00	0.00
Dakota									
6,449.0	0.00	0.00	6,449.0	0.0	0.0	0.0	0.00	0.00	0.00
Rierdon									
6,784.0	0.00	0.00	6,784.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt									
6,901.0	0.00	0.00	6,901.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt Base									
6,985.0	0.00	0.00	6,985.0	0.0	0.0	0.0	0.00	0.00	0.00
Spearfish									
7,250.0	0.00	0.00	7,250.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt									
7,297.0	0.00	0.00	7,297.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt Base									
7,341.0	0.00	0.00	7,341.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt									
7,372.0	0.00	0.00	7,372.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt Base									
7,576.0	0.00	0.00	7,576.0	0.0	0.0	0.0	0.00	0.00	0.00
Broom Creek (Top of Minnelusa Gp.)									
7,653.0	0.00	0.00	7,653.0	0.0	0.0	0.0	0.00	0.00	0.00
Amsden									
7,823.0	0.00	0.00	7,823.0	0.0	0.0	0.0	0.00	0.00	0.00
Tyler									
7,994.0	0.00	0.00	7,994.0	0.0	0.0	0.0	0.00	0.00	0.00
Otter (Base of Minnelusa Gp.)									
8,368.0	0.00	0.00	8,368.0	0.0	0.0	0.0	0.00	0.00	0.00
Kibby Lime									
8,516.0	0.00	0.00	8,516.0	0.0	0.0	0.0	0.00	0.00	0.00
Charles Salt									
9,140.0	0.00	0.00	9,140.0	0.0	0.0	0.0	0.00	0.00	0.00
UB									
9,216.0	0.00	0.00	9,216.0	0.0	0.0	0.0	0.00	0.00	0.00
Base Last Salt									
9,262.0	0.00	0.00	9,262.0	0.0	0.0	0.0	0.00	0.00	0.00
Ratcliffe									
9,440.0	0.00	0.00	9,440.0	0.0	0.0	0.0	0.00	0.00	0.00
Mission Cyn									
9,993.0	0.00	0.00	9,993.0	0.0	0.0	0.0	0.00	0.00	0.00
Lodgepole									
10,209.0	0.00	0.00	10,209.0	0.0	0.0	0.0	0.00	0.00	0.00
Lodgepole Fracture Zone									

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-18/17
Well: Ash Federal 5300 11-18T
Wellbore: Ash Federal 5300 11-18T
Design: Plan #1

Local Co-ordinate Reference: Well Ash Federal 5300 11-18T
TVD Reference: WELL @ 2078.0ft (Original Well Elev)
MD Reference: WELL @ 2078.0ft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,327.4	0.00	0.00	10,327.4	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 12.00									
10,750.0	50.71	90.29	10,697.0	-0.9	175.1	175.1	12.00	12.00	0.00
False Bakken									
10,764.5	52.45	90.29	10,706.0	-0.9	186.5	186.5	12.00	12.00	0.00
Upper Bakken									
10,788.3	55.30	90.29	10,720.0	-1.0	205.6	205.6	12.00	12.00	0.00
Middle Bakken									
10,883.3	66.70	90.29	10,766.0	-1.4	288.6	288.6	12.00	12.00	0.00
Lower Bakken									
10,939.0	73.39	90.29	10,785.0	-1.7	340.9	340.9	12.00	12.00	0.00
Three Forks									
11,075.1	89.72	90.29	10,804.9	-2.4	475.1	475.1	12.00	12.00	0.00
Start 14.9 hold at 11075.1 MD- KOP									
11,089.9	89.72	90.29	10,805.0	-2.5	489.9	489.9	0.02	0.02	0.00
Three Forks Target									
11,090.0	89.72	90.29	10,805.0	-2.5	490.0	490.0	0.00	0.00	0.00
Start 9508.1 hold at 11090.0 MD- EOC - 7"									
20,594.1	89.72	90.29	10,851.4	-50.0	9,993.9	9,994.0	0.00	0.00	0.00

Design Targets

Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (m)	Easting (m)	Latitude	Longitude
Interp @ 10805.0 (Ash F - plan hits target center - Point	0.00	0.00	10,805.0	-2.5	489.9	124,702.40	369,043.74	48° 4' 47.226 N	103° 36' 2.505 W
Ash Federal PBHL - plan hits target center - Point	0.00	0.00	10,851.4	-50.0	9,993.9	124,571.22	371,937.62	48° 4' 46.730 N	103° 33' 42.535 W

Casing Points

Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)
2,130.0	2,130.0	9 5/8"	9.625	13.500
11,090.0	10,805.0	7"	7.000	8.750

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-18/17
Well: Ash Federal 5300 11-18T
Wellbore: Ash Federal 5300 11-18T
Design: Plan #1

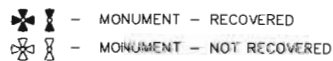
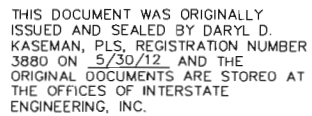
Local Co-ordinate Reference:
TVD Reference:
MD Reference:
North Reference:
Survey Calculation Method:

Well Ash Federal 5300 11-18T
 WELL @ 2078.0ft (Original Well Elev)
 WELL @ 2078.0ft (Original Well Elev)
 True
 Minimum Curvature

Formations						
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)	
2,028.0	2,028.0	Pierre				
4,615.0	4,615.0	Greenhorn				
5,025.0	5,025.0	Mowry				
5,429.0	5,429.0	Dakota				
6,449.0	6,449.0	Rierdon				
6,784.0	6,784.0	Dunham Salt				
6,901.0	6,901.0	Dunham Salt Base				
6,985.0	6,985.0	Spearfish				
7,250.0	7,250.0	Pine Salt				
7,297.0	7,297.0	Pine Salt Base				
7,341.0	7,341.0	Opeche Salt				
7,372.0	7,372.0	Opeche Salt Base				
7,576.0	7,576.0	Broom Creek (Top of Minnelusa Gp.)				
7,653.0	7,653.0	Amsden				
7,823.0	7,823.0	Tyler				
7,994.0	7,994.0	Otter (Base of Minnelusa Gp.)				
8,368.0	8,368.0	Kibby Lime				
8,516.0	8,516.0	Charles Salt				
9,140.0	9,140.0	UB				
9,216.0	9,216.0	Base Last Salt				
9,262.0	9,262.0	Ratcliffe				
9,440.0	9,440.0	Mission Cyn				
9,993.0	9,993.0	Lodgepole				
10,209.0	10,209.0	Lodgepole Fracture Zone				
10,750.0	10,697.0	False Bakken				
10,764.5	10,706.0	Upper Bakken				
10,788.3	10,720.0	Middle Bakken				
10,883.3	10,766.0	Lower Bakken				
10,939.0	10,785.0	Three Forks				
11,089.9	10,805.0	Three Forks Target				

Plan Annotations					
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates			
		+N/-S (ft)	+E/-W (ft)	Comment	
10,327.4	10,327.4	0.0	0.0	Start Build 12.00	
11,075.1	10,804.9	11.9	475.0	Start 14.9 hold at 11075.1 MD- KOP	
11,090.0	10,805.0	12.3	489.9	Start 9508.1 hold at 11090.0 MD- EOC	
20,598.1				TD at 20598.1	

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"
800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



STAKED ON 5/25/12
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
OF FABIAN KUORSTAD OF OASIS PETROLEUM. I CERTIFY THAT
THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY
SUPERVISION AND IS TRUE AND CORRECT TO
THE BEST OF MY KNOWLEDGE AND BELIEF.

RECEIVED LAND SURVEYOR
MARYL D. KASEMAN
LS-3880
DATE: 5/30/12
NORTH DAKOTA



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[illegible]

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 18, T13N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H.	Project No.: 012-09-146
Checked By: J.D.K.	Date: MAY 2012

Interstate Engineering, Inc.
P.O. Box 948
425 East Main Street
Sklody, Montana 59270
Ph (406) 433-5817
Fax (406) 433-5618
www.jengl.com

Other offices in Wisconsin, North Dakota and South Dakota



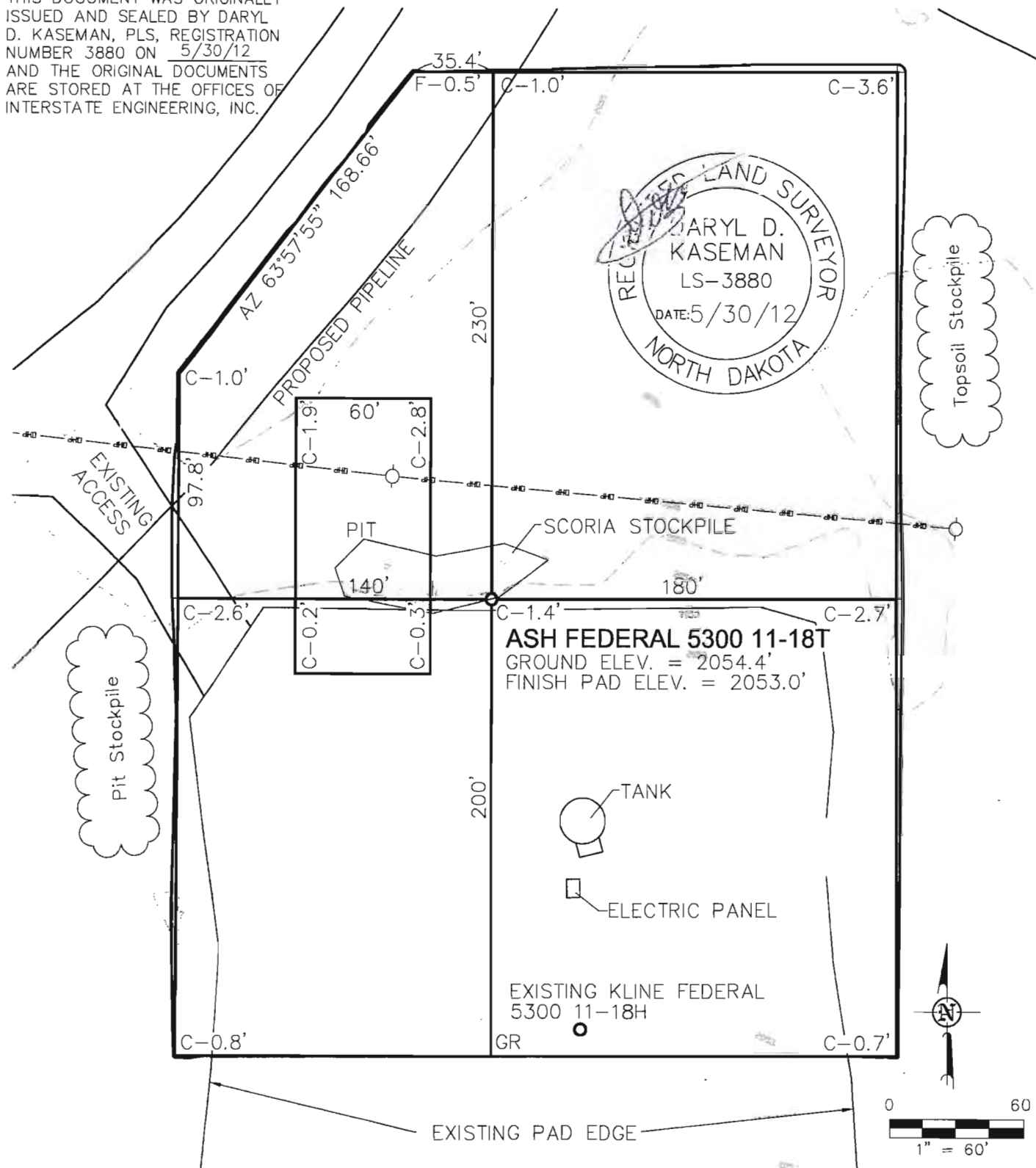
INTERSTATE
ENGINEERING

Professionals you need, people you trust

1/8

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"

THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL
D. KASEMAN, PLS, REGISTRATION
NUMBER 3880 ON 5/30/12
AND THE ORIGINAL DOCUMENTS
ARE STORED AT THE OFFICES OF
INTERSTATE ENGINEERING, INC.



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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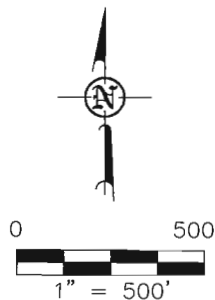
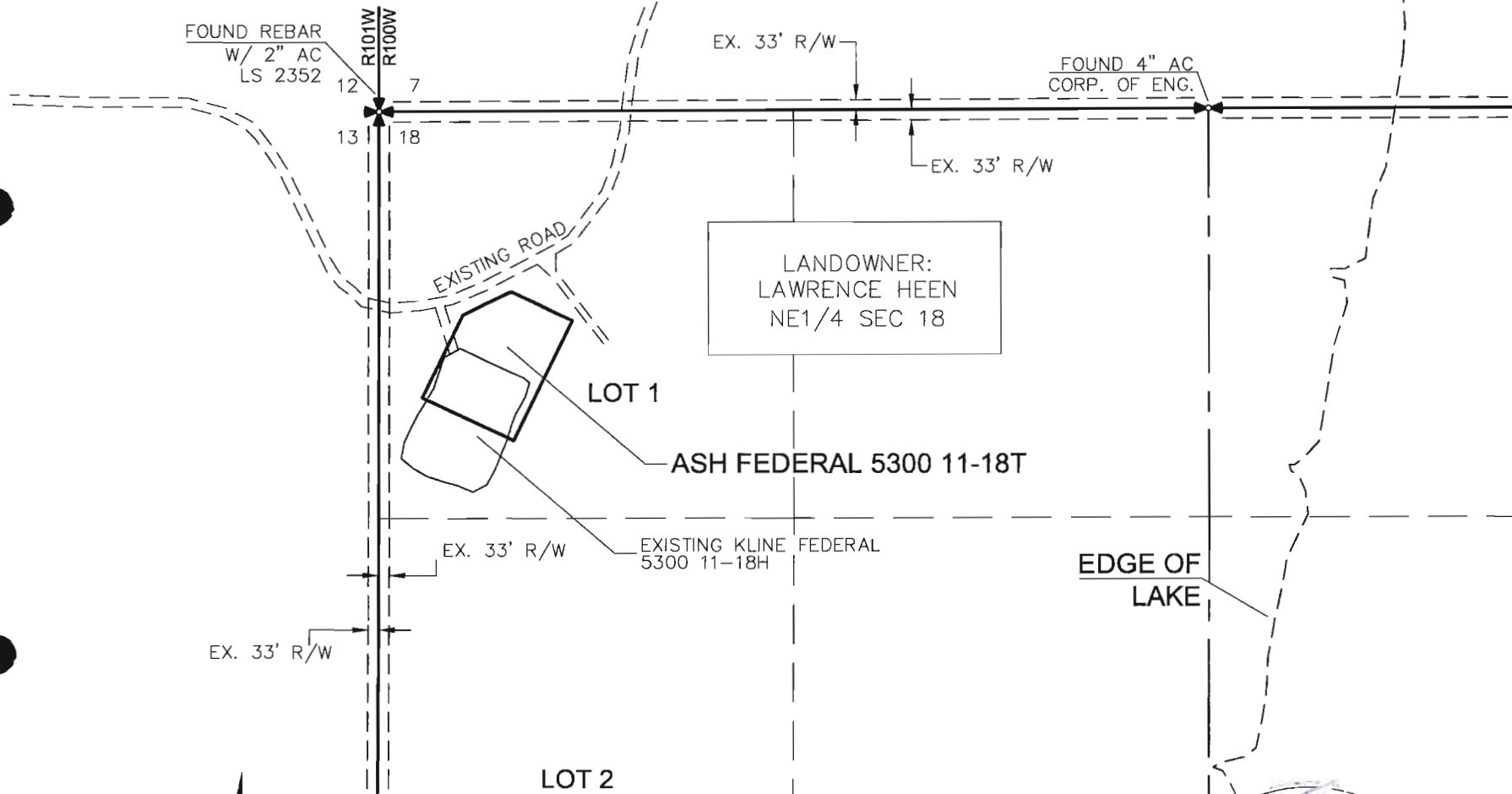
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengi.com

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>B.H.H.</u>	Project No.: <u>S12-09-145</u>
Checked By: <u>D.D.K.</u>	Date: <u>MAY 2012</u>

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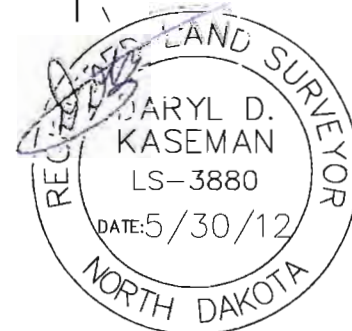
ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "ASH FEDERAL 5300 11-18T"
 800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



LOT 2

THIS DOCUMENT WAS ORIGINALLY
 ISSUED AND SEALED BY DARYL D.
 KASEMAN, PLS, REGISTRATION NUMBER
 3880 ON 5/30/12 AND THE
 ORIGINAL DOCUMENTS ARE STORED AT
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 ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete
 utilities location is recommended before construction.



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 SHEET NO.

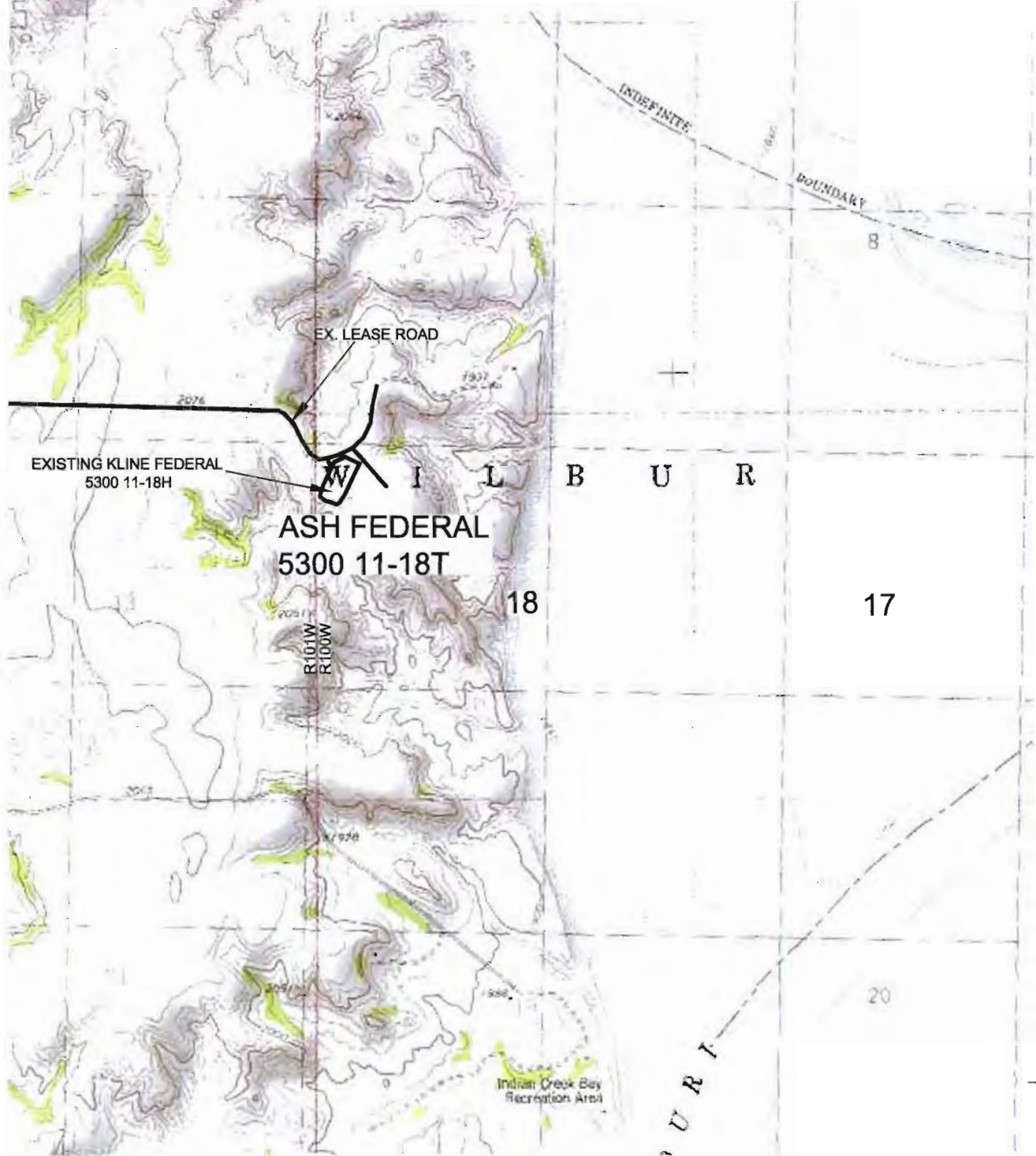


Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-3817
 Fax (406) 433-3818
 www.ieng.com
 Other offices in Louisiana, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 ACCESS APPROACH
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.H.
 Checked By: D.D.K.
 Project No.: 912-09-145
 Date: MAY 2012

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
 ASH FEDERAL 5300 11-18T
 800' FNL/350' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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5/8

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.iengi.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S12-09-145
 Checked By: D.D.K. Date: MAY 2012

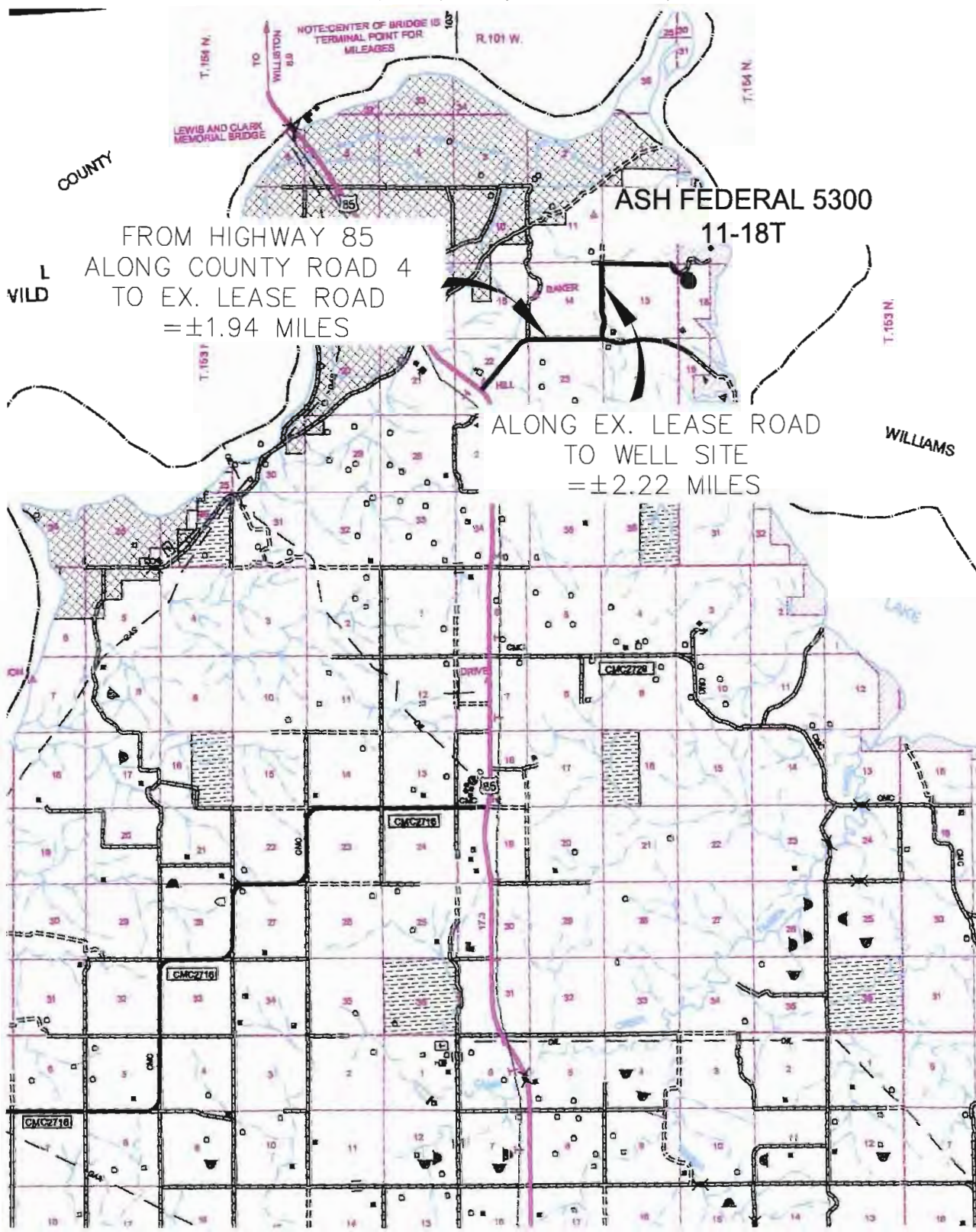
Revision No.	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

6/8

SHEET NO.



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
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Fax (406) 433-5618
www.iengl.com
Our services are provided in Montana, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S12-09-145
Checked By: D.D.K. Date: MAY 2012

Revision No.	Date	By	Description

CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

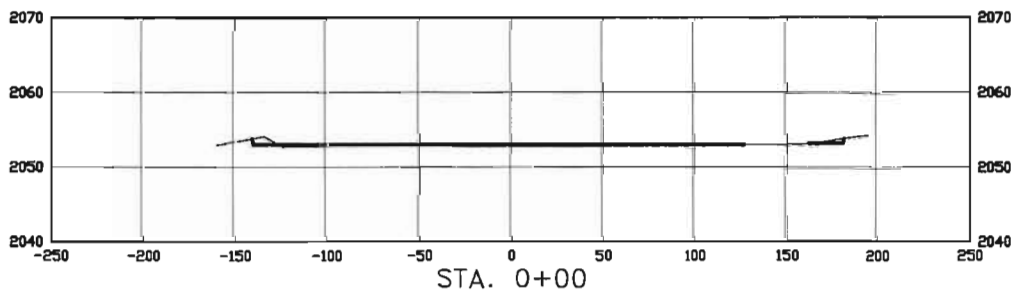
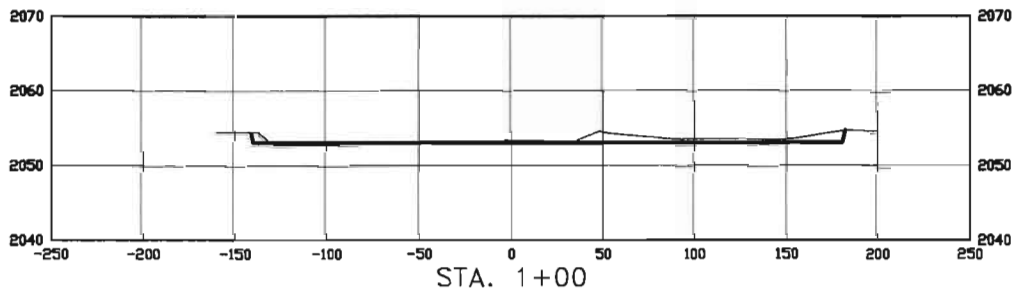
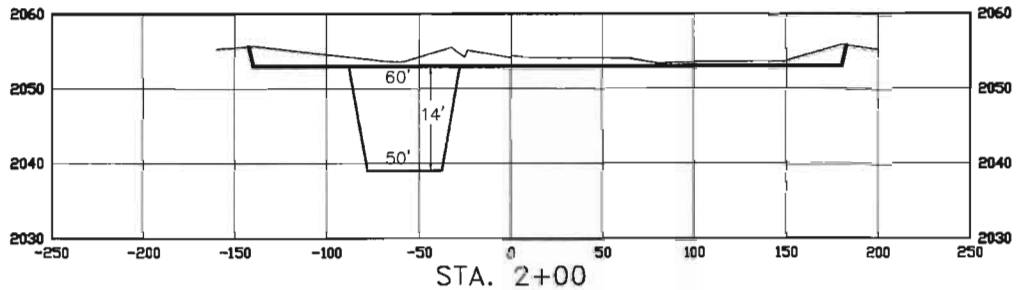
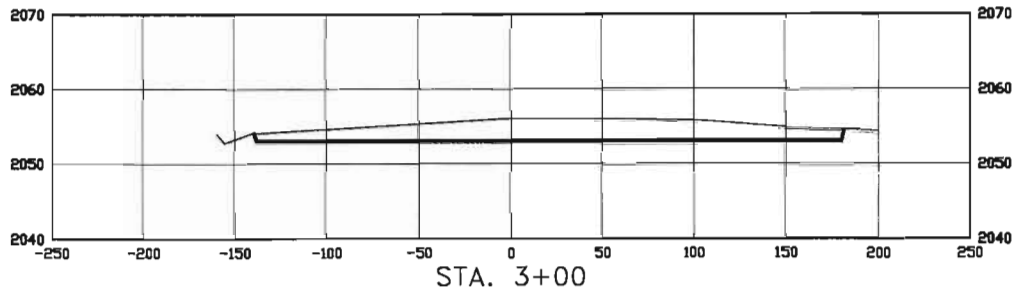
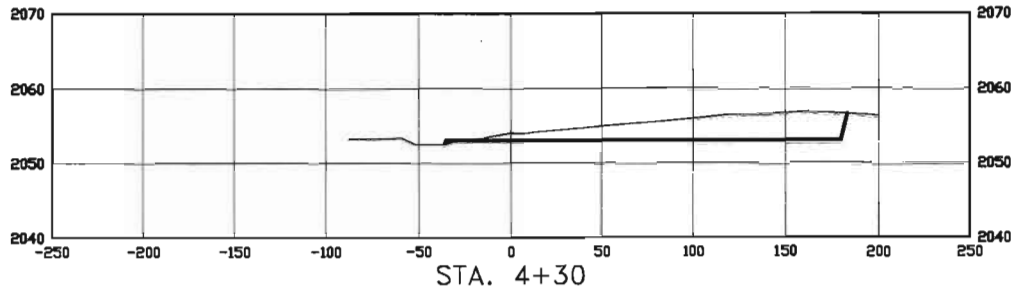
"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS
ORIGINALLY ISSUED AND
SEALED BY DARYL D. KASEMAN,
P.L.S., REGISTRATION NUMBER
3880 ON 5/30/12 AND THE
ORIGINAL DOCUMENTS ARE
STORED AT THE OFFICES OF
INTERSTATE ENGINEERING, INC.

SCALE
HORIZ 1"=100'
VERT 1"=25'



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SHEET NO.



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OASIS PETROLEUM NORTH AMERICA, LLC
PAD CROSS SECTIONS
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

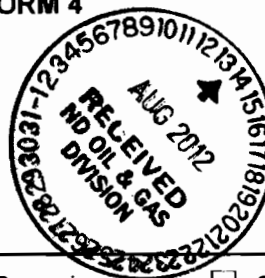
Drawn By: B.H.H. Project No.: 812-09-145
Checked By: D.D.K. Date: MAY 2012

Revision No.	Date	By	Description

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
23230

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date August 13, 2012
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to Rule 43-02-03-31

Well Name and Number Ash Federal 5300 11-18T					
Footages 800 F N L 350 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum/ Kllne 5300 11-18H located within a mile of the subject well.

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Brandi Terry		
Title Regulatory Specialist	Date August 13, 2012		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-13-2012	
By 	
Title Richard A. Suggs	Geologist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

ROBIN E. HESKETH
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN, SUITE 1500
HOUSTON, TX 77002 USA

Date: 7/9/2012

RE: **CORES AND SAMPLES**

Well Name: **ASH FEDERAL 5300 11-18T** Well File No.: **23230**
Location: **LOT1 18-153-100** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **THREE FORKS**

Dear ROBIN E. HESKETH:

North Dakota Century Code (NDCC) Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for the NDCC Section 38-08-04 and North Dakota Administrative Code 43-02-03-38.1.
- 2) Samples shall include all cuttings from:

Base of the Last Charles Salt

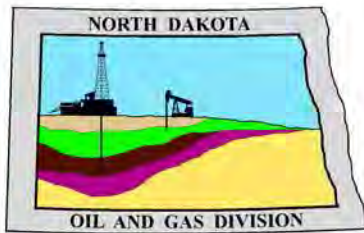
Samples of cuttings shall be taken at 30' maximum intervals through all vertical, build and horizontal sections. Samples must be washed, dried, packed in sample envelopes in correct order with labels showing operator, well name, location and depth, and forwarded in standard boxes to the State Geologist within 30 days of the completion of drilling operations.

- 3) Cores: ALL CORES cut shall be preserved in correct order, properly boxed, and forwarded to the State Geologist within 90 days of completion of drilling operations. Any extension of time must have written approval from the State Geologist.
- 4) All cores, core chips, and samples must be shipped, prepaid, to the State Geologist at the following address:

**ND Geological Survey Core Library
Campus Road and Cornell
Grand Forks, ND 58202**
- 5) NDCC Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Richard A. Suggs
Geologist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

June 28, 2012

Brandi Terry
Regulatory Specialist
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 202
Houston, TX 77002

**RE: HORIZONTAL WELL
ASH FEDERAL 5300 11-18T
LOT1 Section 18-153N-100W
McKenzie County
Well File # 23230**

Dear Brandi:

Pursuant to Commission Order No. 19840, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500' setback** from the north & south boundaries and **200' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 17 & 18-T153N-R100W. **Tool error is not required pursuant to order.**

PERMIT STIPULATIONS: OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. Based on the azimuth of the proposed lateral the maximum legal coordinates from the well head is: 300' N & 9994' E.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov. Survey points shall be of such frequency to accurately determine the entire location of the well bore. Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of a Cement Bond Log from which the presence of cement can be determined in every well in which production or intermediate casing has been set and a Gamma Ray Log must be run from total depth to ground level elevation of the well bore. All logs must be submitted as one paper copy and one digital copy in LAS (Log ASCII) format, or a format approved by the Director. Image logs that include, but are not limited to, Mud Logs, Cement Bond Logs, and Cyberlook Logs, cannot be produced in their entirety as LAS (Log ASCII) files. To create a solution and establish a standard format for industry to follow when submitting image logs, the Director has given approval for the operator to submit an image log as a TIFF (*.tif) formatted file. The TIFF (*.tif) format will be accepted only when the log cannot be produced in its entirety as a LAS (Log ASCII) file format. The digital copy may be submitted on a 3.5" floppy diskette, a standard CD, or attached to an email sent to digitallogs@nd.gov. Thank you for your cooperation.

Sincerely,

David Tabor
Engineering Technician IV



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 06 / 15 / 2012	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9491
Address 1001 Fannin Suite 202		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name ASH FEDERAL				Well Number 5300 11-18T			
Surface Footages 800 F N L 350 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 788 F N L 840 F W L		Qtr-Qtr LOT1	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 12 N From WH 490 E From WH		Azimuth 88.6 °	Longstring Total Depth 11090 Feet MD 10805 Feet TVD				
Bottom Hole Footages From Nearest Section Line 550 F N L 200 F E L		Qtr-Qtr NENE	Section 17	Township 153 N	Range 100 W	County McKenzie	
Bottom Hole Coordinates From Well Head 250 N From WH 9994 E From WH		KOP Lateral 1 10328 Feet MD	Azimuth Lateral 1 89.7 °		Estimated Total Depth Lateral 1 20597 Feet MD 10851 Feet TVD		
Latitude of Well Head 48 ° 04 ' 47.25 "		Longitude of Well Head -103 ° 36 ' 09.72 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) SPACING UNIT: Sections 17 & 18-T153N-R100W		
Ground Elevation 2054 Feet Above S.L.	Acres in Spacing/Drilling Unit 1280	Spacing/Drilling Unit Setback Requirement 500 Feet N/S 200 Feet E/W			Industrial Commission Order 19840		
North Line of Spacing/Drilling Unit 10544 Feet	South Line of Spacing/Drilling Unit 10489 Feet	East Line of Spacing/Drilling Unit 5244 Feet			West Line of Spacing/Drilling Unit 5256 Feet		
Objective Horizons Three Forks						Pierre Shale Top 2028	
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2130 Feet	Cement Volume 638 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11090 Feet MD 10805 Feet TVD		Cement Volume 912 Sacks	Cement Top 3929 Feet	Top Dakota Sand 5429 Feet
Base Last Charles Salt (If Applicable) 9216 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo:KOP-KibbyGR/Res-BSCGR-SurfCND thru DakotaCBL/GR-TOCGR-Surface							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH				
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH				
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W	County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.

Date
06 / 07 / 2012

ePermit

Printed Name
Brandi Terry

Title
Regulatory Specialist

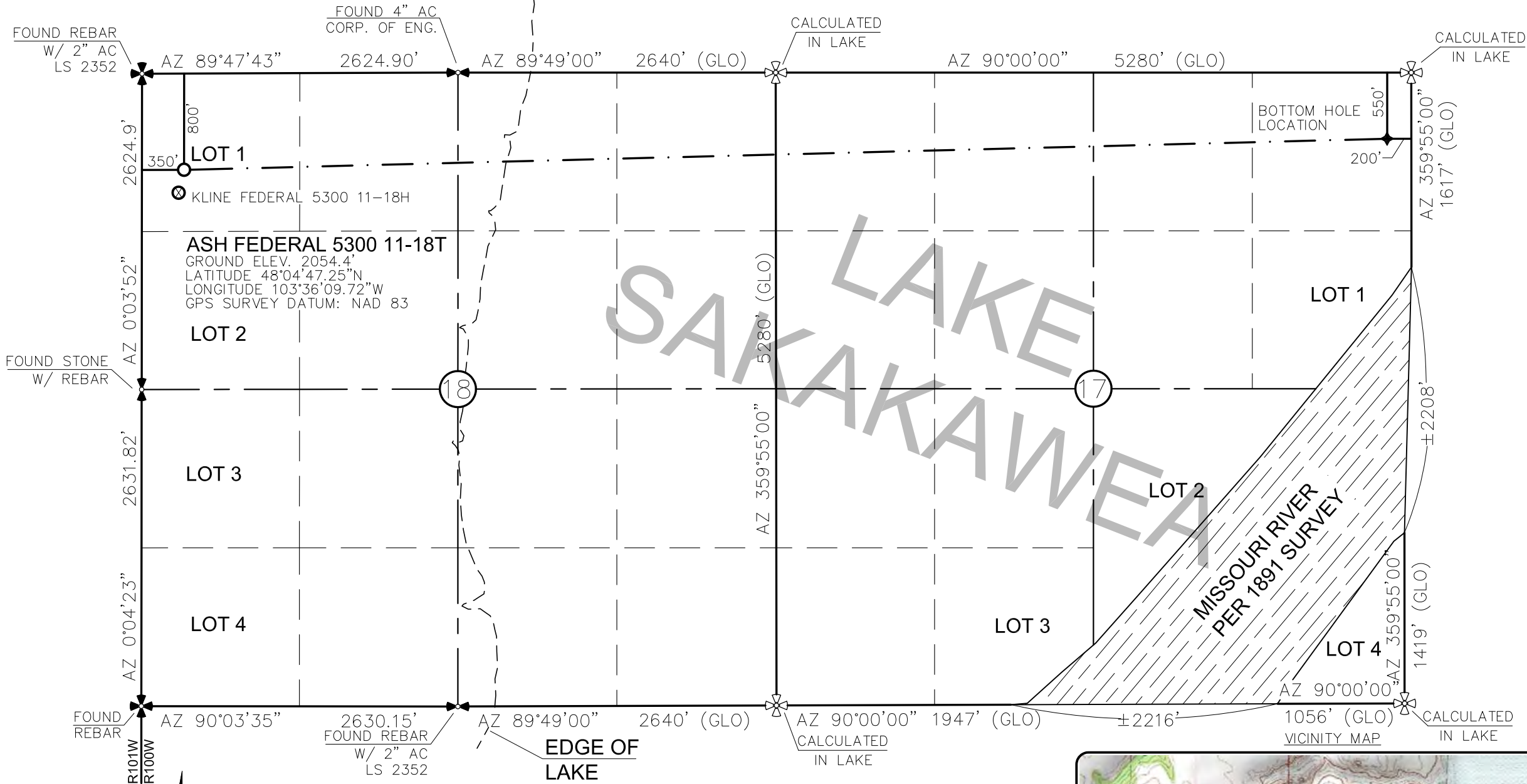
FOR STATE USE ONLY

Permit and File Number 23230	API Number 33 - 053 - 04211
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

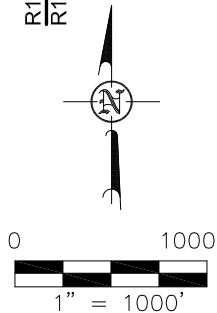
FOR STATE USE ONLY

Date Approved 6 / 28 / 2012
By David Tabor
Title Engineering Technician IV

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"
800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS, REGISTRATION NUMBER
3880 ON 5/30/12 AND THE
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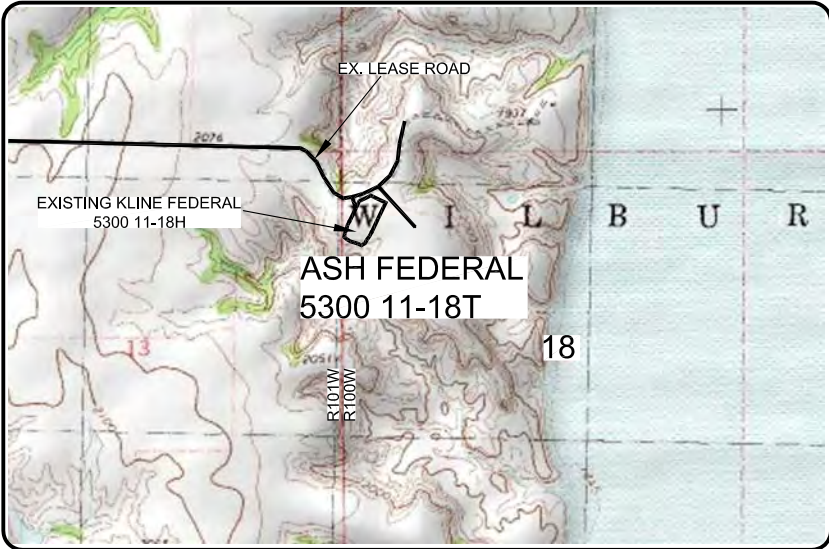
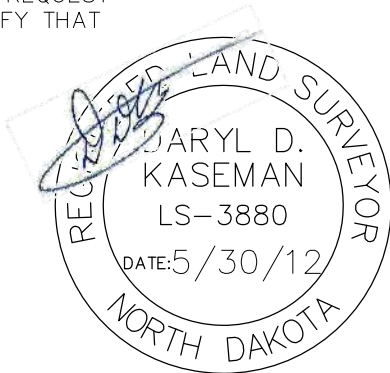
- ⊕ — MONUMENT — RECOVERED
⊗ — MONUMENT — NOT RECOVERED

STAKED ON 5/25/12
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
OF FABIAN KJORSTAD OF OASIS PETROLEUM. I CERTIFY THAT
THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY
SUPERVISION AND IS TRUE AND CORRECT TO
THE BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]

DARYL D. KASEMAN LS-3880

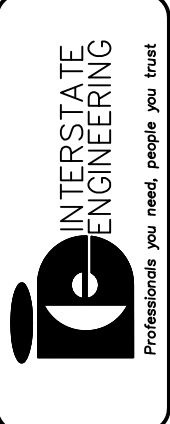


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Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC WELL LOCATION PLAT SECTION 18, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: S12-09-145 Date: MAY 2012
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Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.iengr.com
Other offices in Minnesota, North Dakota and South Dakota



**Oasis Petroleum
Well Summary
Ash 5300 11-18T
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 2,130'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,130'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.02	3520 / 3.53	453 / 2.72	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.0 ppg fluid on backside (2150' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2150' setting depth).
- c) Based on string weight in 9.0 ppg fluid at 2125' TVD plus 100k# overpull. (Buoyed weight equals 66k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 55% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **413 sks** (234 bbls) Conventional system with 94 lb/sk cement, 4% extender, 2% expanding agent, 2% CaCl₂ and 0.25 lb/sk lost circulation control agent

Tail Slurry: **225 sks** (60 bbls) Conventional system with 94 lb/sk cement, 3% NaCl, and .25 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Ash 5300 11-18T
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,600'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,600' – 10,328'	32	HCP-110	LTC	6.094"	6.000"	6,730	8,970	9,870
7"	10,328' – 11,090'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

**Special Drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,600'	6,600'	7", 29#, P-110, LTC, 8rd	8530 / 2.41*	11220 / 1.19	797 / 2.08
6,600' – 10,328'	3,728'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.20*	12460 / 1.29	
6,600' – 10,328'	3,728'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.06**	12460 / 1.29	
10,328' – 11,090'	762'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.51*	11220 / 1.15	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9,000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9.0 ppg fluid on backside—to 10,805' TVD.
- c. Based on string weight in 10 ppg fluid, (281k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess. TOC targeted to 1,500' above Dakota (3,929').

Pre-flush (Spacer): **125 bbls** Saltwater
20 sks Pozmix A
10 bbls Fresh Water

Lead Slurry: **246 sks** (95 bbls) Conventional system with 47 lb/sk cement, 10% NaCl, 34 lb/sk extender, 10% D020 extender, 1% D079 extender, 1% anti-settling agent, 1% fluid loss agent, 0.2% anti-foam agent, 0.6% retarder, and 0.4% dispersant

Tail Slurry: **666 sks** (196 bbls) Conventional system with 94 lb/sk cement, 10% NaCl, 35% Silica, 0.2% fluid loss agent, 0.8% dispersant and 0.3% retarder

**Oasis Petroleum
Well Summary
Ash 5300 11-18T
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

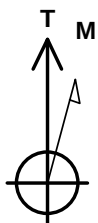
							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	9,976' to 20,448'	11.6	P-110	BTC	4.000"	3.875"	2,270	3,020	3,780

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
9,976' to 20,448'	4-1/2", 11.6 lb, P-110, BTC, 8rd	7560 / 1.41	10690 / 1.10	385 / 1.91	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,851' TVD.
- b) Burst pressure based on 9,000 psi treating pressure with 10.2 ppg internal fluid gradient and 9.0 ppg external fluid gradient @ 10,851' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 101k lbs.) plus 100k lbs overpull.

DRILLING PLAN									
OPERATOR	Oasis Petroleum					COUNTY/STATE	McKenzie Co., ND		
WELL NAME	5300 11-18T					RIG	Nabors B-22		
WELL TYPE	Horizontal Upper Three Forks								
LOCATION	NW/NW 18-153N-100W					Surface Location (survey plat): 800' fml 350' fml			
EST. T.D.	20,598'					GROUND ELEV:	2053 Finished Pad Elev.		Sub Height: 25
TOTAL LATERA 9,508'					KB ELEV: 2078				
PROGNOSIS: Based on 2,078' KB(est)					LOGS: Type Interval				
OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota									
CBL/GR: Above top of cement/GR to base of casing									
MWD GR: KOP to lateral TD									
MARKER DEPTH (Surf Loc) DATUM (Surf Loc)					DEVIATION:				
Pierre	NDIC MAP	2,028	50'		Surf:	3 deg. max., 1 deg / 100'; srvy every 500'			
Greenhorn		4,615	-2,537'		Prod:	5 deg. max., 1 deg / 100'; srvy every 100'			
Mowry		5,025	-2,947'						
Dakota		5,429	-3,351'						
Rierdon		6,449	-4,371'						
Dunham Salt		6,784	-4,706'						
Dunham Salt Base		6,901	-4,823'						
Spearfish		6,985	-4,907'						
Pine Salt		7,250	-5,172'						
Pine Salt Base		7,297	-5,219'						
Opeche Salt		7,341	-5,263'						
Opeche Salt Base		7,372	-5,294'						
Broom Creek (Top of Minnelusa Gp.)		7,576	-5,498'						
Amsden		7,653	-5,575'						
Tyler		7,823	-5,745'						
Otter (Base of Minnelusa Gp.)		7,994	-5,916'						
Kibby Lime		8,368	-6,290'						
Charles Salt		8,516	-6,438'						
UB		9,140	-7,062'						
Base Last Salt		9,216	-7,138'						
Ratcliffe		9,262	-7,184'						
Mission Cyn		9,440	-7,362'						
Lodgepole		9,993	-7,915'						
Lodgepole Fracture Zone		10,209	-8,131'						
False Bakken		10,697	-8,619'						
Upper Bakken		10,706	-8,628'						
Middle Bakken		10,720	-8,642'						
Lower Bakken		10,766	-8,688'						
Three Forks		10,785	-8,707'						
Three Forks Target		10,805	-8,727'						
Base Target/Claystone		10,817	-8,739'						
Dip Rate: -0.28° or 0.50' drop in TVD per 100' drilled									
Max. Anticipated BHP: 4687					Surface Formation: Glacial till				
MUD: Interval Type					WT	Vis	WL	Remarks	
Surface: 0' - 2,130' FW/Gel - Lime Sweeps					8.4-9.0	28-32	NC	Circ Mud Tanks	
Intermediate: 2,130' - 11,090' Invert					9.5-10.4	40-50	30+HtHp	Circ Mud Tanks	
Lateral: 11,090' - 20,598' Salt Water					9.8-10.2	28-32	NC	Circ Mud Tanks	
CASING: Size Wt ppf Hole Depth					Cement	WOC	Remarks		
Surface: 9-5/8" 36# 13-1/2" 2,130'					To Surface	12	100' into Pierre		
Intermediate: 7" 29/32# 8-3/4" 11,090'					3,929'	24	1500' above Dakota		
Production Liner: 4.5" 11.6# 6" 20,598'					TOL @ 10,278'		50' above KOP		
PROBABLE PLUGS, IF REQ'D:									
OTHER: MD TVD FNL/FSL FEL/FWL S-T-R AZI									
Surface: 2,130 2,130 800' FNL 350' FWL Sec 18 T153N-R100W Survey Company:									
KOP: 10,328' 10,328' 800' FNL 350' FWL Sec 18 T153N-R100W Build Rate: 12 deg /100'									
EOC: 11,075' 10,805' 788' FNL 825' FWL Sec 18 T153N-R100W 88.6									
Casing Point: 11,090' 10,805' 788' FNL 840' FWL Sec 18 T153N-R100W 88.6									
Middle Bakken Lateral TD: 20,598' 10,851' 550' FNL 200' FEL Sec 17 T153N-R100W 88.6									
Comments:									
Request Sundry for Open Hole Log Waiver									
Exception well: Kline 5300 11-18H (33-053-03426); 153N/100W/S18; located 190° S and 45° W of Ash Surface location									
36 Stage Plug and Perf									
A frac string will be utilized on this well									
OASIS PETROLEUM									
Geology: B. Gates 5/31/2012					Engineering: M. Brown 6/6/2012				



Azimuths to True North
Magnetic North: 8.47°

Magnetic Field
Strength: 56673.3snT
Dip Angle: 73.07°
Date: 6/6/2012
Model: IGRF200510

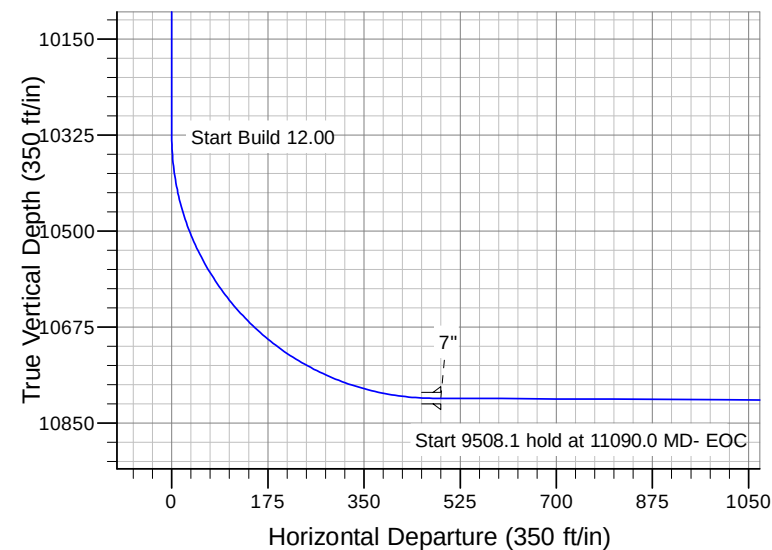
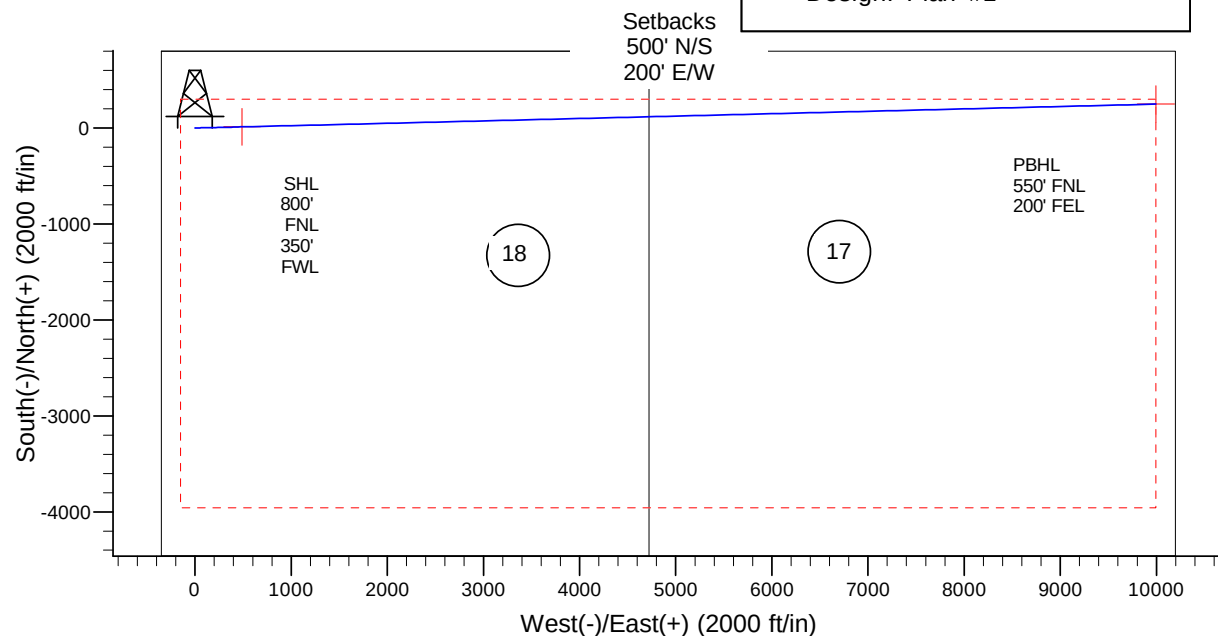


Project: Indian Hills
Site: 153N-100W-18/17
Well: Ash Federal 5300 11-18T
Wellbore: OH
Design: Plan #1

SITE DETAILS: 153N-100W-18/17

Site Centre Latitude: 48° 4' 47.250 N
Longitude: 103° 36' 9.720 W

Positional Uncertainty: 0.0
Convergence: -2.31
Local North: True

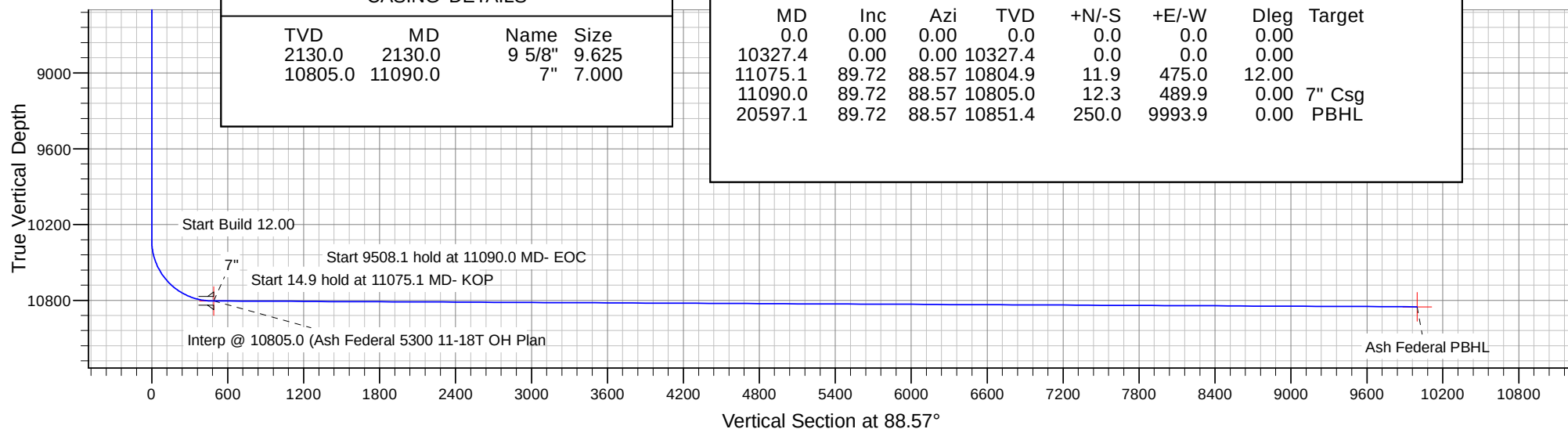


CASING DETAILS

TVD	MD	Name	Size
2130.0	2130.0	9 5/8"	9.625
10805.0	11090.0	7"	7.000

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
10327.4	0.00	0.00	10327.4	0.0	0.0	0.00	
11075.1	89.72	88.57	10804.9	11.9	475.0	12.00	
11090.0	89.72	88.57	10805.0	12.3	489.9	0.00	7" Csg
20597.1	89.72	88.57	10851.4	250.0	9993.9	0.00	PBHL



Oasis

Indian Hills

153N-100W-18/17

Ash Federal 5300 11-18T

OH

Plan: Plan #1

Standard Planning Report

28 June, 2012

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Ash Federal 5300 11-18T
Company:	Oasis	TVD Reference:	WELL @ 2078.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0ft (Original Well Elev)
Site:	153N-100W-18/17	North Reference:	True
Well:	Ash Federal 5300 11-18T	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-18/17			
Site Position:		Northing:	124,709.16 m	Latitude: 48° 4' 47.250 N
From:	Lat/Long	Easting:	368,894.57 m	Longitude: 103° 36' 9.720 W
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence: -2.31 °

Well	Ash Federal 5300 11-18T			
Well Position	+N/-S	0.0 ft	Northing:	124,709.16 m
	+E/-W	0.0 ft	Easting:	368,894.57 m
Position Uncertainty		0.0 ft	Wellhead Elevation:	Ground Level: 2,053.0 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	6/6/2012	8.47	73.07	56,673

Design	Plan #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)
	0.0	0.0	0.0	88.57

Plan Sections										
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
10,327.4	0.00	0.00	10,327.4	0.0	0.0	0.00	0.00	0.00	0.00	
11,075.1	89.72	88.57	10,804.9	11.9	475.0	12.00	12.00	0.00	88.57	
11,090.0	89.72	88.57	10,805.0	12.3	489.9	0.00	0.00	0.00	0.00	Interp @ 10805.0 (As
20,597.1	89.72	88.57	10,851.4	250.0	9,993.9	0.00	0.00	0.00	0.00	Ash Federal PBHL

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Ash Federal 5300 11-18T
Company:	Oasis	TVD Reference:	WELL @ 2078.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0ft (Original Well Elev)
Site:	153N-100W-18/17	North Reference:	True
Well:	Ash Federal 5300 11-18T	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
2,028.0	0.00	0.00	2,028.0	0.0	0.0	0.0	0.00	0.00	0.00
Pierre									
2,130.0	0.00	0.00	2,130.0	0.0	0.0	0.0	0.00	0.00	0.00
9 5/8"									
4,615.0	0.00	0.00	4,615.0	0.0	0.0	0.0	0.00	0.00	0.00
Greenhorn									
5,025.0	0.00	0.00	5,025.0	0.0	0.0	0.0	0.00	0.00	0.00
Mowry									
5,429.0	0.00	0.00	5,429.0	0.0	0.0	0.0	0.00	0.00	0.00
Dakota									
6,449.0	0.00	0.00	6,449.0	0.0	0.0	0.0	0.00	0.00	0.00
Rierdon									
6,784.0	0.00	0.00	6,784.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt									
6,901.0	0.00	0.00	6,901.0	0.0	0.0	0.0	0.00	0.00	0.00
Dunham Salt Base									
6,985.0	0.00	0.00	6,985.0	0.0	0.0	0.0	0.00	0.00	0.00
Spearfish									
7,250.0	0.00	0.00	7,250.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt									
7,297.0	0.00	0.00	7,297.0	0.0	0.0	0.0	0.00	0.00	0.00
Pine Salt Base									
7,341.0	0.00	0.00	7,341.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt									
7,372.0	0.00	0.00	7,372.0	0.0	0.0	0.0	0.00	0.00	0.00
Opeche Salt Base									
7,576.0	0.00	0.00	7,576.0	0.0	0.0	0.0	0.00	0.00	0.00
Broom Creek (Top of Minnelusa Gp.)									
7,653.0	0.00	0.00	7,653.0	0.0	0.0	0.0	0.00	0.00	0.00
Amsden									
7,823.0	0.00	0.00	7,823.0	0.0	0.0	0.0	0.00	0.00	0.00
Tyler									
7,994.0	0.00	0.00	7,994.0	0.0	0.0	0.0	0.00	0.00	0.00
Otter (Base of Minnelusa Gp.)									
8,368.0	0.00	0.00	8,368.0	0.0	0.0	0.0	0.00	0.00	0.00
Kibby Lime									
8,516.0	0.00	0.00	8,516.0	0.0	0.0	0.0	0.00	0.00	0.00
Charles Salt									
9,140.0	0.00	0.00	9,140.0	0.0	0.0	0.0	0.00	0.00	0.00
UB									
9,216.0	0.00	0.00	9,216.0	0.0	0.0	0.0	0.00	0.00	0.00
Base Last Salt									
9,262.0	0.00	0.00	9,262.0	0.0	0.0	0.0	0.00	0.00	0.00
Ratcliffe									
9,440.0	0.00	0.00	9,440.0	0.0	0.0	0.0	0.00	0.00	0.00
Mission Cyn									
9,993.0	0.00	0.00	9,993.0	0.0	0.0	0.0	0.00	0.00	0.00
Lodgepole									
10,209.0	0.00	0.00	10,209.0	0.0	0.0	0.0	0.00	0.00	0.00
Lodgepole Fracture Zone									

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Ash Federal 5300 11-18T
Company:	Oasis	TVD Reference:	WELL @ 2078.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0ft (Original Well Elev)
Site:	153N-100W-18/17	North Reference:	True
Well:	Ash Federal 5300 11-18T	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Planned Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
10,327.4	0.00	0.00	10,327.4	0.0	0.0	0.0	0.00	0.00	0.00
Start Build 12.00									
10,750.1	50.71	88.57	10,697.0	4.4	175.1	175.1	12.00	12.00	0.00
False Bakken									
10,764.5	52.45	88.57	10,706.0	4.7	186.4	186.5	12.00	12.00	0.00
Upper Bakken									
10,788.3	55.30	88.57	10,720.0	5.1	205.6	205.7	12.00	12.00	0.00
Middle Bakken									
10,883.4	66.71	88.57	10,766.0	7.2	288.6	288.7	12.00	12.00	0.00
Lower Bakken									
10,939.1	73.40	88.57	10,785.0	8.5	340.9	341.0	12.00	12.00	0.00
Three Forks									
11,075.1	89.72	88.57	10,804.9	11.9	475.0	475.1	12.00	12.00	0.00
Start 14.9 hold at 11075.1 MD- KOP									
11,090.0	89.72	88.57	10,805.0	12.3	489.9	490.0	0.00	0.00	0.00
Start 9508.1 hold at 11090.0 MD- EOC - 7"									
11,094.9	89.72	88.57	10,805.0	12.4	494.8	494.9	0.00	0.00	0.00
Three Forks Target									
20,597.1	89.72	88.57	10,851.4	250.0	9,993.9	9,997.0	0.00	0.00	0.00

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (ft)	+N/-S (ft)	+E/-W (ft)	Northing (m)	Easting (m)	Latitude	Longitude
- hit/miss target									
- Shape									
Interp @ 10805.0 (Ash F	0.00	0.00	10,805.0	12.3	489.9	124,706.88	369,043.91	48° 4' 47.371 N	103° 36' 2.505 W
- plan hits target center									
- Point									
Ash Federal PBHL	0.00	0.00	10,851.4	250.0	9,993.9	124,662.58	371,941.30	48° 4' 49.691 N	103° 33' 42.533 W
- plan hits target center									
- Point									

Casing Points					
Measured Depth (ft)	Vertical Depth (ft)	Name		Casing Diameter (in)	Hole Diameter (in)
2,130.0	2,130.0	9 5/8"		9.625	13.500
11,090.0	10,805.0	7"		7.000	8.750

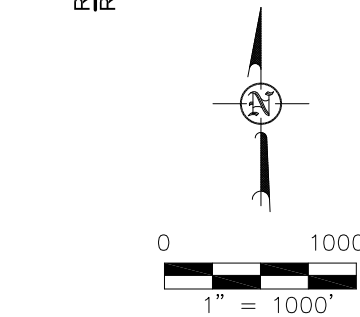
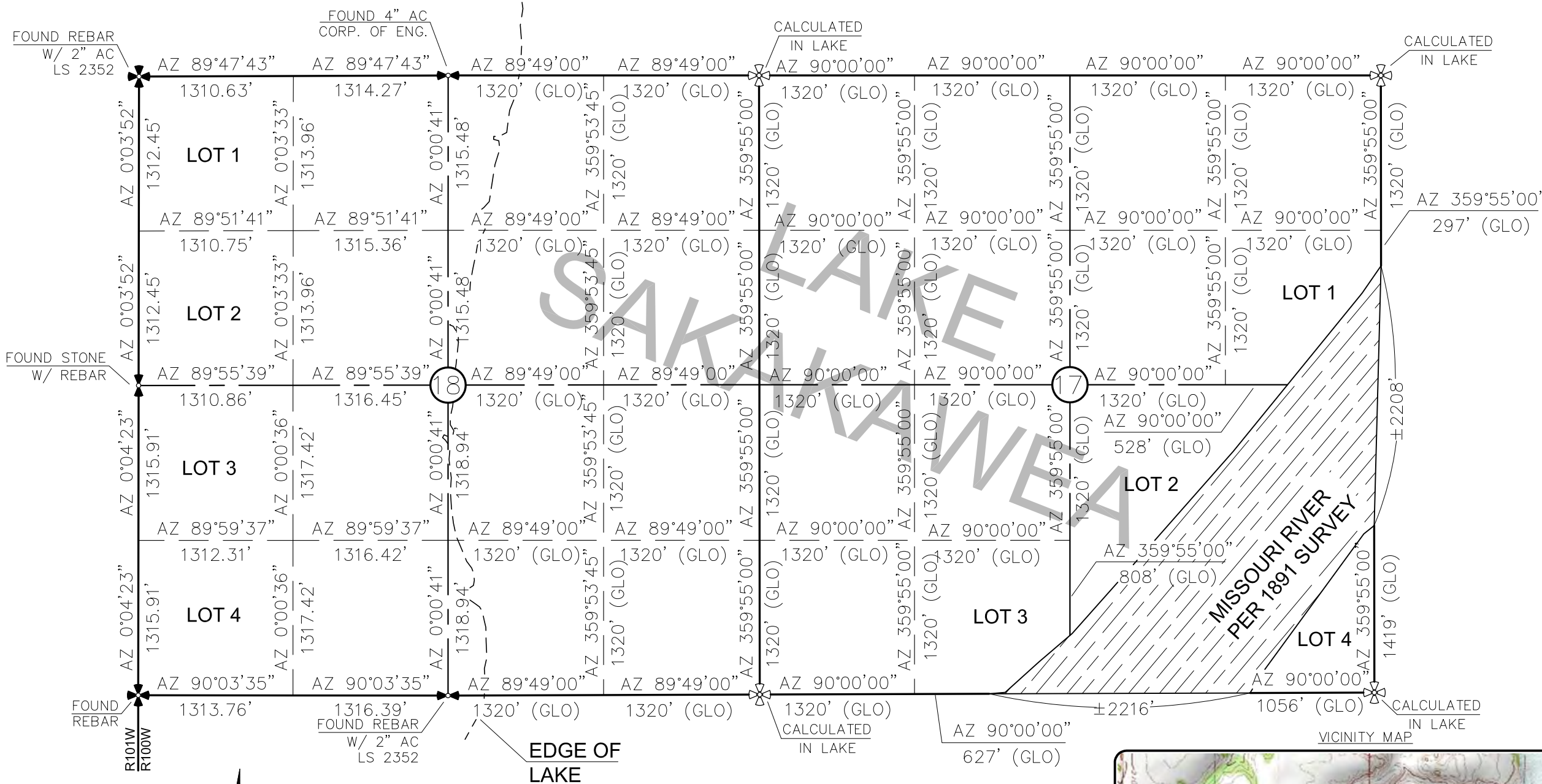
Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Ash Federal 5300 11-18T
Company:	Oasis	TVD Reference:	WELL @ 2078.0ft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2078.0ft (Original Well Elev)
Site:	153N-100W-18/17	North Reference:	True
Well:	Ash Federal 5300 11-18T	Survey Calculation Method:	Minimum Curvature
Wellbore:	OH		
Design:	Plan #1		

Formations						
Measured Depth (ft)	Vertical Depth (ft)	Name	Lithology	Dip (°)	Dip Direction (°)	
2,028.0	-50.0	Pierre				
4,615.0	2,537.0	Greenhorn				
5,025.0	2,947.0	Mowry				
5,429.0	3,351.0	Dakota				
6,449.0	4,371.0	Rierdon				
6,784.0	4,706.0	Dunham Salt				
6,901.0	4,823.0	Dunham Salt Base				
6,985.0	4,907.0	Spearfish				
7,250.0	5,172.0	Pine Salt				
7,297.0	5,219.0	Pine Salt Base				
7,341.0	5,263.0	Opeche Salt				
7,372.0	5,294.0	Opeche Salt Base				
7,576.0	5,498.0	Broom Creek (Top of Minnelusa Gp.)				
7,653.0	5,575.0	Amsden				
7,823.0	5,745.0	Tyler				
7,994.0	5,916.0	Otter (Base of Minnelusa Gp.)				
8,368.0	6,290.0	Kibby Lime				
8,516.0	6,438.0	Charles Salt				
9,140.0	7,062.0	UB				
9,216.0	7,138.0	Base Last Salt				
9,262.0	7,184.0	Ratcliffe				
9,440.0	7,362.0	Mission Cyn				
9,993.0	7,915.0	Lodgepole				
10,209.0	8,131.0	Lodgepole Fracture Zone				
10,750.1	8,619.0	False Bakken				
10,764.5	8,628.0	Upper Bakken				
10,788.3	8,642.0	Middle Bakken				
10,883.4	8,688.0	Lower Bakken				
10,939.1	8,707.0	Three Forks				
11,094.9	8,727.0	Three Forks Target				

Plan Annotations					
Measured Depth (ft)	Vertical Depth (ft)	Local Coordinates			
		+N/-S (ft)	+E/-W (ft)	Comment	
10,327.4	10,327.4	0.0	0.0	Start Build 12.00	
11,075.1	10,804.9	11.9	475.0	Start 14.9 hold at 11075.1 MD- KOP	
11,090.0	10,805.0	12.3	489.9	Start 9508.1 hold at 11090.0 MD- EOC	
20,598.1				TD at 20598.1	

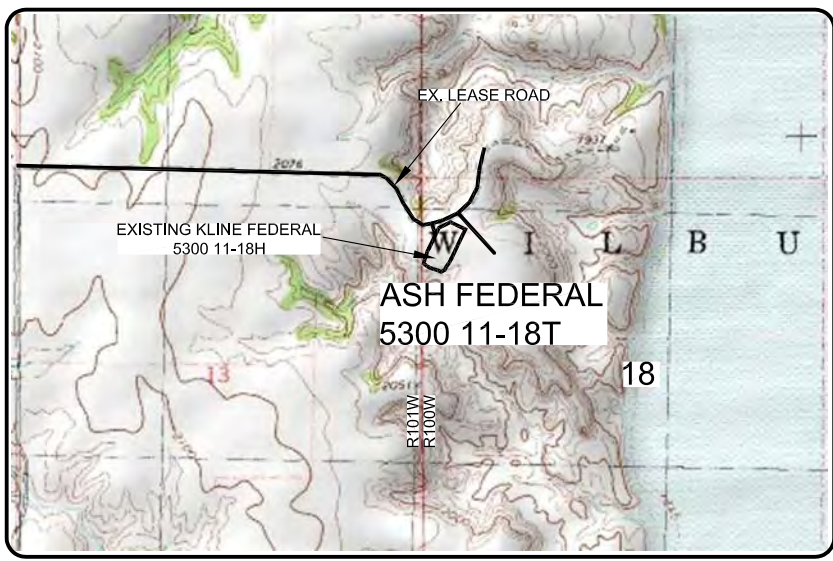
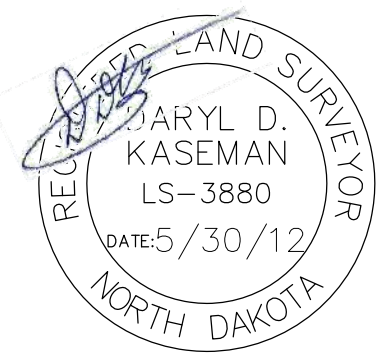
SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"ASH FEDERAL 5300 11-18T"
800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTIONS 17 & 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
- MONUMENT - NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS, REGISTRATION NUMBER
3880 ON 5/30/12 AND THE
ORIGINAL DOCUMENTS ARE STORED AT
THE OFFICES OF INTERSTATE
ENGINEERING, INC.

ALL AZIMUTHS ARE BASED ON G.P.S.
OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
WAS 1891. THE CORNERS FOUND ARE AS
INDICATED AND ALL OTHERS ARE COMPUTED FROM
THOSE CORNERS FOUND AND BASED ON G.L.O.
DATA. THE MAPPING ANGLE FOR THIS AREA IS
APPROXIMATELY 0°03'.



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Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTIONS 17 & 18, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: S12-09-145 Date: MAY 2012
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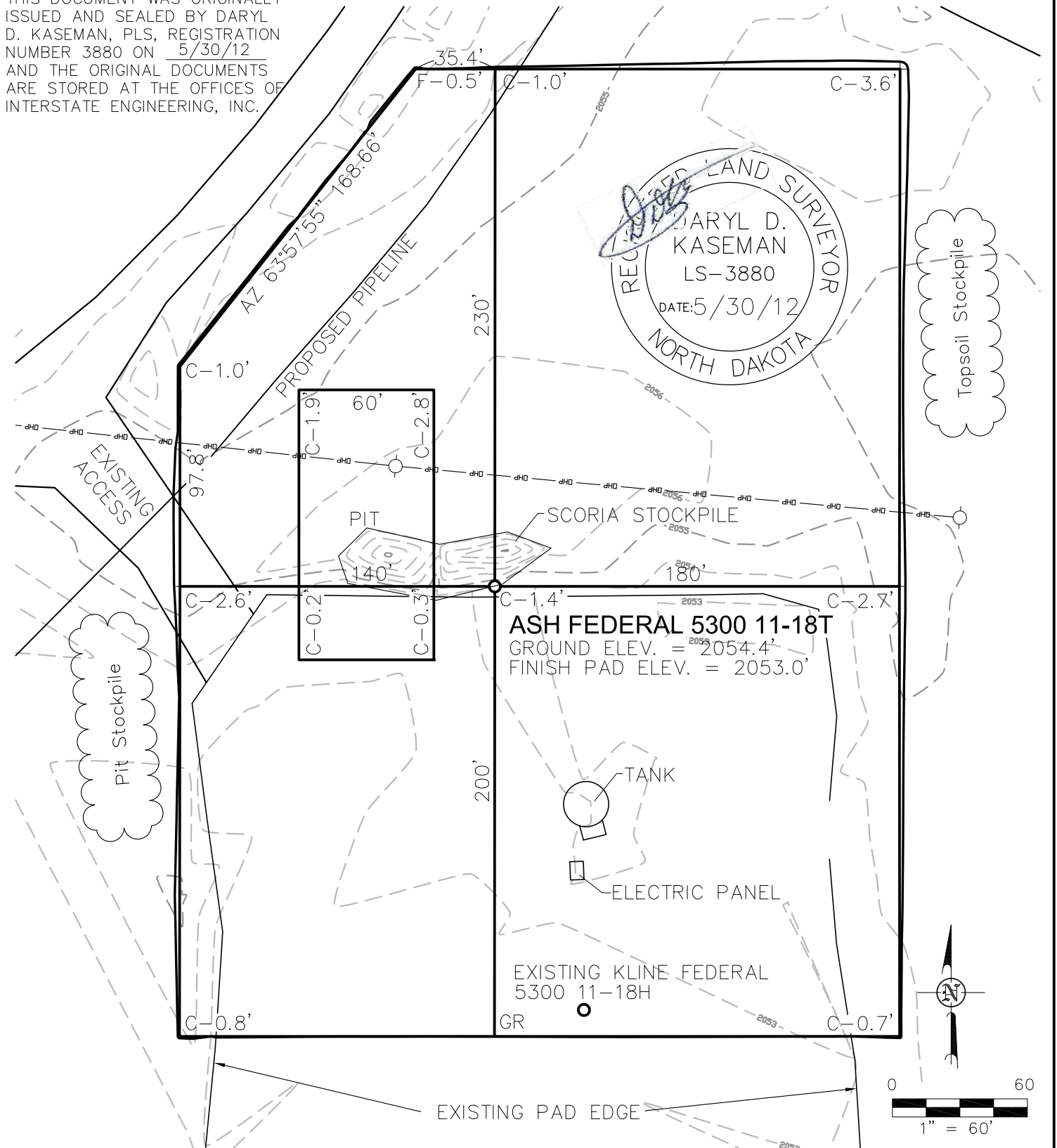
PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

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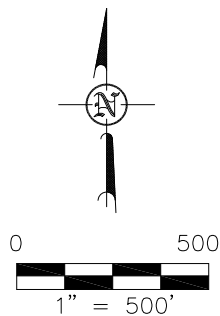
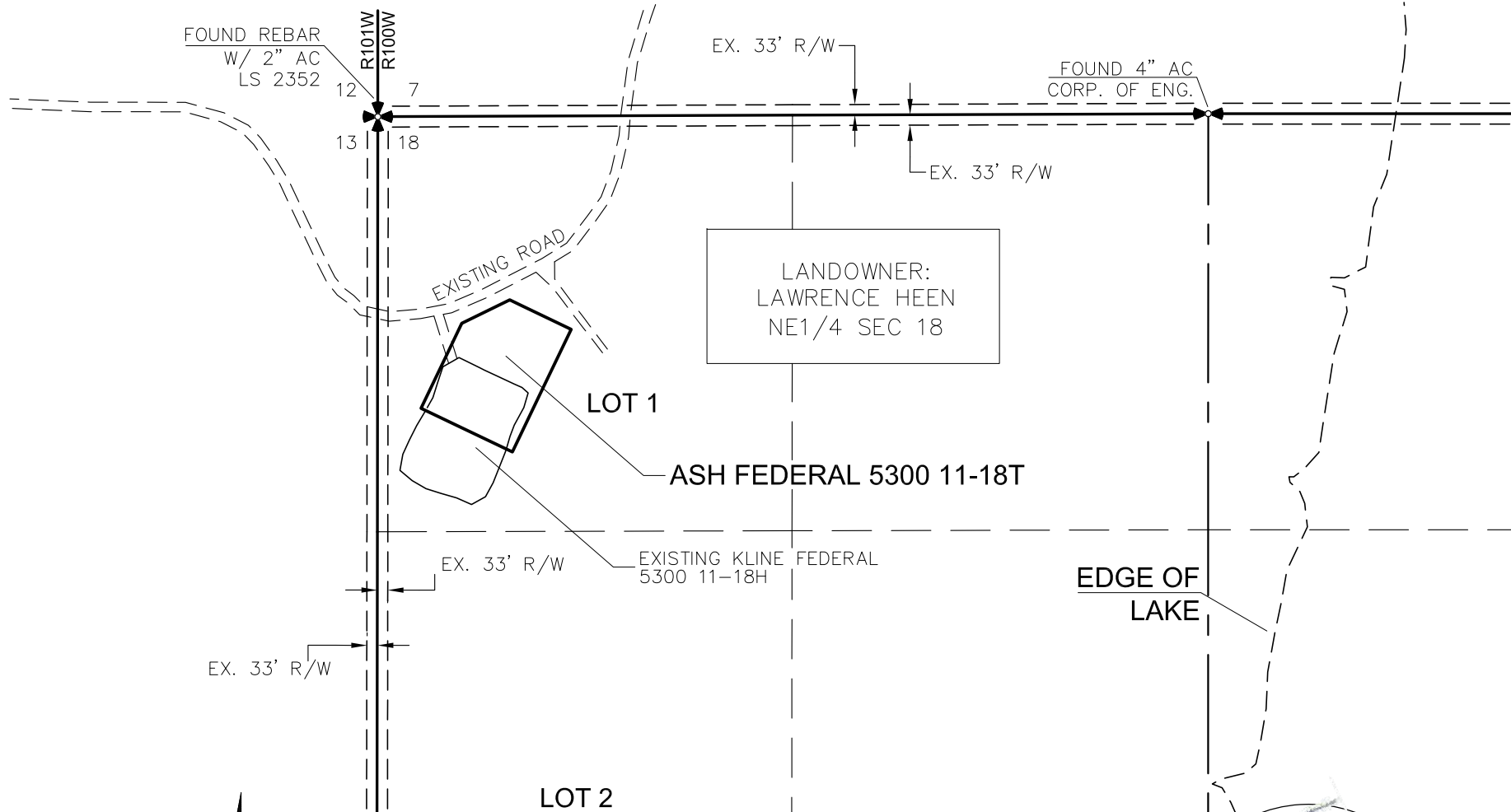
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PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S12-09-145
Checked By: D.D.K. Date: MAY 2012

Revision No.	Date	By	Description

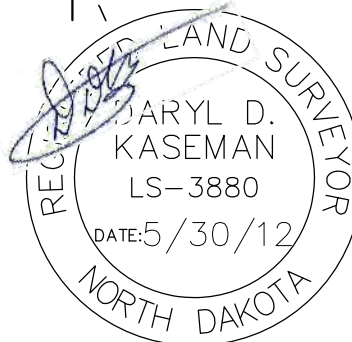
ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "ASH FEDERAL 5300 11-18T"
 800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



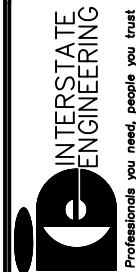
LOT 2

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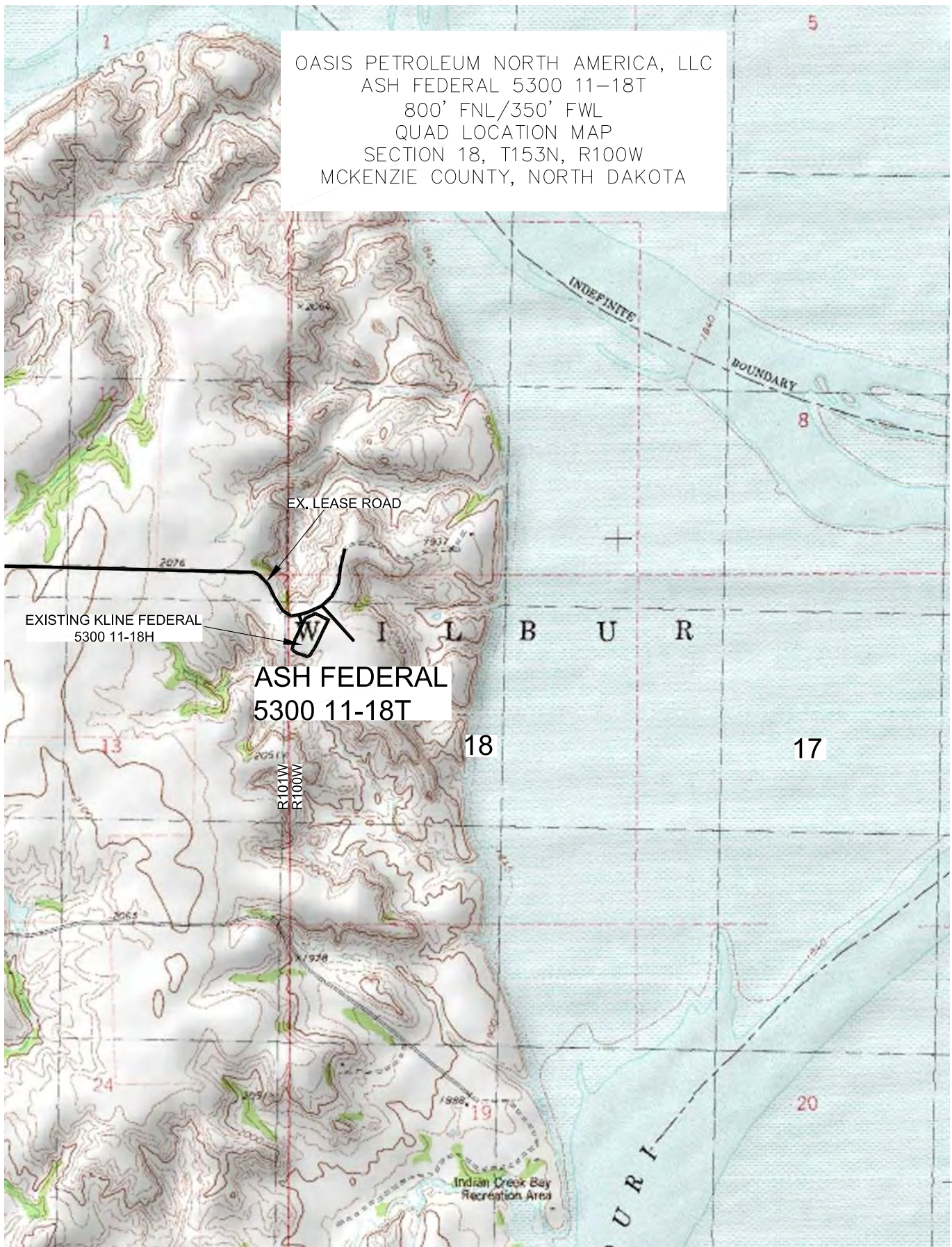
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 ACCESS APPROACH
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 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.J.H. Project No.: S1209-145
 Checked By: D.D.K. Date: MAY 2012

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
 ASH FEDERAL 5300 11-18T
 800' FNL/350' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

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 Checked By: D.D.K. Date: MAY 2012

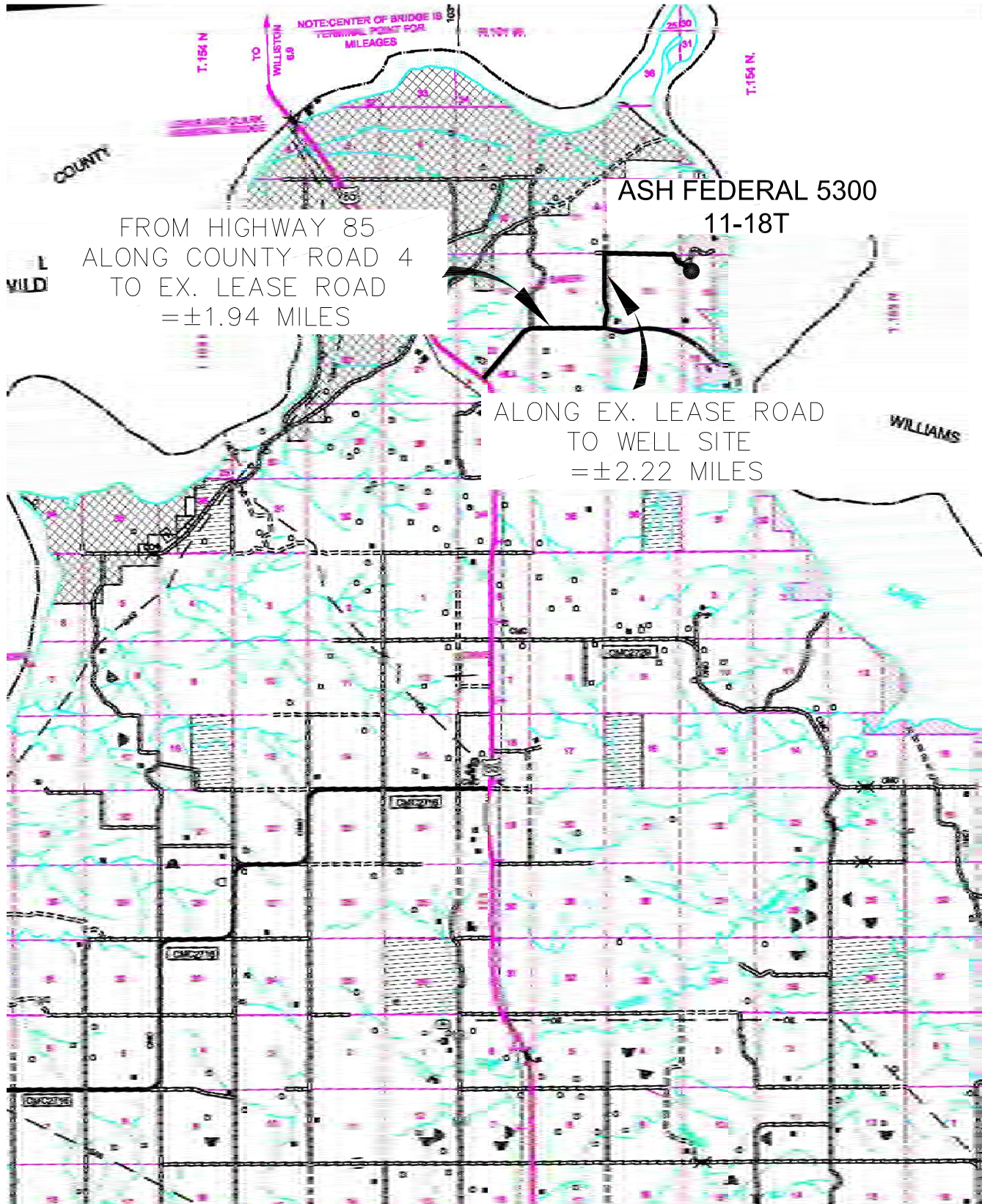
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COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"ASH FEDERAL 5300 11-18T"

800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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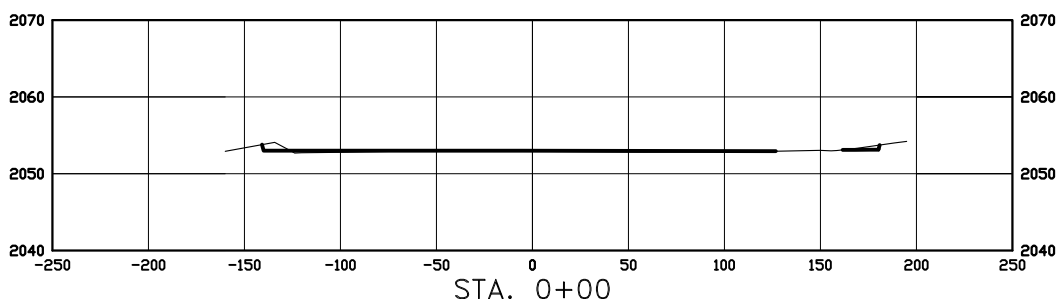
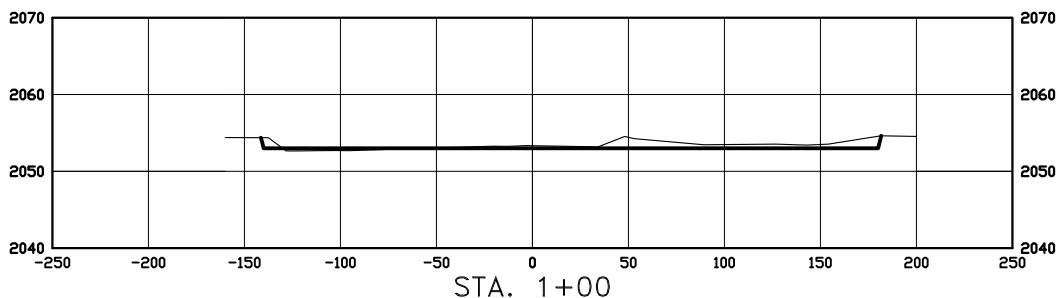
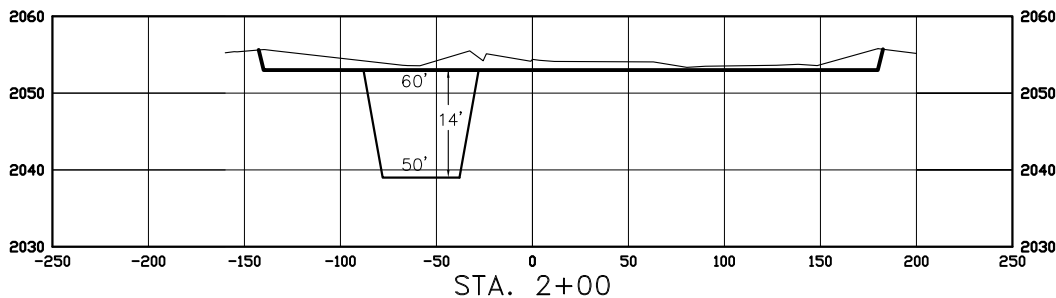
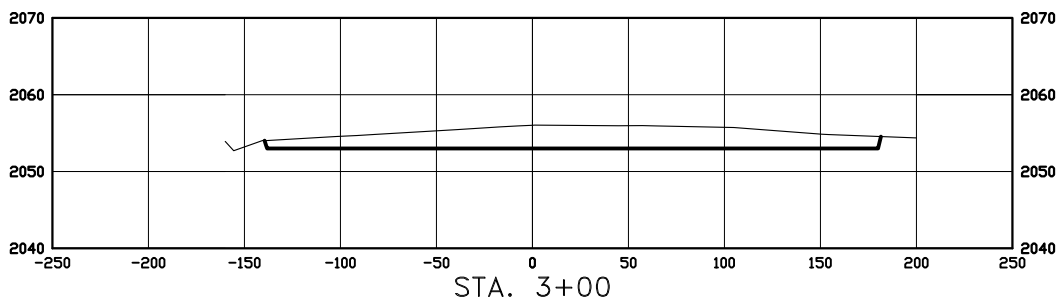
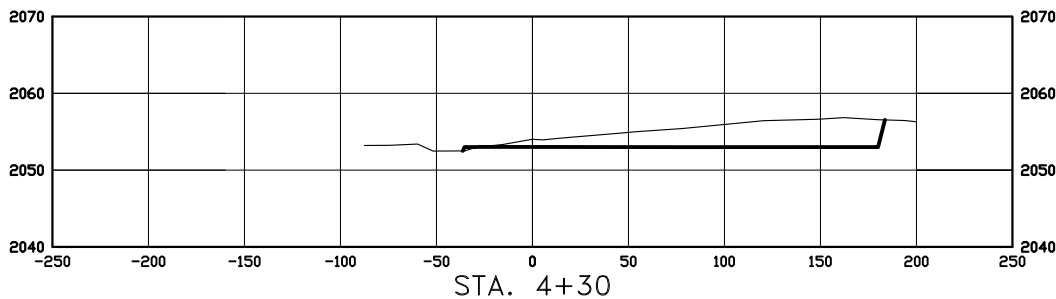
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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S12-09-145
Checked By: D.D.K. Date: MAY 2012

Revision No.	Date	By	Description

CROSS SECTIONS
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "ASH FEDERAL 5300 11-18T"
 800 FEET FROM NORTH LINE AND 350 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE
 HORIZ 1"=100'
 VERT 1"=25'

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